

Solids at Rest

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Disclaimer

Discussions taken from Alonso Sepúlveda [1], Landau & Lifschits [2], Lautrup [3] books.

1 Stress

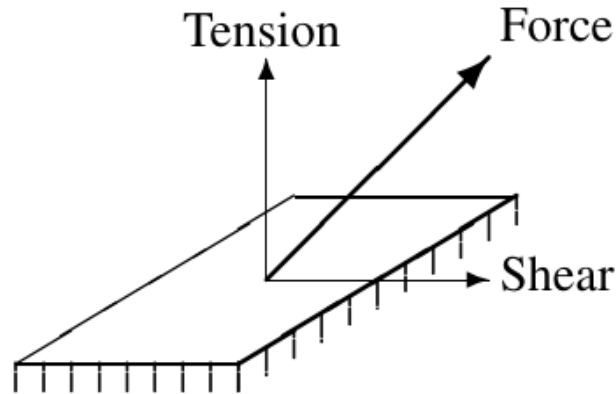


Figure 1: The force on a small piece of a surface can be resolved into a normal pressure force and a tangential shear force. Figure taken from Ref. [3].

- In fluids at rest, the only contact force is pressure, but in solids and viscous fluids in motion, additional *shear stresses* arise—forces acting tangentially to contact surfaces.
- Shear stress is the shear force per unit area and plays a crucial role in maintaining the structural integrity of solids.
- Friction in everyday life is a manifestation of shear stress at contact surfaces between materials.
- Materials are broadly classified as:
 - **Fluids**, which respond to stress by *flowing*.
 - **Solids**, which respond by *deforming*.
- Elastic deformation grows linearly with stress but can lead to permanent plastic deformation or rupture at high stress levels.
- The study of stress applies to all materials—solids, fluids, and intermediate forms—including artificial and exotic substances with special technological uses.
- Contact forces depend not only on spatial position but also on the orientation of the surface; therefore, a more general description using the *stress tensor* is required.
- The stress tensor, consisting of nine components, describes the complete range of contact forces acting within a material.
- Understanding the concept and properties of stress is fundamental to continuum mechanics and central to this chapter.

1.1 Friction

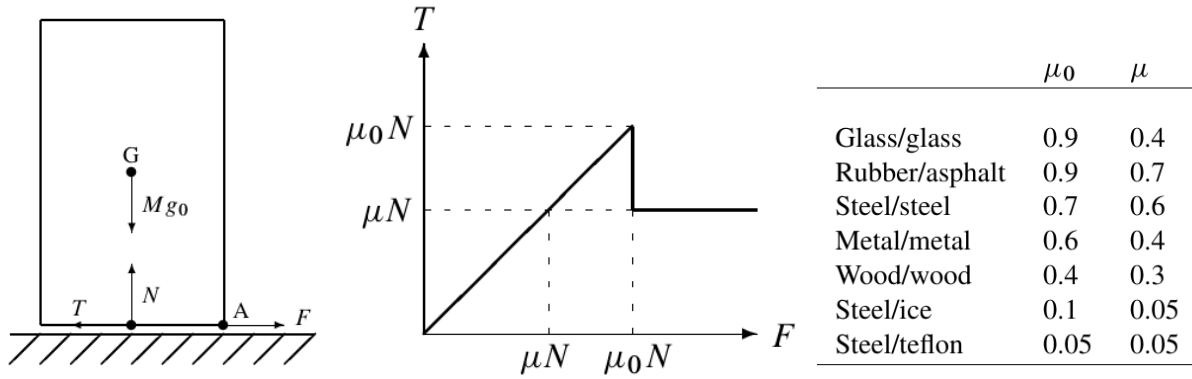


Figure 2: Left: Balance of forces on a crate at rest on a horizontal floor. Center: Sketch of tangential reaction (traction) T as a function of applied force F . Up to $F = \mu_0 N$, the traction adjusts itself to the applied force, $T = F$. At $F = \mu_0 N$, the tangential reaction drops abruptly to a lower value, $T = \mu N$, and stays there regardless of the applied force. Right: Typical friction coefficients for various combinations of materials. Figures taken from Ref. [3].

- **Friction as a shear force:**

- Friction between solid bodies is a type of shear force.
- It enables us to hold, grab, drag, and rub objects without them slipping.
- Everyday tasks, from stirring coffee to lighting a fire, involve working against friction.

- **Static friction:**

- Occurs when an object is at rest and resists being set into motion.
- The static frictional force T balances the applied force F up to a maximum value (see Fig. 2):

$$T < \mu_0 N,$$

where μ_0 is the *coefficient of static friction* and N is the normal load.

- The coefficient μ_0 is dimensionless and typically around 0.5 or higher in everyday materials.
- Its magnitude depends on the materials in contact and the roughness of their surfaces.

- **Dynamic (kinetic) friction:**

- Once motion starts, friction still acts but with a smaller magnitude than static friction.
- The frictional force is proportional to the normal load:

$$T = \mu N,$$

where μ is the *coefficient of dynamic friction*, satisfying $\mu < \mu_0$.

- After motion begins, a smaller force $F = \mu N$ maintains constant speed.

- Work done against dynamic friction is converted into heat:

$$P = FU,$$

where P is the power and U is the velocity.

- Static and dynamic friction are both independent of the size of the contact area, so that a crate on legs is as hard to drag as one without. The best way to diminish the force necessary to drag the crate is to place it on wheels.

- **Comparison:**

- Static friction prevents motion; dynamic friction resists motion once it starts.
- Static friction does not produce work, while dynamic friction continuously dissipates energy as heat.

The law of static friction dates back to Amontons (1699), while the description of kinetic friction is attributed to Coulomb (1779). Despite its apparent simplicity and its presence in everyday phenomena, friction between solid surfaces remains a complex subject, and a universally accepted microscopic explanation is still lacking [4, 5].

1.2 Stress fields

- Stress fields describe the local distribution of normal and tangential forces acting on a surface.
- Measured in pascals ($\text{Pa} = \text{N m}^{-2}$), analogous to pressure.
- Types of stresses:
 - *Tension stress*: acts along the normal and pulls outward.
 - *Pressure stress*: acts along the normal and pushes inward.
 - *Shear stress*: acts tangentially along the surface.

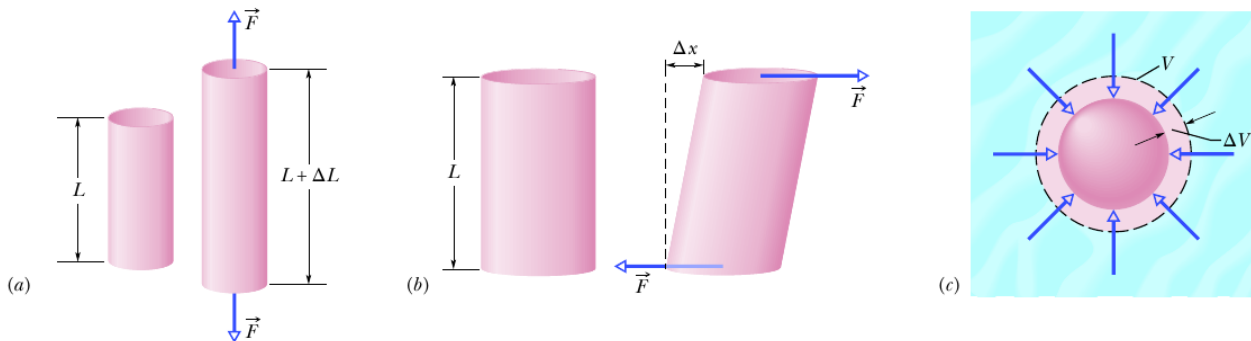


Figure 3: (a) A cylinder subject to *tensile stress* stretches by an amount ΔL . (b) A cylinder subject to *shearing stress* deforms by an amount Δx , somewhat like a pack of playing cards would. (c) A solid sphere subject to uniform *hydraulic stress* from a fluid shrinks in volume by an amount ΔV . All the deformations shown are greatly exaggerated.

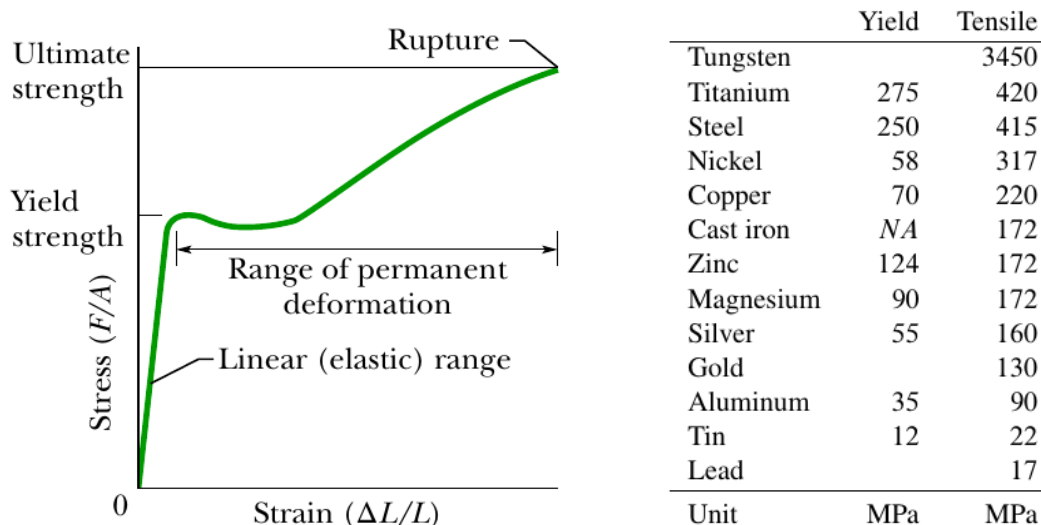


Figure 4: A stress–strain curve for a steel test specimen. The specimen deforms permanently when the stress is equal to the yield strength of the specimen’s material. It ruptures when the stress is equal to the ultimate strength of the material. Figure taken from Ref..

1.2.1 External and internal stresses:

- *External stresses* act between a body and its environment.
- *Internal stresses* act across imaginary surfaces within the body and exist even in the absence of external forces.
- Internal stresses depend on material composition, geometry, temperature, and external loading.
- Rapid temperature changes (e.g., fast cooling) can permanently “freeze” internal stresses, especially in brittle materials like glass.

1.2.2 Tensile strength and yield stress:

- When external forces grow large, materials may deform plastically or fracture.
- *Tensile strength*: the maximum stress a material can withstand before failure.
- *Yield stress*: the stress at which plastic deformation begins.
- *Ductile materials* (e.g., copper) can stretch without breaking; *brittle materials* (e.g., cast iron) fail suddenly.
- Typical tensile strengths:
 - Metals: several hundred megapascals.
 - Carbon fibers: several gigapascals.
 - Carbon nanotubes: up to 50 GPa.

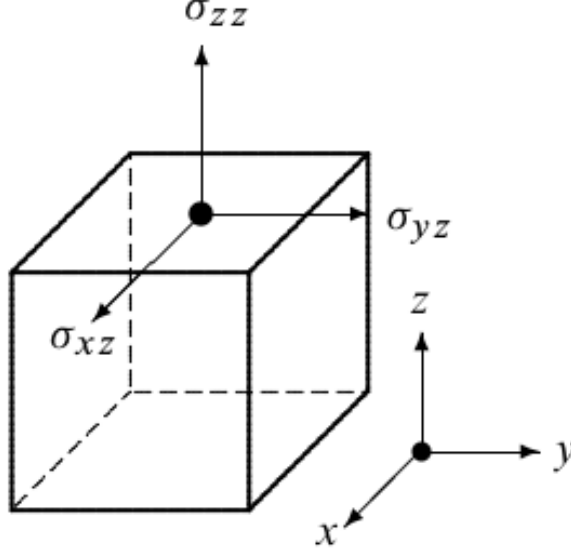


Figure 5: Components of stress acting on a surface element dS_z in the xy -plane.

1.3 The stress tensor

- Stresses at a point are described by nine components σ_{ij} in Cartesian coordinates:

$$\sigma_{ij} = \frac{dF_i}{dS_j}. \quad (1)$$

- These include three normal stresses (σ_{xx} , σ_{yy} , σ_{zz}) and six shear stresses.
- By convention, the sign is chosen such that a positive value of σ_{yy} corresponds to a pull or tension.

1.3.1 Cauchy's stress hypothesis

This asserts that the force

$$d\vec{F} = (dF_x, dF_y, dF_z) \quad (2)$$

on an arbitrary surface element,

$$d\vec{S} = (dS_x, dS_y, dS_z), \quad (3)$$

is of the form

$$\begin{aligned} dF_x &= \sigma_{xx}dS_x + \sigma_{xy}dS_y + \sigma_{xz}dS_z, \\ dF_y &= \sigma_{yx}dS_x + \sigma_{yy}dS_y + \sigma_{yz}dS_z, \\ dF_z &= \sigma_{zx}dS_x + \sigma_{zy}dS_y + \sigma_{zz}dS_z. \end{aligned} \quad (4)$$

The components of the force are expressed compactly as

$$dF_i = \sum_j \sigma_{ij} dS_j. \quad (5)$$

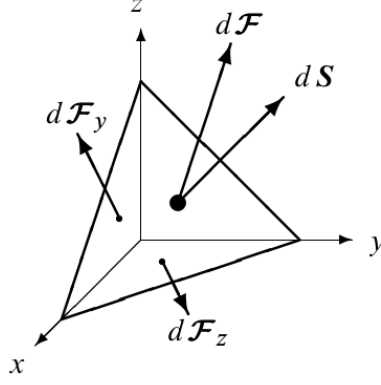


Figure 6: The tiny triangle and its projections form a tetrahedron. The (vector) force acting on the triangle in the xy -plane is called $d\vec{F}_z$, and the force acting on the triangle in the zx -plane is $d\vec{F}_y$. The force $d\vec{F}_x$ acting on the triangle in the yz -plane is hidden from view. The force $d\vec{F}$ acts on the skew triangle. Figure taken from Ref. [3].

- The nine stress components form the *stress tensor field* $\sigma_{ij}(x, t)$.
- The stress tensor is written as

$$\vec{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix}. \quad (6)$$

- The stress acting on a surface element $d\vec{S}$ is given by

$$d\vec{F} = \vec{\sigma} \cdot d\vec{S}. \quad (7)$$

- Writing $d\vec{S} = \hat{n} dS$, the stress per unit area becomes

$$\frac{d\vec{F}}{dS} = \vec{\sigma} \cdot \hat{n}, \quad (8)$$

called the *stress vector*.

- Although $\vec{\sigma} \cdot \hat{n}$ behaves like a vector, it is not a true vector field since it depends on the surface orientation.

Proof of Cauchy's stress hypothesis: Let's take again a surface element in the shape of a tiny triangle (see Fig. 6) with area vector

$$d\vec{S} = (dS_x, dS_y, dS_z).$$

This triangle and its projections on the coordinate planes form together a little body in the shape of a right tetrahedron. Let the external (vector) forces acting on the three triangular faces of the tetrahedron be denoted, respectively,

$$d\vec{F}_x, d\vec{F}_y, \text{ and } d\vec{F}_z.$$

Adding the external force $d\vec{F}$ acting on the skew face and a possible volume force $\vec{f}dV$, Newton's Second Law for the small tetrahedron becomes

$$\vec{a}_{\text{CM}} dM = \vec{f} dV + d\vec{F}_x + d\vec{F}_y + d\vec{F}_z + d\vec{F}, \quad (9)$$

where $\vec{a}_{\text{CM}}$ is the center-of-mass acceleration of the tetrahedron and $dM = \rho dV$ its mass.

The volume of the tetrahedron scales like the third power of its linear size, whereas the surface area only scales like the second power. Making the tetrahedron progressively smaller, the body force term and the acceleration term will vanish faster than the surface terms. In the limit of a truly infinitesimal tetrahedron, only the surface terms survive, so that we must have

$$d\vec{F}_x + d\vec{F}_y + d\vec{F}_z + d\vec{F} = 0.$$

Taking into account that the area projections dS_x , dS_y , and dS_z point into the tetrahedron, we define the stress vectors along the coordinate directions as

$$\vec{\sigma}_x = -\frac{d\vec{F}_x}{dS_x}, \quad \vec{\sigma}_y = -\frac{d\vec{F}_y}{dS_y}, \quad \vec{\sigma}_z = -\frac{d\vec{F}_z}{dS_z}.$$

Consequently,

$$d\vec{F} = \vec{\sigma}_x dS_x + \vec{\sigma}_y dS_y + \vec{\sigma}_z dS_z. \quad (10)$$

This shows that the force on an arbitrary surface element may be written as a linear combination of three basic stress vectors along the coordinate axes. Introducing the nine coordinates of the three stress vectors,

$$\vec{\sigma}_x = (\sigma_{xx}, \sigma_{yx}, \sigma_{zx}), \quad \vec{\sigma}_y = (\sigma_{xy}, \sigma_{yy}, \sigma_{zy}), \quad \vec{\sigma}_z = (\sigma_{xz}, \sigma_{yz}, \sigma_{zz}),$$

we arrive at Cauchy's hypothesis.

1.4 Total force

Including a volume force density f_i , the total force on a body of volume V with surface S becomes

$$F_i = \int_V f_i dV + \oint_S \sum_j \sigma_{ij} dS_j. \quad (11)$$

Using Gauss' theorem, this may be written as a single volume integral,

$$F_i = \int_V f_i^* dV, \quad (12)$$

where

$$f_i^* = f_i + \sum_j \nabla_j \sigma_{ij}, \quad (13)$$

is the *effective force density*. The effective force is not just a formal quantity, because the total force on a material particle is $d\vec{F} = \vec{f}^* dV$. In matrix notation this reads

$$\vec{f}^* = \vec{f} + \vec{\nabla} \cdot \vec{\sigma}^T, \quad (14)$$

where $\vec{\sigma}^T$ is the transposed matrix, defined as $\sigma_{ji}^T = \sigma_{ij}$.

In other words, the **total contact force** on a small box-shaped material particle (see Fig. 7) is calculated from the differences of the stress vectors acting on the sides. Suppressing the dependence on y and z , the resultant contact force on dS_x becomes

$$d\vec{F} = (\vec{\sigma}_x(x + dx) - \vec{\sigma}_x(x)) dS_x \approx \nabla_x \vec{\sigma}_x dV.$$

Adding the similar contributions from dS_y and dS_z , one arrives at the total surface force,

$$d\vec{F} = (\nabla_x \vec{\sigma}_x + \nabla_y \vec{\sigma}_y + \nabla_z \vec{\sigma}_z) dV = \nabla \cdot \vec{\sigma}^T dV.$$

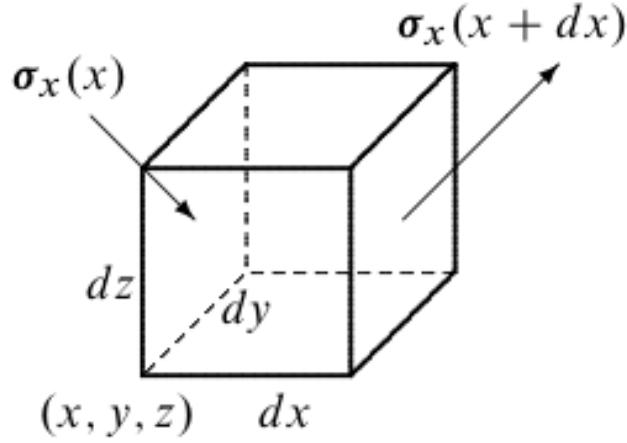


Figure 7: The total contact force on a small box-shaped material particle.

1.4.1 Mechanical Pressure

- **Hydrostatic equilibrium:** When the only contact force is pressure, the stress tensor is isotropic:

$$\boldsymbol{\sigma} = -p \mathbf{1}, \quad \text{or equivalently,} \quad \sigma_{ij} = -p \delta_{ij},$$

where $\mathbf{1}$ is the unit matrix and δ_{ij} is the Kronecker delta.

- **General case:** In real materials, the stress tensor usually has off-diagonal (shear) components. The diagonal components (σ_{xx} , σ_{yy} , σ_{zz}) differ and correspond to the normal stresses, often interpreted as “pressures”:

$$p_x = -\sigma_{xx}, \quad p_y = -\sigma_{yy}, \quad p_z = -\sigma_{zz}.$$

These components do *not* form a true vector, since they do not transform properly under rotation.

The *mechanical pressure* is defined as the average of the normal stresses:

$$p = \frac{1}{3}(p_x + p_y + p_z) = -\frac{1}{3}(\sigma_{xx} + \sigma_{yy} + \sigma_{zz}). \quad (15)$$

This definition ensures that pressure is a scalar field, invariant under Cartesian coordinate transformations, because it depends on the *trace* of the stress tensor:

$$\text{Tr } \vec{\sigma} = \sigma_{xx} + \sigma_{yy} + \sigma_{zz}. \quad (16)$$

The mechanical pressure corresponds to the isotropic part of the stress tensor. It represents the mean normal stress acting equally in all directions and is the only scalar combination of stress components that remains invariant under coordinate changes.

1.5 Mechanical equilibrium

In mechanical equilibrium, the total force on any body must vanish, for if it does not the body will begin to move. So the general condition is that $\vec{F} = 0$ for all volumes V . In particular, requiring that the force on each and every material particle must vanish, we arrive at *Cauchy's equilibrium equation(s)*:

$$f_i^* = f_i + \sum_j \nabla_j \sigma_{ij} = 0. \quad (17)$$

In spite of their simplicity, these partial differential equations govern mechanical equilibrium in all kinds of continuous matter, be it solid, fluid, or otherwise. For a fluid at rest, where pressure is the only stress component, we have $\sigma_{ij} = -p \delta_{ij}$, and recover the equation of hydrostatic equilibrium, $f_i - \nabla_i p = 0$.

The three individual equations contained in Cauchy's equilibrium equation read

$$\begin{aligned} f_x + \nabla_x \sigma_{xx} + \nabla_y \sigma_{xy} + \nabla_z \sigma_{xz} &= 0, \\ f_y + \nabla_x \sigma_{yx} + \nabla_y \sigma_{yy} + \nabla_z \sigma_{yz} &= 0, \\ f_z + \nabla_x \sigma_{zx} + \nabla_y \sigma_{zy} + \nabla_z \sigma_{zz} &= 0. \end{aligned} \quad (18)$$

These equations are in themselves not sufficient to determine the state of continuous matter, but must be supplemented by suitable *constitutive equations* connecting stress and state. For fluids at rest, the equation of state serves this purpose by relating hydrostatic pressure to mass density and temperature. In elastic solids, the constitutive equations are more complicated and relate stress to displacement.

1.5.1 Symmetry

There is one very general condition which may normally be imposed, namely the symmetry of the stress tensor:

$$\sigma_{ij} = \sigma_{ji} \text{ or } \vec{\sigma}^T = \vec{\sigma}. \quad (19)$$

Symmetry only affects the shear stress components, requiring

$$\sigma_{xy} = \sigma_{yx}, \quad \sigma_{yz} = \sigma_{zy}, \quad \sigma_{zx} = \sigma_{xz}, \quad (20)$$

and thus reduces the number of independent stress components from nine to six.

Being thus a symmetric matrix, the stress tensor may be diagonalized. The eigenvectors define the principal directions of stress and the eigenvalues the principal tensions or stresses. In the principal basis, there are no off-diagonal elements, i.e., shear stresses, only pressures. The principal basis is generally different from point to point in space.

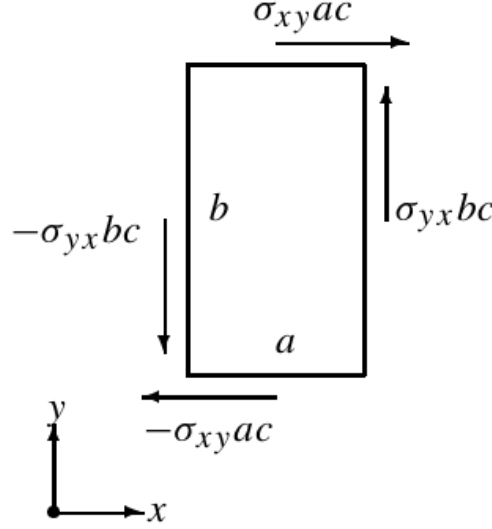


Figure 8: An asymmetric stress tensor will produce a non-vanishing moment of force on a small box (the z - direction not shown). Figure taken from Ref. [3].

There is in fact no general proof of symmetry of the stress tensor, but only some theoretical arguments that allow us to choose the stress tensor to be symmetric in all normal materials. Here we shall present a simple argument, only valid in complete mechanical equilibrium.

Consider a material particle in the shape of a tiny rectangular box with sides a , b , and c (see Fig. 8). The force acting in the y -direction on a face in the x -plane is $\sigma_{yx}bc$ whereas the force acting in the x -plane on a face in the y -plane is $\sigma_{xy}ac$. On the opposite faces the contact forces have opposite sign in mechanical equilibrium (their difference is, as we have seen, of order abc). Since the total force vanishes, the total moment of force on the box may be calculated around any point we wish. Using the lower left corner, we get

$$\mathcal{M}_z = a \sigma_{yx}bc - b \sigma_{xy}ac = (\sigma_{yx} - \sigma_{xy})abc.$$

This shows that if the stress tensor is asymmetric, $\sigma_{xy} \neq \sigma_{yx}$, there will be a resultant moment on the box. In mechanical equilibrium this cannot be allowed, since such a moment would begin to rotate the box, and consequently the stress tensor must be symmetric. Conversely, when the stress tensor is symmetric, mechanical equilibrium of the forces alone guarantees that all local moments of force will vanish.

Cylindrical coordinates:

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{1}{r} \frac{\partial \sigma_{r\theta}}{\partial \theta} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} + \frac{\partial \sigma_{rz}}{\partial z} + \rho f_r = \rho a_r, \quad (21)$$

$$\frac{\partial \sigma_{r\theta}}{\partial r} + \frac{1}{r} \frac{\partial \sigma_{\theta\theta}}{\partial \theta} + \frac{2\sigma_{r\theta}}{r} + \frac{\partial \sigma_{\theta z}}{\partial z} + \rho f_\theta = \rho a_\theta, \quad (22)$$

$$\frac{\partial \sigma_{rz}}{\partial r} + \frac{1}{r} \frac{\partial \sigma_{\theta z}}{\partial \theta} + \frac{\partial \sigma_{zz}}{\partial z} + \frac{\sigma_{rz}}{r} + \rho f_z = \rho a_z. \quad (23)$$

Spherical coordinates:

$$\frac{1}{r^2} \frac{\partial(r^2 \sigma_{rr})}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sigma_{r\theta} \sin \theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial \sigma_{r\phi}}{\partial \phi} - \frac{\sigma_{\theta\theta} + \sigma_{\phi\phi}}{r} + \rho f_r = \rho a_r, \quad (24)$$

$$\frac{1}{r^3} \frac{\partial(r^3 \sigma_{r\theta})}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sigma_{\theta\theta} \sin \theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial \sigma_{\theta\phi}}{\partial \phi} - \frac{\sigma_{\phi\phi} \cot \theta}{r} + \rho f_\theta = \rho a_\theta, \quad (25)$$

$$\frac{1}{r^3} \frac{\partial(r^3 \sigma_{r\phi})}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sigma_{\theta\phi} \sin \theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial \sigma_{\phi\phi}}{\partial \phi} + \frac{\sigma_{\theta\phi} \cot \theta}{r} + \rho f_\phi = \rho a_\phi. \quad (26)$$

Example. The stress field for an infinite elastic space loaded by a concentrated force at the origin (the Kelvin problem) is given in cylindrical coordinates by

$$\sigma_{rr} = A \left(\frac{z}{R^3} - \frac{3r^2 z}{R^5} \right), \quad \sigma_{\theta\theta} = \frac{Az}{R^3}, \quad (27)$$

$$\sigma_{zz} = -A \left(\frac{z}{R^3} + \frac{3z^3}{R^5} \right), \quad \sigma_{rz} = \sigma_{zr} = -A \left(\frac{r}{R^3} + \frac{3rz^2}{R^5} \right), \quad (28)$$

$$\sigma_{r\theta} = \sigma_{\theta r} = 0, \quad \sigma_{\theta z} = \sigma_{z\theta} = 0, \quad (29)$$

where $R^2 = r^2 + z^2$ and A is a constant related to the load. Verify that the stress field is in equilibrium away from the origin in the absence of body forces.

Solution. Since $R^2 = r^2 + z^2$,

$$\frac{\partial R}{\partial r} = \frac{r}{R}, \quad \frac{\partial R}{\partial z} = \frac{z}{R}. \quad (30)$$

For the radial equilibrium equation, we obtain

$$\frac{\partial \sigma_{rr}}{\partial r} = A \left(-\frac{9rz}{R^5} + \frac{15r^3 z}{R^7} \right), \quad (31)$$

$$\frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} = -\frac{3Arz}{R^5}, \quad (32)$$

$$\frac{\partial \sigma_{rz}}{\partial z} = A \left(-\frac{3rz}{R^5} + \frac{15rz^3}{R^7} \right). \quad (33)$$

Consequently,

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} + \frac{\partial \sigma_{rz}}{\partial z} = A \left(-\frac{15rz}{R^5} + \frac{15rz(r^2 + z^2)}{R^7} \right) = 0. \quad (34)$$

The azimuthal equilibrium equation is satisfied because $\sigma_{r\theta} = \sigma_{\theta z} = 0$ and the stress field is independent of θ . For the axial equation,

$$\frac{\partial \sigma_{rz}}{\partial r} = -A \left(\frac{1}{R^3} - \frac{3r^2}{R^5} + \frac{3z^2}{R^5} - \frac{15r^2 z^2}{R^7} \right), \quad (35)$$

$$\frac{\sigma_{rz}}{r} = -A \left(\frac{1}{R^3} + \frac{3z^2}{R^5} \right), \quad (36)$$

$$\frac{\partial \sigma_{zz}}{\partial z} = -A \left(\frac{1}{R^3} + \frac{6z^2}{R^5} - \frac{15z^4}{R^7} \right). \quad (37)$$

Adding these terms and using $R^2 = r^2 + z^2$ gives

$$\frac{\partial \sigma_{rz}}{\partial r} + \frac{\sigma_{rz}}{r} + \frac{\partial \sigma_{zz}}{\partial z} = 0. \quad (38)$$

Thus, all three equilibrium equations hold for $R > 0$. The origin is excluded because it is the location of the concentrated force.

1.5.2 Boundary Conditions

- **Cauchy's equation of mechanical equilibrium** requires appropriate boundary conditions on the surface of a body, since it is valid only within the body's volume.
- The **stress tensor** $\vec{\sigma}$ is a local physical quantity that can be assumed continuous in regions where the material properties vary smoothly.
- Across real boundaries or interfaces, where material properties may change abruptly, Newton's Third Law implies that the two sides exert equal and opposite forces on each other (in the absence of surface tension).

- The **stress vector**

$$\vec{\sigma} \cdot \hat{n} = \left\{ \sum_j \sigma_{ij} n_j \right\}$$

must therefore be continuous across a surface with normal $\hat{n}$.

- This condition is expressed as the vanishing of the surface discontinuity of the stress vector:

$$[\vec{\sigma} \cdot \hat{n}] = 0.$$

where the brackets $[\]$ denote the difference between the two sides of the surface.

- This does *not* imply that all stress tensor components are continuous—only three linear combinations (those corresponding to the stress vector) are constrained to be continuous.
- The **mechanical pressure** is, therefore, not necessarily continuous across interfaces. In general continuum mechanics, the mechanical pressure loses the simple, intuitive interpretation it had in hydrostatics.

Example Consider the following stress field in a thick-walled elastic cylinder:

$$\sigma_{rr} = A + \frac{B}{r^2}, \quad \sigma_{\theta\theta} = A - \frac{B}{r^2}, \quad \sigma_{r\theta} = \sigma_{rz} = \sigma_{\theta z} = \sigma_{zz} = 0, \quad (39)$$

where A and B are constants.

- Verify that this stress field satisfies the equations of equilibrium in the absence of body forces.
- Find the traction vector on a cylindrical surface $r = a$.
- Determine A and B if the inner surface $r = r_i$ is subject to a uniform pressure p_i , while the outer surface $r = r_o$ is free of traction.

Solution.

- (a) Only the radial equilibrium equation needs to be checked. Indeed,

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{1}{r} \frac{\partial \sigma_{r\theta}}{\partial \theta} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} + \frac{\partial \sigma_{rz}}{\partial z} = -\frac{2B}{r^3} + 0 + \frac{2B}{r^3} + 0 = 0. \quad (40)$$

The other equilibrium equations are satisfied as well.

- (b) On a surface whose unit normal is $\mathbf{n} = \mathbf{e}_r$, the traction vector is

$$\mathbf{t}(\mathbf{n}) = \boldsymbol{\sigma} \mathbf{n} = \left(A + \frac{B}{a^2} \right) \mathbf{e}_r. \quad (41)$$

- (c) The boundary conditions are

$$\sigma_{rr}(r_i) = -p_i, \quad \sigma_{rr}(r_o) = 0. \quad (42)$$

Therefore,

$$A + \frac{B}{r_i^2} = -p_i, \quad A + \frac{B}{r_o^2} = 0, \quad (43)$$

which gives

$$A = \frac{p_i r_i^2}{r_o^2 - r_i^2}, \quad B = -\frac{p_i r_i^2 r_o^2}{r_o^2 - r_i^2}. \quad (44)$$

Hence,

$$\sigma_{rr} = \frac{p_i r_i^2}{r_o^2 - r_i^2} \left(1 - \frac{r_o^2}{r^2} \right), \quad \sigma_{\theta\theta} = \frac{p_i r_i^2}{r_o^2 - r_i^2} \left(1 + \frac{r_o^2}{r^2} \right). \quad (45)$$

Example. The equilibrium stress field in an elastic spherical shell under internal and external pressure, in the absence of body forces, has the form

$$\sigma_{rr} = A - \frac{2B}{r^3}, \quad \sigma_{\theta\theta} = \sigma_{\phi\phi} = A + \frac{B}{r^3}, \quad \sigma_{r\theta} = \sigma_{r\phi} = \sigma_{\theta\phi} = 0. \quad (46)$$

- (a) Verify that this field satisfies the equations of equilibrium in the absence of body forces.
(b) Find the traction vector on a spherical surface $r = a$.
(c) Determine A and B if the inner surface $r = r_i$ is subject to a uniform pressure p_i , while the outer surface $r = r_o$ is free of traction.

Solution.

- (a) The radial equilibrium equation reduces to

$$\frac{1}{r^2} \frac{d}{dr} (r^2 \sigma_{rr}) - \frac{\sigma_{\theta\theta} + \sigma_{\phi\phi}}{r} = 0. \quad (47)$$

Its left-hand side is

$$\frac{1}{r^2} \frac{d}{dr} \left(A r^2 - \frac{2B}{r} \right) - \frac{2}{r} \left(A + \frac{B}{r^3} \right) = \left(\frac{2A}{r} + \frac{2B}{r^4} \right) - \left(\frac{2A}{r} + \frac{2B}{r^4} \right) = 0. \quad (48)$$

By spherical symmetry, the angular equilibrium equations are also satisfied.

(b) For $\mathbf{n} = \mathbf{e}_r$, the traction vector on $r = a$ is

$$\mathbf{t}(\mathbf{n}) = \boldsymbol{\sigma}\mathbf{n} = \left(A - \frac{2B}{a^3}\right) \mathbf{e}_r. \quad (49)$$

(c) The boundary conditions give

$$A - \frac{2B}{r_i^3} = -p_i, \quad A - \frac{2B}{r_o^3} = 0. \quad (50)$$

Solving for the constants,

$$A = \frac{p_i r_i^3}{r_o^3 - r_i^3}, \quad B = \frac{p_i r_i^3 r_o^3}{2(r_o^3 - r_i^3)}. \quad (51)$$

Thus,

$$\sigma_{rr} = \frac{p_i r_i^3}{r_o^3 - r_i^3} \left(1 - \frac{r_o^3}{r^3}\right), \quad \sigma_{\theta\theta} = \sigma_{\phi\phi} = \frac{p_i r_i^3}{r_o^3 - r_i^3} \left(1 + \frac{r_o^3}{2r^3}\right). \quad (52)$$

1.6 Indicial notation

1. Given

$$[S_{ij}] = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 3 & 0 & 3 \end{pmatrix}, \quad [a_i] = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix},$$

evaluate:

- (a) $S_{ii} = \sum_{i=1}^3 S_{ii}$
- (b) $S_{ij}S_{ij} = \sum_{i=1}^3 \sum_{j=1}^3 S_{ij}S_{ij}$
- (c) $S_{ij}S_{ji} = \sum_{i=1}^3 \sum_{j=1}^3 S_{ij}S_{ji}$
- (d) $S_{jk}S_{kj} = \sum_{j=1}^3 \sum_{k=1}^3 S_{jk}S_{kj}$
- (e) $a_m a_m = \sum_{m=1}^3 a_m a_m$
- (f) $S_{mn}a_m a_n = \sum_{m=1}^3 \sum_{n=1}^3 S_{mn}a_m a_n$
- (g) $S_{nm}a_m a_n = \sum_{n=1}^3 \sum_{m=1}^3 S_{nm}a_m a_n$

2. Determine which of the following equations has an identical meaning with

$$a_i = \sum_{j=1}^3 Q_{ij}a'_j.$$

- (a) $a_p = \sum_{m=1}^3 Q_{pm}a'_m$
- (b) $a_p = \sum_{q=1}^3 Q_{qp}a'_q$
- (c) $a_m = \sum_{n=1}^3 a'_n Q_{mn}$

3. Given

$$[a_i] = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \quad [B_{ij}] = \begin{pmatrix} 2 & 3 & 0 \\ 0 & 5 & 1 \\ 0 & 2 & 1 \end{pmatrix},$$

demonstrate the equivalence:

$$(a) \quad b_i = \sum_{j=1}^3 B_{ij} a_j, \quad [\vec{b}] = [B][\vec{a}]$$

$$(b) \quad s = \sum_{i=1}^3 \sum_{j=1}^3 B_{ij} a_i a_j, \quad s = [\vec{a}]^T [B] [\vec{a}]$$

4. Write in indicial notation:

$$(a) \quad [A] = [B][C].$$

$$(b) \quad [D] = [B]^T [C].$$

$$(c) \quad [E] = [B]^T [C][F].$$

5. Write in indicial notation:

$$(a) \quad s = A_1^2 + A_2^2 + A_3^2$$

$$(b) \quad \frac{\partial^2 \phi}{\partial x_1^2} + \frac{\partial^2 \phi}{\partial x_2^2} + \frac{\partial^2 \phi}{\partial x_3^2}$$

6. Given

$$S_{ij} = a_i a_j, \quad S'_{ij} = a'_i a'_j,$$

where

$$a'_i = \sum_{m=1}^3 Q_{mi} a_m, \quad a'_j = \sum_{n=1}^3 Q_{nj} a_n,$$

and

$$\sum_{k=1}^3 Q_{ik} Q_{jk} = \delta_{ij},$$

show that

$$S'_{ii} = \sum_{i=1}^3 S_{ii}.$$

7. Write in long form:

$$a_i = \frac{\partial v_i}{\partial t} + \sum_{j=1}^3 v_j \frac{\partial v_i}{\partial x_j}.$$

8. Given

$$T_{ij} = 2\mu E_{ij} + \lambda \left(\sum_{k=1}^3 E_{kk} \right) \delta_{ij},$$

show that

$$(a) \quad T_{ij} E_{ij} = 2\mu \sum_{i,j} E_{ij} E_{ij} + \lambda (\sum_k E_{kk})^2$$

$$(b) \quad T_{ij} T_{ij} = 4\mu^2 \sum_{i,j} E_{ij} E_{ij} + (4\mu\lambda + 3\lambda^2) (\sum_k E_{kk})^2$$

9. Given

$$a_i = \sum_{j=1}^3 T_{ij} b_j, \quad a'_i = \sum_{j=1}^3 T'_{ij} b'_j,$$

where

$$a_i = \sum_{m=1}^3 Q_{im} a'_m, \quad T_{ij} = \sum_{m,n=1}^3 Q_{im} Q_{jn} T'_{mn},$$

(a) show that

$$\sum_{m,n=1}^3 Q_{im} T'_{mn} b'_n = \sum_{m,n=1}^3 Q_{im} Q_{jn} T'_{mn} b_j$$

(b) if $\sum_{k=1}^3 Q_{ik} Q_{mk} = \delta_{im}$, show that

$$T'_{kn} (b'_n - \sum_{j=1}^3 Q_{nj} b_j) = 0$$

10. Given

$$[\vec{a}] = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad [\vec{b}] = \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix},$$

evaluate d_k , where

$$d_k = \sum_{i=1}^3 \sum_{j=1}^3 \varepsilon_{ijk} a_i b_j,$$

and show that

$$d_k = (\vec{a} \times \vec{b}) \cdot \vec{e}_k.$$

11. (a) If

$$\sum_{i=1}^3 \sum_{j=1}^3 \varepsilon_{ijk} T_{ij} = 0,$$

show that

$$T_{ij} = T_{ji}.$$

(b) Show that

$$\sum_{i=1}^3 \delta_{ij} \varepsilon_{ijk} = 0.$$

12. Verify the following equation:

$$\sum_{m=1}^3 \varepsilon_{ijm} \varepsilon_{klm} = \delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk}.$$

Hint: There are six cases to be considered:

(1) $i = j$

(2) $i = k$

(3) $i = l$

(4) $j = k$

(5) $j = l$

(6) $k = l$

13. Use the identity

$$\sum_{m=1}^3 \varepsilon_{ijm} \varepsilon_{klm} = \delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk}$$

as a shortcut to obtain the following results:

(a)

$$\sum_{l=1}^3 \sum_{m=1}^3 \varepsilon_{ilm} \varepsilon_{jlm} = 2\delta_{ij}$$

(b)

$$\sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{ijk} \varepsilon_{ijk} = 6$$

14. Use the identity

$$\sum_{m=1}^3 \varepsilon_{ijm} \varepsilon_{klm} = \delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk}$$

to show that

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}.$$

15. Show that

(a) if

$$T_{ij} = -T_{ji},$$

then

$$\sum_{i=1}^3 \sum_{j=1}^3 T_{ij} a_i a_j = 0$$

(b) if

$$T_{ij} = -T_{ji}, \quad S_{ij} = S_{ji},$$

then

$$\sum_{i=1}^3 \sum_{j=1}^3 T_{ij} S_{ij} = 0$$

16. Let

$$T_{ij} = \frac{1}{2}(S_{ij} + S_{ji}), \quad R_{ij} = \frac{1}{2}(S_{ij} - S_{ji}),$$

show that

$$T_{ij} = T_{ji}, \quad R_{ij} = -R_{ji}, \quad S_{ij} = T_{ij} + R_{ij}.$$

17. Let $f(x_1, x_2, x_3)$ be a function of x_1 , x_2 , and x_3 , and let $v_i(x_1, x_2, x_3)$ be three functions of x_1 , x_2 , and x_3 . Express the total differential df and dv_i in indicial notation:

$$df = \sum_{i=1}^3 \frac{\partial f}{\partial x_i} dx_i, \quad dv_i = \sum_{j=1}^3 \frac{\partial v_i}{\partial x_j} dx_j.$$

18. Let $|A_{ij}|$ denote the determinant of the matrix $[A_{ij}]$. Show that

$$|A_{ij}| = \sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{ijk} A_{1i} A_{2j} A_{3k}.$$

1.7 Dyads

1.7.1 Dyadic product

Between orthonormal unit vectors in three-dimensional space, three types of products can be defined.

Scalar

$$\hat{e}_i \cdot \hat{e}_j = \delta_{ij}.$$

Since $\vec{A} = \sum_{i=1}^3 A_i \hat{e}_i$ and $\vec{B} = \sum_{i=1}^3 B_i \hat{e}_i$, it follows that:

$$\vec{A} \cdot \vec{B} = \sum_{i=1}^3 \sum_{j=1}^3 \hat{e}_i \cdot \hat{e}_j A_i B_j = \sum_{i=1}^3 A_i B_i.$$

Vector

$$\hat{e}_i \times \hat{e}_j = \sum_{k=1}^3 \varepsilon_{ijk} \hat{e}_k, \tag{53}$$

such that:

$$\vec{A} \times \vec{B} = \sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{ijk} A_i B_j \hat{e}_k.$$

ε_{ijk} is the Levi-Civita symbol, defined by $\varepsilon_{123} = 1$ and antisymmetric under exchange of contiguous indices. For example,

$$\varepsilon_{123} = -\varepsilon_{213} = \varepsilon_{132}, \quad \varepsilon_{111} = 0.$$

Note on Levi-Civita

From the identity

$$\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C},$$

it follows, substituting $\vec{A} = \sum_{i=1}^3 A_i \hat{e}_i$, etc., and taking into account Eq. (53):

$$\sum_{i,j,l,m=1}^3 \left[\sum_{k=1}^3 \varepsilon_{ijk} \varepsilon_{lmk} - \delta_{il} \delta_{jm} + \delta_{im} \delta_{jl} \right] A_j B_l C_m \hat{e}_i = 0. \quad (54)$$

Thus,

$$\sum_{k=1}^3 \varepsilon_{ijk} \varepsilon_{lmk} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}. \quad (55)$$

This equation may also be written as:

$$\sum_{k=1}^3 \varepsilon_{ijk} \varepsilon_{lmk} = \begin{vmatrix} \delta_{il} & \delta_{im} \\ \delta_{jl} & \delta_{jm} \end{vmatrix}. \quad (56)$$

It is easy to show that:

$$\sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{ijk} \varepsilon_{ljk} = \sum_{j=1}^3 \begin{vmatrix} \delta_{il} & \delta_{ij} \\ \delta_{jl} & \delta_{jj} \end{vmatrix} = 2\delta_{il}, \quad (57)$$

$$\sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{ijk} \varepsilon_{ijk} = 6, \quad (58)$$

$$\sum_{j=1}^3 \delta_{ij} \varepsilon_{ijk} = 0. \quad (59)$$

Equation above is the sum over k (with $n = k$) of the expression:

$$\varepsilon_{ijk} \varepsilon_{lmn} = \begin{vmatrix} \delta_{il} & \delta_{im} & \delta_{in} \\ \delta_{jl} & \delta_{jm} & \delta_{jn} \\ \delta_{kl} & \delta_{km} & \delta_{kn} \end{vmatrix}. \quad (60)$$

The Levi-Civita symbol can be used to express determinants. Indeed,

$$|A| = \sum_{l=1}^3 \sum_{m=1}^3 \sum_{n=1}^3 \varepsilon_{lmn} A_{1l} A_{2m} A_{3n}. \quad (61)$$

Using this expression, it is straightforward to show that:

$$|AB| = |A||B|.$$

In Special Relativity, the symbol $\varepsilon_{\mu\nu\sigma\rho}$, where μ, ν, σ, ρ take values between 0 and 3, is defined analogously:

$$\varepsilon_{0123} = 1,$$

and it is antisymmetric under exchange of any pair of adjacent indices. It can be shown that

$$\varepsilon^{\mu\nu\sigma\rho} = -1,$$

and, with δ_μ^α the Kronecker delta, one has:

$$\varepsilon_{\mu\nu\sigma\rho}\varepsilon^{\alpha\beta\gamma\delta} = \begin{vmatrix} \delta_\mu^\alpha & \delta_\mu^\beta & \delta_\mu^\gamma & \delta_\mu^\delta \\ \delta_\nu^\alpha & \delta_\nu^\beta & \delta_\nu^\gamma & \delta_\nu^\delta \\ \delta_\sigma^\alpha & \delta_\sigma^\beta & \delta_\sigma^\gamma & \delta_\sigma^\delta \\ \delta_\rho^\alpha & \delta_\rho^\beta & \delta_\rho^\gamma & \delta_\rho^\delta \end{vmatrix}. \quad (62)$$

Problem: Using the identity above, show that:

$$(\vec{A} \times \vec{B}) \cdot (\vec{A} \times \vec{B}) = A^2 B^2 - (\vec{A} \cdot \vec{B})^2, \quad (63)$$

$$(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C}). \quad (64)$$

Dyadic

It yields a bilinear form in the basis vectors:

$$\vec{A}\vec{B} = \left(\sum_{i=1}^3 A_i \hat{e}_i \right) \left(\sum_{j=1}^3 B_j \hat{e}_j \right) = \sum_{i=1}^3 \sum_{j=1}^3 A_i B_j \hat{e}_i \hat{e}_j.$$

$\vec{A}\vec{B}$ is known as a dyad. It is a bilinear form in $\hat{e}_i$. In general, without reference to the product of two vectors, a dyad is defined as:

$$\mathbf{T} = \sum_{i=1}^3 \sum_{j=1}^3 T_{ij} \hat{e}_i \hat{e}_j.$$

$$\tilde{\mathbf{T}} = \sum_{i=1}^3 \sum_{j=1}^3 T_{ji} \hat{e}_i \hat{e}_j$$

is the transpose of the dyad $\mathbf{T}$, and T_{ij} are its components.

A dyad of particular interest is the identity (or unit), written as:

$$\mathbf{I} = \sum_{i=1}^3 \hat{e}_i \hat{e}_i = \sum_{i=1}^3 \sum_{j=1}^3 \delta_{ij} \hat{e}_i \hat{e}_j.$$

In Cartesian, cylindrical, and spherical coordinates, it is written as:

$$\mathbf{I} = \hat{\mathbf{i}}\hat{\mathbf{i}} + \hat{\mathbf{j}}\hat{\mathbf{j}} + \hat{\mathbf{k}}\hat{\mathbf{k}} \quad (65)$$

$$= \hat{e}_\rho \hat{e}_\rho + \hat{e}_\phi \hat{e}_\phi + \hat{e}_z \hat{e}_z \quad (66)$$

$$= \hat{e}_r \hat{e}_r + \hat{e}_\theta \hat{e}_\theta + \hat{e}_\phi \hat{e}_\phi. \quad (67)$$

Thus, the components of the identity dyad are the elements of the Kronecker delta. It is true that:

$$\mathbf{I} \cdot \vec{A} = \vec{A} \cdot \mathbf{I} = \vec{A},$$

and, in general,

$$\mathbf{T} \cdot \vec{A} \neq \vec{A} \cdot \mathbf{T},$$

unless $\mathbf{T}$ is symmetric, i.e., $T_{ij} = T_{ji}$.

The scalar product between two dyads is a dyad:

$$\begin{aligned}\mathbf{T} \cdot \mathbf{M} &= \sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 \sum_{\ell=1}^3 T_{ij} M_{k\ell} (\hat{e}_i \hat{e}_j \cdot \hat{e}_k \hat{e}_\ell) \\ &= \sum_{i,j,k,\ell=1}^3 T_{ij} M_{k\ell} \hat{e}_i \hat{e}_\ell \delta_{jk} = \sum_{i=1}^3 \sum_{j=1}^3 \sum_{\ell=1}^3 T_{ij} M_{j\ell} \hat{e}_i \hat{e}_\ell.\end{aligned}$$

A double scalar product between dyads can also be defined according to:

$$\hat{e}_i \hat{e}_j : \hat{e}_k \hat{e}_\ell = \delta_{il} \delta_{jk},$$

such that:

$$\mathbf{T} : \mathbf{M} = \sum_{i=1}^3 \sum_{j=1}^3 T_{ij} M_{ji},$$

which is a scalar.

It can be easily shown that:

- The divergence of a dyad is a vector.
- The gradient of a vector is a dyad.
- The curl of a dyad is a dyad.
- $\vec{a} \cdot \mathbf{M} \cdot \vec{b} = \vec{b} \cdot \mathbf{M} \cdot \vec{a} = \mathbf{M} : (\vec{b} \vec{a})$.

Example

Consider the tensor (the dyadic product of $\vec{c}$ and $\vec{d}$):

$$\mathbf{T} = \vec{c} \vec{d}.$$

- Obtain the determinant of $\mathbf{T}$.
- Show that if $\mathbf{T} \cdot \vec{a} = \vec{b}$, then $\mathbf{T} \cdot (\vec{a} + \vec{h}) = \vec{b}$, where $\vec{h}$ is any vector perpendicular to $\vec{d}$.

Solution

(a)

$$[\mathbf{T}] = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} \begin{pmatrix} d_1 & d_2 & d_3 \end{pmatrix} = \begin{pmatrix} c_1 d_1 & c_1 d_2 & c_1 d_3 \\ c_2 d_1 & c_2 d_2 & c_2 d_3 \\ c_3 d_1 & c_3 d_2 & c_3 d_3 \end{pmatrix} \quad (68)$$

$$\det(\mathbf{T}) = c_1 c_2 c_3 d_1 d_2 d_3 \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0. \quad (69)$$

That is, $\mathbf{T}$ is a singular tensor, for which an inverse does not exist.

(b)

$$\mathbf{T} \cdot (\vec{a} + \vec{h}) = (\vec{c} \vec{d}) \cdot (\vec{a} + \vec{h}) \quad (70)$$

$$= \vec{c}(\vec{d} \cdot \vec{a}) + \vec{c}(\vec{d} \cdot \vec{h}). \quad (71)$$

Since

$$\vec{d} \cdot \vec{h} = 0 \quad (\vec{h} \perp \vec{d}),$$

it follows that

$$\mathbf{T} \cdot (\vec{a} + \vec{h}) = \vec{c}(\vec{d} \cdot \vec{a}) = (\vec{c} \vec{d}) \cdot \vec{a} = \mathbf{T} \cdot \vec{a} = \vec{b}. \quad (72)$$

That is, all vectors $\vec{a} + \vec{h}$ transform into the same vector $\vec{b}$, so the transformation is not one-to-one.

Exercise

Show that $\mathbf{T}$ is invariant under coordinate transformations.

From the transformation rules it follows that:

$$\mathbf{T}' = \sum_{i=1}^3 \sum_{j=1}^3 \hat{e}'_i \hat{e}'_j T'_{ij} \quad (73)$$

$$= \sum_{i,j,k,l,m,n=1}^3 a_{ik} a_{jl} \hat{e}_k \hat{e}_l a_{im} a_{jn} T_{mn} \quad (74)$$

$$= \sum_{k=1}^3 \sum_{l=1}^3 \hat{e}_k \hat{e}_l T_{kl} = \mathbf{T}. \quad (75)$$

1.7.2 Principal Scalar Invariants of a Tensor

The characteristic equation of a tensor $\mathbf{T}$,

$$|T_{ij} - \lambda \delta_{ij}| = 0,$$

can be written as:

$$\lambda^3 - I_1 \lambda^2 + I_2 \lambda - I_3 = 0, \quad (76)$$

where

$$I_1 = T_{11} + T_{22} + T_{33} = \sum_{i=1}^3 T_{ii} = \text{tr } \mathbf{T}, \quad (77)$$

$$I_2 = \begin{vmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{vmatrix} + \begin{vmatrix} T_{22} & T_{23} \\ T_{32} & T_{33} \end{vmatrix} + \begin{vmatrix} T_{11} & T_{13} \\ T_{31} & T_{33} \end{vmatrix} \quad (78)$$

$$= \frac{1}{2} (T_{ii} T_{jj} - T_{ij} T_{ji}) = \frac{1}{2} [(\text{tr } \mathbf{T})^2 - \text{tr}(\mathbf{T}^2)], \quad (79)$$

$$I_3 = \begin{vmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{vmatrix} = \det(\mathbf{T}). \quad (80)$$

Since, by definition, the eigenvalues of $\mathbf{T}$ do not depend on the choice of basis vectors, the coefficients of the characteristic equation do not depend on any particular basis. They are called the *principal scalar invariants* of $\mathbf{T}$.

We note that, in terms of the eigenvalues $\lambda_1, \lambda_2, \lambda_3$, the invariants take the form:

$$I_1 = \lambda_1 + \lambda_2 + \lambda_3, \quad (81)$$

$$I_2 = \lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1, \quad (82)$$

$$I_3 = \lambda_1\lambda_2\lambda_3. \quad (83)$$

Example

For the tensor $\mathbf{T}$

$$[\mathbf{T}] = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & -3 \end{pmatrix},$$

first find the principal scalar invariants and then evaluate the eigenvalues using the characteristic equation.

Solution

Thus,

$$I_1 = 2 + 3 - 3 = 2, \quad (84)$$

$$I_2 = \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} + \begin{vmatrix} 3 & 4 \\ 4 & -3 \end{vmatrix} + \begin{vmatrix} 2 & 0 \\ 0 & -3 \end{vmatrix} \quad (85)$$

$$= (6) + (-9 - 16) + (-6) = -25, \quad (86)$$

$$I_3 = \det(\mathbf{T}) = -50. \quad (87)$$

These values give the characteristic equation:

$$\lambda^3 - 2\lambda^2 - 25\lambda + 50 = 0. \quad (88)$$

or

$$(\lambda - 2)(\lambda - 5)(\lambda + 5) = 0. \quad (89)$$

Thus, the eigenvalues are

$$\lambda = 2, \quad \lambda = 5, \quad \lambda = -5.$$

1.7.3 Dyadic Integral Theorems

Just as vector identities have their corresponding extension to dyads, there are also integral theorems involving dyads. For example, starting from the divergence theorem,

$$\int \nabla \cdot \vec{A} dV = \oint d\vec{S} \cdot \vec{A}, \quad (90)$$

and taking $\vec{A} = \mathbf{T} \cdot \vec{a}$, where $\mathbf{T}$ is a dyadic field and $\vec{a}$ is a constant vector (in Cartesian coordinates), it is possible to show that Gauss's theorem for the dyad $\mathbf{T}$ takes the form:

$$\int \nabla \cdot \mathbf{T} dV = \oint d\vec{S} \cdot \mathbf{T}. \quad (91)$$

It is also true that:

$$\int \nabla \times \mathbf{T} dV = \oint d\vec{S} \times \mathbf{T}, \quad (92)$$

$$\int d\vec{S} \cdot (\nabla \times \mathbf{T}) = \oint d\vec{\ell} \cdot \mathbf{T}. \quad (93)$$

The last equation corresponds to the dyadic version of Stokes' theorem.

It can be shown that the three integral theorems are valid in orthogonal curvilinear coordinates.

We now develop some integral theorems that are useful in the study of dyadic Green functions.

From the divergence theorem for the dyad $\mathbf{T}$,

$$\int \nabla \cdot \mathbf{T} dV = \oint d\vec{S} \cdot \mathbf{T},$$

it follows:

(A) For $\mathbf{T} = \vec{A}\vec{B}$:

$$\int [\vec{B}(\nabla \cdot \vec{A}) + (\vec{A} \cdot \nabla)\vec{B}] dV = \oint (\hat{n} \cdot \vec{A})\vec{B} dS. \quad (94)$$

(B) For $\mathbf{T} = \vec{A} \times \vec{U}$:

$$\int [(\nabla \times \vec{A}) \cdot \vec{U} - \vec{A} \cdot (\nabla \times \vec{U})] dV = \oint \hat{n} \cdot (\vec{A} \times \vec{U}) dS \quad (95)$$

$$= \oint (\hat{n} \times \vec{A}) \cdot \vec{U} dS. \quad (96)$$

In the last step we used the identity

$$\vec{B} \cdot (\vec{A} \times \vec{U}) = \vec{B} \times (\vec{A} \cdot \vec{U}).$$

(C) For $\mathbf{T} = \phi \mathbf{G}$:

$$\int [\phi \nabla \cdot \mathbf{G} + (\nabla \phi) \cdot \mathbf{G}] dV = \oint \hat{n} \cdot \mathbf{G} \phi dS. \quad (97)$$

1.7.4 Problems

1. Show that, in general,

$$\vec{A} \cdot \tilde{\mathbf{T}} = \mathbf{T} \cdot \vec{A}$$

if and only if $\mathbf{T}$ is a symmetric dyad, that is, if

$$\mathbf{T} = \tilde{\mathbf{T}}, \quad \text{with} \quad \tilde{\mathbf{T}} = \sum_{i=1}^3 \sum_{j=1}^3 \hat{e}_i \hat{e}_j T_{ji}.$$

2. Show that if $\mathbf{T}$ is an antisymmetric dyad, then

$$\vec{A} \cdot \mathbf{T} = -\mathbf{T} \cdot \vec{A}, \quad \text{and} \quad \vec{A} \cdot \mathbf{T} \cdot \vec{A} = 0.$$

3. Show that $\vec{A} \cdot \mathbf{T} \cdot \vec{B}$ is a scalar.

4. Using the derivatives of the unit vectors in spherical and cylindrical coordinates show that

$$\nabla \vec{r} = \mathbf{I}.$$

5. Using the transformation rules of the unit vectors between Cartesian and spherical coordinates, show that

$$\mathbf{I} = \hat{\mathbf{i}}\hat{\mathbf{i}} + \hat{\mathbf{j}}\hat{\mathbf{j}} + \hat{\mathbf{k}}\hat{\mathbf{k}} = \hat{e}_r\hat{e}_r + \hat{e}_\theta\hat{e}_\theta + \hat{e}_\phi\hat{e}_\phi.$$

This proves the invariance of the identity dyad under coordinate transformations.

6. Show the following theorem: every vector $\vec{A}$ in 3D can be decomposed, with respect to a plane whose normal is $\hat{n}$, into two parts, one perpendicular to the plane and of magnitude $\vec{A} \cdot \hat{n}$, and another $\vec{A}_t$ lying in the plane:

$$\vec{A} = (\vec{A} \cdot \hat{n})\hat{n} + \vec{A}_t = (\vec{A} \cdot \hat{n})\hat{n} + \vec{A} \cdot (\mathbf{I} - \hat{n}\hat{n}).$$

7. Evaluate $\nabla \vec{A}$ if

$$\vec{A} = 3y\hat{\mathbf{i}} + 2z\hat{\mathbf{j}} + x\hat{\mathbf{k}}.$$

8. In Cartesian coordinates show that:

•

$$\nabla \vec{r} = \mathbf{I}$$

•

$$\mathbf{I} \cdot \mathbf{I} = \mathbf{I}$$

•

$$\mathbf{I} : \mathbf{I} = 3$$

•

$$\vec{A} \cdot (\mathbf{T} \cdot \vec{B}) = (\vec{A} \cdot \mathbf{T}) \cdot \vec{B}$$

• If $\mathbf{T}$ is a constant diagonal dyad, the expression

$$\vec{r} \cdot \mathbf{T} \cdot \vec{r} = 1$$

describes ellipsoids or one- or two-sheeted hyperboloids.

$$\vec{r} \cdot \mathbf{T} \cdot \vec{r} = 0$$

represents a cone, if the signs are not all the same.

- The gradient of a vector is a dyad.
- The divergence of a dyad is a vector.
- The curl of a dyad is a dyad.

- The gradient of the gradient of a scalar is a dyad.

9. Show the following identities using Cartesian coordinates (they are also valid in orthogonal curvilinear coordinates):

-

$$\nabla \cdot (\vec{A}\vec{B}) = \vec{B}(\nabla \cdot \vec{A}) + (\vec{A} \cdot \nabla)\vec{B}$$

-

$$\nabla \cdot (\phi \mathbf{T}) = \phi \nabla \cdot \mathbf{T} + (\nabla \phi) \cdot \mathbf{T}$$

-

$$\nabla \times (\vec{A}\vec{B}) = (\nabla \times \vec{A})\vec{B} - (\vec{A} \times \nabla)\vec{B}$$

-

$$\nabla \cdot (\mathbf{T} \times \vec{A}) = (\nabla \cdot \mathbf{T}) \times \vec{A} - (\tilde{\mathbf{T}} \cdot \nabla) \times \vec{A}$$

where $\tilde{\mathbf{T}}$ is the transpose of $\mathbf{T}$:

$$\tilde{\mathbf{T}} = \sum_{i=1}^3 \sum_{j=1}^3 \hat{e}_i \hat{e}_j T_{ji}$$

-

$$\nabla \cdot (\vec{A} \times \mathbf{T}) = (\nabla \times \vec{A}) \cdot \mathbf{T} - \vec{A} \cdot (\nabla \times \mathbf{T})$$

-

$$\nabla \cdot (\vec{A} \cdot \mathbf{T}) = \vec{A} \cdot (\nabla \cdot \tilde{\mathbf{T}}) + \mathbf{T} : (\nabla \vec{A})$$

-

$$\nabla \cdot (\mathbf{T} \cdot \vec{A}) = (\nabla \cdot \mathbf{T}) \cdot \vec{A} + \tilde{\mathbf{T}} : (\nabla \vec{A})$$

-

$$\nabla(\mathbf{T} \cdot \vec{A}) = (\nabla \mathbf{T}) \cdot \vec{A} + \nabla \vec{A} \cdot \mathbf{T}$$

-

$$\nabla(\mathbf{T} : \mathbf{R}) = \nabla \mathbf{T} : \mathbf{R} + \nabla \mathbf{R} : \mathbf{T}$$

-

$$\nabla \cdot (\mathbf{T} \cdot \mathbf{R}) = (\nabla \cdot \mathbf{T}) \cdot \mathbf{R} + \tilde{\mathbf{T}} \nabla : \mathbf{R}$$

-

$$\nabla \times (\vec{A} \cdot \mathbf{T}) = (\nabla \times \tilde{\mathbf{T}}) \cdot \vec{A} + (\nabla \times \vec{A}) \cdot \mathbf{T}$$

-

$$\nabla(\vec{A} \cdot \mathbf{T} \cdot \vec{B}) = \nabla \vec{A} \cdot \mathbf{T} \cdot \vec{B} + \nabla \tilde{\mathbf{T}} : \vec{A}\vec{B} + \nabla \vec{B} \cdot \tilde{\mathbf{T}} \cdot \vec{A}$$

-

$$\nabla(r^n \vec{r}) = \mathbf{I} r^n + n r^{n-2} \vec{r} \vec{r}$$

10. In electrically anisotropic media the relation between the electric field $\vec{E}$ and the displacement vector $\vec{D}$ has the form:

$$\vec{D} = \mathbf{E} \cdot \vec{E},$$

where $\mathbf{E}$ is the permittivity dyad. Show that

$$D_i = \sum_{j=1}^3 \varepsilon_{ij} E_j,$$

and that in isotropic media

$$\mathbf{E} = \varepsilon \mathbf{I},$$

where ε is the scalar permittivity.

11. Prove that the components T_{12}, T_{23}, T_{31} of a dyad transform under reflection about the x -axis as:

$$T'_{12} = -T_{12}, \quad T'_{23} = T_{23}, \quad T'_{31} = -T_{31}.$$

This implies that these three components of the dyad transform as the components of an axial vector. In fact, if $\mathbf{T}$ is an antisymmetric dyad, it will have only three nonzero components, from which we can construct an axial vector $\vec{A}$ in the form:

$$A_1 = T_{23}, \quad A_2 = T_{31}, \quad A_3 = T_{12},$$

or, in general,

$$A_i = \frac{1}{2} \sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{ijk} T_{jk}.$$

12. The electric quadrupole of a charge distribution $\rho(\vec{r})$ is a symmetric dyad $\mathbf{Q}$ defined by

$$\mathbf{Q} = \int \rho(\vec{r}) [3\vec{r}\vec{r} - r^2 \mathbf{I}] dV,$$

where $\mathbf{I}$ is the identity dyad.

- Show that the trace of $\mathbf{Q}$ is zero:

$$\sum_{i=1}^3 Q_{ii} = 0.$$

- The electrostatic potential of the distribution is written as:

$$\phi(\vec{r}) = \frac{\vec{r} \cdot \mathbf{Q} \cdot \vec{r}}{2r^5}.$$

Calculate the electrostatic field.

1.8 Exercises

1. With reference to a rectangular Cartesian coordinate system, the matrix of the stress tensor at a point in a body is

$$[T] = \begin{bmatrix} 2 & 4 & 3 \\ 4 & 0 & 0 \\ 3 & 0 & -1 \end{bmatrix} \text{ MPa.} \quad (98)$$

- (a) Find the stress vector and the magnitude of the normal stress on a plane passing through the point and parallel to

$$x_1 + 2x_2 + 2x_3 - 6 = 0.$$

- (b) If

$$\mathbf{e}'_1 = \frac{1}{3}(2\mathbf{e}_1 + 2\mathbf{e}_2 + \mathbf{e}_3), \quad \mathbf{e}'_2 = \frac{1}{\sqrt{2}}(\mathbf{e}_1 - \mathbf{e}_2),$$

find T'_{12} .

2. In the absence of body forces, does the following stress distribution satisfy the equations of equilibrium? Here ν is a constant.

$$T_{11} = x_2^2 + \nu(x_1^2 - x_2^2), \quad T_{12} = -2\nu x_1 x_2, \quad (99)$$

$$T_{22} = x_1^2 + \nu(x_2^2 - x_1^2), \quad T_{23} = T_{13} = 0, \quad (100)$$

$$T_{33} = \nu(x_1^2 + x_2^2). \quad (101)$$

3. The state of stress at a point in a body is

$$[\sigma] = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 0 \end{bmatrix} \text{ MPa.} \quad (102)$$

For each coordinate plane, with normal along $\mathbf{e}_1$, $\mathbf{e}_2$, or $\mathbf{e}_3$: (a) find the normal stress; (b) find the magnitude of the shear stress.

4. The state of stress at a point in a body is

$$[\sigma] = \begin{bmatrix} 2 & -1 & 3 \\ -1 & 4 & 0 \\ 3 & 0 & -1 \end{bmatrix} \text{ MPa.} \quad (103)$$

- (a) Find the traction vector on the plane whose normal is in the direction $2\mathbf{e}_1 + 2\mathbf{e}_2 + \mathbf{e}_3$.
(b) Determine the magnitudes of the normal and shear stresses on this plane.

5. Repeat above problem for a plane through the point and parallel to

$$x_1 - 2x_2 + 3x_3 = 4. \quad (104)$$

6. The stress distribution in a body is

$$[\sigma] = \begin{bmatrix} 0 & 100x_1 & -100x_2 \\ 100x_1 & 0 & 0 \\ -100x_2 & 0 & 0 \end{bmatrix} \text{ MPa.} \quad (105)$$

Find the traction vector on the plane through $(1/2, \sqrt{3}/2, 3)$ that is tangent there to the cylindrical surface $x_1^2 + x_2^2 = 1$.

7. At a point in a continuum, $\sigma_{11} = 1$ MPa, $\sigma_{22} = -1$ MPa, and all other $\sigma_{ij} = 0$.
- Show that the only plane with zero traction has its normal in the $\mathbf{e}_3$ direction.
 - Give three planes on which the normal stress vanishes.

8. Consider the following stress distribution in a circular cylindrical bar:

$$[\sigma] = \begin{bmatrix} 0 & -\alpha x_3 & \alpha x_2 \\ -\alpha x_3 & 0 & 0 \\ \alpha x_2 & 0 & 0 \end{bmatrix}. \quad (106)$$

- Find the traction vector on (i) the lateral surface $x_2^2 + x_3^2 = 4$, (ii) the end face $x_1 = 0$, and (iii) the end face $x_1 = l$.
- Find the resultant force and moment on the end face $x_1 = l$.

9. Given the stress distribution

$$[\sigma] = \begin{bmatrix} x_1 + x_2 & \sigma_{12}(x_1, x_2) & 0 \\ \sigma_{12}(x_1, x_2) & x_1 - 2x_2 & 0 \\ 0 & 0 & x_2 \end{bmatrix}, \quad (107)$$

find σ_{12} such that the distribution is in equilibrium without body forces and the traction vector on the plane $x_1 = 1$ is

$$\mathbf{t} = (1 + x_2)\mathbf{e}_1 + (5 - x_2)\mathbf{e}_2. \quad (108)$$

10. Consider the stress tensor

$$[\sigma] = \alpha \begin{bmatrix} x_2 & -x_3 & 0 \\ -x_3 & 0 & -x_2 \\ 0 & -x_2 & \sigma_{33} \end{bmatrix}. \quad (109)$$

Find an expression for σ_{33} such that the equations of equilibrium hold in the presence of the body force $\vec{f} = -g\mathbf{e}_3$, where g is constant.

11. In the absence of body forces, a body's equilibrium stress distribution is

$$\sigma_{11} = Ax_2, \quad \sigma_{12} = \sigma_{21} = x_1, \quad \sigma_{22} = Bx_1 + Cx_2, \quad \sigma_{33} = \frac{\sigma_{11} + \sigma_{22}}{2}, \quad (110)$$

with all other $\sigma_{ij} = 0$. The boundary plane $x_1 - x_2 = 0$ is free of traction. (a) Find C . (b) Find A and B .

12. In the absence of body forces, do the following stress components satisfy the equations of equilibrium?

$$\sigma_{11} = \alpha[x_2^2 + \nu(x_1^2 - x_2^2)], \quad \sigma_{22} = \alpha[x_1^2 + \nu(x_2^2 - x_1^2)], \quad (111)$$

$$\sigma_{33} = \alpha\nu(x_1^2 + x_2^2), \quad \sigma_{12} = \sigma_{21} = -2\alpha\nu x_1 x_2, \quad (112)$$

$$\sigma_{13} = \sigma_{31} = 0, \quad \sigma_{23} = \sigma_{32} = 0. \quad (113)$$

13. Repeat above problem for

$$[\sigma] = \alpha \begin{bmatrix} x_1 + x_2 & 2x_1 - x_2 & 0 \\ 2x_1 - x_2 & x_1 - 3x_2 & 0 \\ 0 & 0 & x_1 \end{bmatrix}. \quad (114)$$

14. Suppose the stress distribution has the form of a plane stress state:

$$[\sigma] = \begin{bmatrix} \sigma_{11}(x_1, x_2) & \sigma_{12}(x_1, x_2) & 0 \\ \sigma_{12}(x_1, x_2) & \sigma_{22}(x_1, x_2) & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (115)$$

- (a) If this stress field is in equilibrium, can the body forces depend on x_3 ?
(b) Introduce a function $\varphi(x_1, x_2)$ such that

$$\sigma_{11} = \frac{\partial^2 \varphi}{\partial x_2^2}, \quad \sigma_{22} = \frac{\partial^2 \varphi}{\partial x_1^2}, \quad \sigma_{12} = -\frac{\partial^2 \varphi}{\partial x_1 \partial x_2}. \quad (116)$$

Show that, in the absence of body forces, the equilibrium equations hold for any φ with continuous third derivatives.

15. Verify that the following stress field satisfies the z -equation of equilibrium in the absence of body forces:

$$\sigma_{rr} = A \left(\frac{z}{R^3} - \frac{3r^2 z}{R^5} \right), \quad \sigma_{\theta\theta} = \frac{Az}{R^3}, \quad (117)$$

$$\sigma_{zz} = -A \left(\frac{z}{R^3} + \frac{3z^3}{R^5} \right), \quad \sigma_{rz} = \sigma_{zr} = -A \left(\frac{r}{R^3} + \frac{3rz^2}{R^5} \right), \quad (118)$$

$$\sigma_{r\theta} = \sigma_{\theta z} = 0, \quad R^2 = r^2 + z^2. \quad (119)$$

16. Given the following stress field in cylindrical coordinates:

$$\sigma_{rr} = -\frac{3Pzr^2}{2\pi R^5}, \quad \sigma_{zz} = -\frac{3Pz^3}{2\pi R^5}, \quad \sigma_{rz} = \sigma_{zr} = -\frac{3Pz^2 r}{2\pi R^5}, \quad (120)$$

$$\sigma_{\theta\theta} = \sigma_{r\theta} = \sigma_{\theta z} = 0, \quad R^2 = r^2 + z^2, \quad (121)$$

verify that the stress field satisfies the equations of equilibrium in the absence of body forces.

2 Strain

- All materials deform under external forces, but they respond differently:
 - **Elastic materials** return to their original shape once the force is removed.
 - **Plastic materials** retain permanent deformation.
 - **Viscoelastic materials** behave elastically under quick deformation but flow like viscous fluids over time.
- Elasticity is valid only within a limited range of forces; beyond that, materials become plastic or may fracture.
- Deformation displaces the material from its original position:
 - Small deformations are easier to analyze mathematically than large ones.
 - Large deformations produce complex, non-uniform displacements (e.g., crumpled paper).
 - Coordinate systems deform with the body, becoming curvilinear.
- The theory of finite deformation is mathematically complex, similar to curvilinear coordinate systems. In most engineering applications, deformations are small and can be approximated as linear.
- The description of deformation introduces the **strain tensor**, representing the local deformation or strain of a material.

2.1 Displacements

2.1.1 Uniform scaling

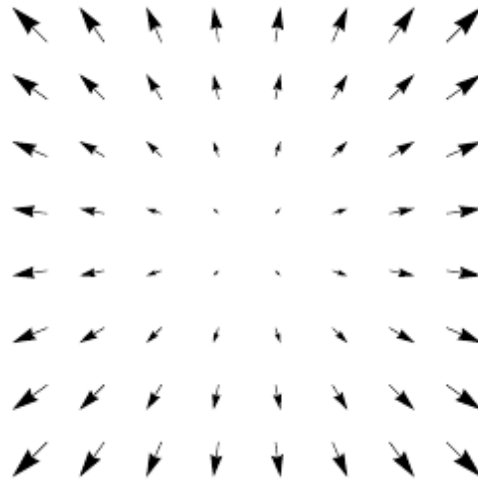


Figure 9: Uniform dilatation. The arrows indicate how material particles are displaced. Figure taken from Ref. [3].

The prime example of deformation is a *uniform scaling* in which the coordinates of all material particles in a body are multiplied with the scale factor κ . A material particle originally situated in the point $\vec{X}$ is thus displaced to the point

$$\vec{x} = \kappa \vec{X}. \quad (122)$$

It is emphasized that *both* $\vec{X}$ and $\vec{x}$ refer to the same coordinate system. Uniform scaling with $\kappa > 1$ is also called uniform *dilatation*, whereas scaling with $0 < \kappa < 1$ is called uniform *compression*. Negative scaling with $\kappa < 0$ is physically impossible.

The only point that does not change place during uniform scaling is the origin of the coordinate system. Although it superficially looks as if the origin plays a special role, this is not really the case. All relative positions of material particles scale in the same way, because

$$\vec{x} - \vec{y} = \kappa(\vec{X} - \vec{Y}), \quad (123)$$

independent of the origin of the coordinate system. There is no special center for a uniform scaling, either geometrically or physically. The origin of the coordinate system is simply an *anchor point* for the mathematical description of scaling.

2.1.2 Linear displacements

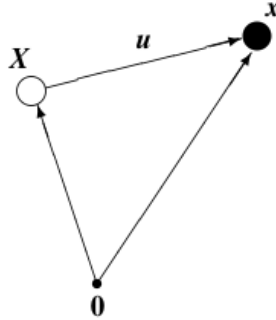


Figure 10: Geometry of displacement. The particle that originally was located at $\vec{X}$ has been displaced to $\vec{x}$ by the displacement vector $\vec{u}$. Figure taken from Ref. [3].

Under a displacement (see Fig. 10), the center-of-mass of a material particle is moved from its original position $\vec{X}$ to its actual position $\vec{x}$. The *displacement vector* is always defined as the difference between the actual and the original coordinates,

$$\vec{u} = \vec{x} - \vec{X}. \quad (124)$$

For the case of uniform scaling, the displacement vector becomes

$$\vec{u} = (\kappa - 1)\vec{X} = \left(1 - \frac{1}{\kappa}\right)\vec{x}. \quad (125)$$

Mathematically we are completely free to express the displacement as a function of either the original position X or the actual position x of the material particle. For scaling, the displacement is in both cases a linear function of the coordinates.

More generally, a linear displacement (and its inverse) takes the form

$$\vec{x} = \mathbf{A} \cdot \vec{X} + \vec{b}, \quad \vec{X} = \mathbf{A}^{-1} \cdot (\vec{x} - \vec{b}), \quad (126)$$

where $\mathbf{A}$ is a non-singular constant matrix and $\vec{b}$ is a constant vector. As for scaling, the displacement vector may be expressed as a function of either the original or the actual positions:

$$\vec{u} = (\mathbf{A} - 1) \cdot \vec{X} + \vec{b} = (1 - \mathbf{A}^{-1}) \cdot \vec{x} + \mathbf{A}^{-1} \cdot \vec{b}. \quad (127)$$

The general linear displacement may be resolved into simpler types: translation along a coordinate axis, rotation by a fixed angle around a coordinate axis, and scaling by a fixed factor along a coordinate axis (see Fig. 11). Physically impossible reflections are excluded. The matrix $\mathbf{A}$ is not restricted to be orthogonal.

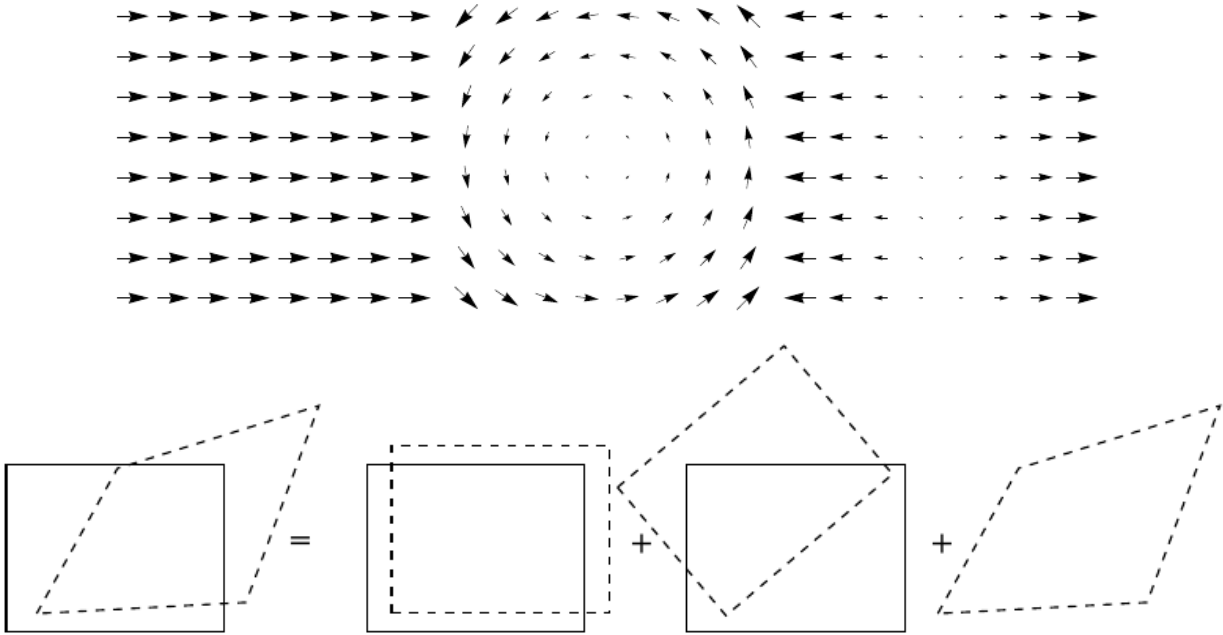


Figure 11: Arrow plots of the displacement fields for simple translation, rotation, and dilatation. Figure taken from Ref. [3]. Bottom: The most general change of a deformable body can be decomposed into translation, rotation, and deformation. Figure taken from Ref. [1].

2.1.3 Simple Translation

A rigid body translation of the material through a distance b along the x -axis is described by

$$x = X + b, \quad y = Y, \quad z = Z. \quad (128)$$

The displacement vector becomes

$$u_x = b, \quad u_y = 0, \quad u_z = 0. \quad (129)$$

Since the geometric relationships in a body are unchanged, this is not a deformation.

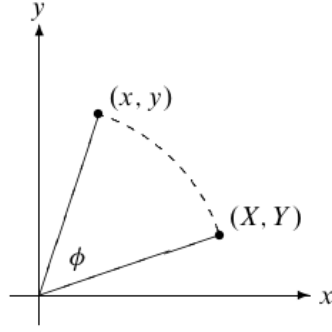


Figure 12: A rigid body rotation through an angle ϕ moves the material particle at (X, Y) to (x, y) . Figure taken from Ref. [3].

2.1.4 Simple Rotation

A rigid body rotation (see Fig. 12) through the angle ϕ around the z -axis takes the form

$$x = X \cos \phi - Y \sin \phi, \quad X = x \cos \phi + y \sin \phi, \quad (130)$$

$$y = X \sin \phi + Y \cos \phi, \quad Y = -x \sin \phi + y \cos \phi, \quad (131)$$

$$z = Z, \quad Z = z. \quad (132)$$

The displacement vector components are

$$u_x = -X(1 - \cos \phi) - Y \sin \phi = x(1 - \cos \phi) - y \sin \phi, \quad (133)$$

$$u_y = X \sin \phi - Y(1 - \cos \phi) = x \sin \phi - y(1 - \cos \phi), \quad (134)$$

$$u_z = 0 = 0. \quad (135)$$

Since all distances in the body remain unchanged, this is not a deformation.

2.1.5 Simple Scaling Along x

Multiplying all x -coordinates by the factor κ gives

$$x = \kappa X, \quad y = Y, \quad z = Z. \quad (136)$$

The displacement vector becomes

$$u_x = (\kappa - 1)X = kx, \quad (137)$$

$$u_y = 0, \quad (138)$$

$$u_z = 0, \quad (139)$$

where $k = 1 - 1/\kappa$. Simple dilatation corresponds to $k > 0$ and simple compression to $k < 0$. Uniform scaling is a combination of three such scalings along the coordinate axes. Scaling is a true deformation.

2.2 Displacement field

2.2.1 Displacement Field Representations

In continuum mechanics, each material point initially at X moves to a new position x under deformation. The **displacement field** is defined as

$$\vec{u}(x) = \vec{x} - \vec{X}(\vec{x}), \quad (140)$$

where $\vec{X}(x)$ denotes the original position of the particle now at $\vec{x}$. This is known as the *Euler representation*. Alternatively, in the *Lagrange representation*, the position is written as $\vec{x} = \vec{x}(\vec{X})$, and

$$\vec{u} = \vec{x}(\vec{X}) - \vec{X}. \quad (141)$$

Both representations are equivalent for small, slowly varying deformations.

2.2.2 Local and Infinitesimal Deformation

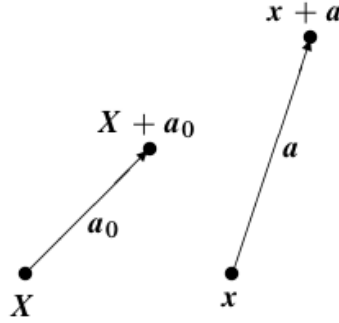


Figure 13: Displacement of a tiny material needle from $\vec{a}_0$ to $\vec{a}$. It may be translated, rotated, and scaled. Only the latter corresponds to a true deformation. Figure taken from Ref. [3].

A true deformation involves local changes in length and angle between neighboring points, not just rigid translations or rotations.

For a small line element or “needle” $\vec{a}_0$ connecting two material points, the deformation changes it to $\vec{a}$. Using $\vec{X}(\vec{x}) = \vec{x} - \vec{u}(\vec{x})$, one finds

$$\vec{a}_0 = \vec{X}(\vec{x} + \vec{a}) - \vec{X}(\vec{x}) = \vec{a} - \vec{u}(\vec{x} + \vec{a}) + \vec{u}(\vec{x}). \quad (142)$$

Expanding $\vec{u}(\vec{x} + \vec{a})$ to first order in $\vec{a}$ gives

$$\vec{u}(\vec{x} + \vec{a}) = \vec{u}(\vec{x}) + (\vec{a} \cdot \nabla) \vec{u}(\vec{x}) + \mathcal{O}(\vec{a}^2), \quad (143)$$

so that the infinitesimal change in the line element is

$$\delta \vec{a} \equiv \vec{a} - \vec{a}_0 = (\vec{a} \cdot \nabla) \vec{u}(\vec{x}). \quad (144)$$

In index notation:

$$\delta a_i = \sum_j a_j \nabla_j u_i. \quad (145)$$

This defines the **displacement gradients** $\{\nabla_j u_i\}$, and in dyadic form:

$$\delta \vec{a} = (\vec{a} \cdot \nabla) \vec{u} = \vec{a} \cdot (\nabla \vec{u}) = (\nabla \vec{u})^T \cdot \vec{a}. \quad (146)$$

2.2.3 Slowly Varying Displacement Field

The displacement field is said to be *slowly varying* when

$$|\nabla_j u_i(x)| \ll 1, \quad (147)$$

and small relative to the body size L ,

$$|\vec{u}(x)| \ll L. \quad (148)$$

This ensures that deformations are infinitesimal, though not necessarily rigid.

Example Small rotations: For small rotation angle $\phi \ll 1$, the displacement field of a simple rotation becomes

$$\vec{u} = (-y, x, 0) \phi,$$

and the gradient matrix becomes

$$\{\nabla_j u_i\} = \begin{pmatrix} 0 & -\phi & 0 \\ \phi & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

to lowest order in ϕ . The change in $\vec{a}$ becomes

$$\delta a_x = -\phi a_y, \quad \delta a_y = \phi a_x.$$

This may also be written as a cross product,

$$\delta \vec{a} = \vec{\phi} \times \vec{a}, \quad \text{where} \quad \vec{\phi} = \phi \vec{e}_z.$$

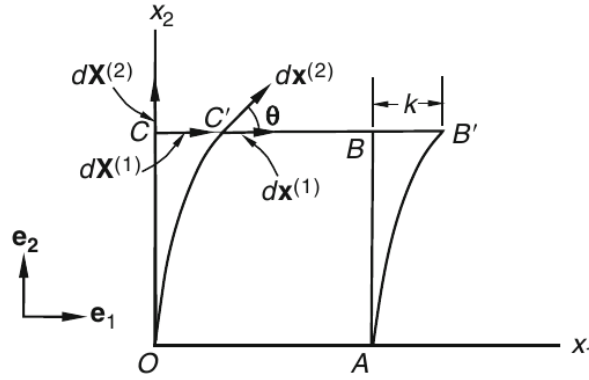


Figure 14: Unit square $OABC$

Example. Given the displacement components

$$u_1 = kx_2^2, \quad u_2 = u_3 = 0, \quad (149)$$

- Sketch the deformed shape of the unit square $OABC$ shown in Fig. 14.
- Find the deformed vectors $d\mathbf{x}^{(1)}$ and $d\mathbf{x}^{(2)}$ corresponding to the material line elements

$$d\mathbf{X}^{(1)} = dx_1 \mathbf{e}_1, \quad d\mathbf{X}^{(2)} = dx_2 \mathbf{e}_2$$

at the point C .

- Determine the stretch of each element in part (b) and the change in angle between them.

Solution.

(a) Along the material line OA , $X_2 = 0$, so $u_1 = u_2 = u_3 = 0$; this line is not displaced. Along CB , $X_2 = 1$, so $u_1 = k$; this line moves k units in the x_1 direction. Along OC and AB , the displacement $u_1 = kX_2^2$ makes the lines parabolic. The resulting boundary is $OAB'C'$, as shown in Fig. 14.

(b) At the material point C , where $X_2 = 1$, the displacement gradient is

$$[\nabla \mathbf{u}] = \left[\begin{array}{ccc} 0 & 2kx_2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right] \bigg|_{X_2=1} = \left[\begin{array}{ccc} 0 & 2k & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right]. \quad (150)$$

Thus,

$$d\mathbf{x}^{(1)} = d\mathbf{X}^{(1)} + (\nabla \mathbf{u})d\mathbf{X}^{(1)} = dX_1 \mathbf{e}_1, \quad (151)$$

while

$$\begin{aligned} d\mathbf{x}^{(2)} &= d\mathbf{X}^{(2)} + (\nabla \mathbf{u})d\mathbf{X}^{(2)} \\ &= dX_2 \mathbf{e}_2 + 2k dX_2 \mathbf{e}_1 = dX_2 (2k \mathbf{e}_1 + \mathbf{e}_2). \end{aligned} \quad (152)$$

(c) The deformed lengths are

$$|d\mathbf{x}^{(2)}| = dX_2 \sqrt{1 + 4k^2}. \quad (153)$$

Consequently, the stretches are

$$\frac{|d\mathbf{x}^{(1)}|}{|d\mathbf{X}^{(1)}|} = 1, \quad \frac{|d\mathbf{x}^{(2)}|}{|d\mathbf{X}^{(2)}|} = \sqrt{1 + 4k^2}. \quad (154)$$

If θ is the angle between the deformed elements,

$$\cos \theta = \frac{d\mathbf{x}^{(1)} \cdot d\mathbf{x}^{(2)}}{|d\mathbf{x}^{(1)}| |d\mathbf{x}^{(2)}|} = \frac{2k}{\sqrt{1 + 4k^2}}. \quad (155)$$

For $|k| \ll 1$, the stretches are both 1 to first order in k , while $\cos \theta \simeq 2k$. If γ denotes the decrease from the original right angle, then

$$\theta = \frac{\pi}{2} - \gamma, \quad \cos \theta = \sin \gamma \simeq \gamma, \quad \gamma \simeq 2k. \quad (156)$$

2.2.4 Cauchy's Strain Tensor

The strain tensor quantifies local geometric changes due to deformation. The change in the scalar product of two material line elements $\vec{a}$ and $\vec{b}$ is

$$\delta(\vec{a} \cdot \vec{b}) = (\delta \vec{a}) \cdot \vec{b} + \vec{a} \cdot (\delta \vec{b}) = (\vec{a} \cdot \nabla) \vec{u} \cdot \vec{b} + (\vec{b} \cdot \nabla) \vec{u} \cdot \vec{a} \quad (157)$$

$$= \sum_{ij} (\nabla_i u_j + \nabla_j u_i) a_i b_j \quad (158)$$

$$= 2 \sum_{ij} U_{ij} a_i b_j = 2 \vec{a} \cdot \vec{U} \cdot \vec{b}, \quad (159)$$

where the **Cauchy (infinitesimal) strain tensor** is defined as:

$$U_{ij} = \frac{1}{2}(\nabla_i u_j + \nabla_j u_i), \quad (160)$$

or, in matrix notation,

$$\vec{U} = \frac{1}{2}(\nabla \vec{u} + (\nabla \vec{u})^T). \quad (161)$$

The strain tensor contains all the information about the local geometric changes caused by the displacement and is accordingly a good measure of local deformation. All bodily translations and rotations have been automatically taken out, and any displacement that is a combination of translations and rotations must consequently yield a vanishing strain tensor. It should, however, be emphasized that Cauchy's expression is only valid for small displacement gradients. When that is not the case, a more complicated expression must be used, involving the square of the displacement gradients .

2.2.5 Compatibility conditions

When the three displacement functions u_1 , u_2 , and u_3 are given, the six strain components can be determined wherever the partial derivatives $\partial u_i / \partial x_j$ exist. Conversely, six arbitrarily prescribed strain components ($U_{11}, U_{22}, U_{33}, U_{12}, U_{13}, U_{23}$) do not, in general, correspond to a displacement field. Such a field would have to satisfy the six strain-displacement relations

$$\frac{\partial u_1}{\partial x_1} = U_{11}, \quad (162)$$

$$\frac{\partial u_2}{\partial x_2} = U_{22}, \quad (163)$$

$$\frac{\partial u_3}{\partial x_3} = U_{33}, \quad (164)$$

$$\frac{1}{2} \left(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} \right) = U_{12}, \quad (165)$$

$$\frac{1}{2} \left(\frac{\partial u_1}{\partial x_3} + \frac{\partial u_3}{\partial x_1} \right) = U_{13}, \quad (166)$$

$$\frac{1}{2} \left(\frac{\partial u_2}{\partial x_3} + \frac{\partial u_3}{\partial x_2} \right) = U_{23}. \quad (167)$$

For example, suppose that

$$U_{11} = kx_2^2, \quad U_{22} = U_{33} = U_{12} = U_{13} = U_{23} = 0. \quad (168)$$

The first strain-displacement relation gives

$$\frac{\partial u_1}{\partial x_1} = kx_2^2, \quad \text{hence} \quad u_1 = kx_1x_2^2 + f(x_2, x_3), \quad (169)$$

while the second gives

$$\frac{\partial u_2}{\partial x_2} = 0, \quad \text{hence} \quad u_2 = g(x_1, x_3), \quad (170)$$

where f and g are arbitrary functions of integration. Since $U_{12} = 0$, we must also have

$$\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} = 0. \quad (171)$$

Substitution yields

$$2kx_1x_2 + \frac{\partial f(x_2, x_3)}{\partial x_2} + \frac{\partial g(x_1, x_3)}{\partial x_1} = 0. \quad (172)$$

Neither of the last two terms can cancel a term proportional to x_1x_2 throughout a region. Thus, for $k \neq 0$, no displacement field corresponds to these prescribed strains: the strain components are *incompatible*.

We now state the compatibility theorem. If the components $U_{ij}(x_1, x_2, x_3)$ have continuous second partial derivatives in a simply connected region, the necessary and sufficient conditions for the existence of a single-valued displacement field satisfying the six strain–displacement relations are

$$\frac{\partial^2 U_{11}}{\partial x_2^2} + \frac{\partial^2 U_{22}}{\partial x_1^2} = 2 \frac{\partial^2 U_{12}}{\partial x_1 \partial x_2}, \quad (173)$$

$$\frac{\partial^2 U_{22}}{\partial x_3^2} + \frac{\partial^2 U_{33}}{\partial x_2^2} = 2 \frac{\partial^2 U_{23}}{\partial x_2 \partial x_3}, \quad (174)$$

$$\frac{\partial^2 U_{33}}{\partial x_1^2} + \frac{\partial^2 U_{11}}{\partial x_3^2} = 2 \frac{\partial^2 U_{31}}{\partial x_3 \partial x_1}, \quad (175)$$

$$\frac{\partial^2 U_{11}}{\partial x_2 \partial x_3} = \frac{\partial}{\partial x_1} \left(-\frac{\partial U_{23}}{\partial x_1} + \frac{\partial U_{31}}{\partial x_2} + \frac{\partial U_{12}}{\partial x_3} \right), \quad (176)$$

$$\frac{\partial^2 U_{22}}{\partial x_3 \partial x_1} = \frac{\partial}{\partial x_2} \left(-\frac{\partial U_{31}}{\partial x_2} + \frac{\partial U_{12}}{\partial x_3} + \frac{\partial U_{23}}{\partial x_1} \right), \quad (177)$$

$$\frac{\partial^2 U_{33}}{\partial x_1 \partial x_2} = \frac{\partial}{\partial x_3} \left(-\frac{\partial U_{12}}{\partial x_3} + \frac{\partial U_{23}}{\partial x_1} + \frac{\partial U_{31}}{\partial x_2} \right). \quad (178)$$

The above conditions can be expressed in a compact way as

$$\nabla_i \nabla_j U_{kl} + \nabla_k \nabla_l U_{ij} = \nabla_i \nabla_l U_{kj} + \nabla_k \nabla_j U_{il}. \quad (179)$$

These are the *equations of compatibility*, also called the *integrability conditions*.

To illustrate why they are necessary, consider

$$\frac{\partial u_1}{\partial x_1} = U_{11}, \quad \frac{\partial u_2}{\partial x_2} = U_{22}. \quad (180)$$

Differentiating twice gives

$$\frac{\partial^2 U_{11}}{\partial x_2^2} = \frac{\partial^3 u_1}{\partial x_2^2 \partial x_1}, \quad \frac{\partial^2 U_{22}}{\partial x_1^2} = \frac{\partial^3 u_2}{\partial x_1^2 \partial x_2}. \quad (181)$$

Interchanging the order of differentiation, we obtain

$$\begin{aligned} \frac{\partial^2 U_{11}}{\partial x_2^2} + \frac{\partial^2 U_{22}}{\partial x_1^2} &= \frac{\partial^2}{\partial x_1 \partial x_2} \left(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} \right) \\ &= 2 \frac{\partial^2 U_{12}}{\partial x_1 \partial x_2}, \end{aligned} \quad (182)$$

which is the first compatibility equation. The other five equations can be established similarly.

Example. Does the following strain field represent a compatible strain field?

$$[U] = k \begin{bmatrix} 2x_1 & x_1 + 2x_2 & 0 \\ x_1 + 2x_2 & 2x_1 & 0 \\ 0 & 0 & 2x_3 \end{bmatrix}. \quad (183)$$

Solution. All strain components are linear functions of x_1, x_2, x_3 , so their second derivatives vanish and the compatibility equations are satisfied. The components are smooth; hence, in a simply connected region, a single-valued displacement field exists. One such field is

$$u_1 = k(x_1^2 + x_2^2), \quad u_2 = k(2x_1x_2 + x_1^2), \quad u_3 = kx_3^2. \quad (184)$$

Example. For the strain field

$$U_{11} = -\frac{x_2}{x_1^2 + x_2^2}, \quad U_{12} = \frac{x_1}{2(x_1^2 + x_2^2)}, \quad U_{22} = U_{33} = U_{23} = U_{13} = 0, \quad (185)$$

does there exist a single-valued continuous displacement field?

Solution. Of the six compatibility conditions, only the first needs to be checked; the others are automatically satisfied. We have

$$\frac{\partial U_{11}}{\partial x_2} = -\frac{(x_1^2 + x_2^2) - 2x_2^2}{(x_1^2 + x_2^2)^2} = \frac{x_2^2 - x_1^2}{(x_1^2 + x_2^2)^2}, \quad \frac{\partial U_{22}}{\partial x_1} = 0, \quad (186)$$

and

$$2\frac{\partial U_{12}}{\partial x_1} = \frac{(x_1^2 + x_2^2) - 2x_1^2}{(x_1^2 + x_2^2)^2} = \frac{x_2^2 - x_1^2}{(x_1^2 + x_2^2)^2} = \frac{\partial U_{11}}{\partial x_2}. \quad (187)$$

Consequently,

$$\frac{\partial^2 U_{11}}{\partial x_2^2} + \frac{\partial^2 U_{22}}{\partial x_1^2} = 2\frac{\partial^2 U_{12}}{\partial x_1 \partial x_2}. \quad (188)$$

Thus, the compatibility equations are satisfied wherever $x_1^2 + x_2^2 \neq 0$.

2.2.6 Components of the Strain Tensor

For a general displacement field:

$$U_{xx} = \nabla_x u_x, \quad U_{yy} = \nabla_y u_y, \quad U_{zz} = \nabla_z u_z, \quad (189)$$

$$U_{yz} = U_{zy} = \frac{1}{2}(\nabla_y u_z + \nabla_z u_y), \quad U_{zx} = U_{xz} = \frac{1}{2}(\nabla_z u_x + \nabla_x u_z), \quad U_{xy} = U_{yx} = \frac{1}{2}(\nabla_x u_y + \nabla_y u_x). \quad (190)$$

(a) Rectangular coordinates:

$$[U] = \begin{bmatrix} \frac{\partial u_1}{\partial x_1} & \frac{1}{2} \left(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} \right) & \frac{1}{2} \left(\frac{\partial u_1}{\partial x_3} + \frac{\partial u_3}{\partial x_1} \right) \\ \frac{1}{2} \left(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} \right) & \frac{\partial u_2}{\partial x_2} & \frac{1}{2} \left(\frac{\partial u_2}{\partial x_3} + \frac{\partial u_3}{\partial x_2} \right) \\ \frac{1}{2} \left(\frac{\partial u_1}{\partial x_3} + \frac{\partial u_3}{\partial x_1} \right) & \frac{1}{2} \left(\frac{\partial u_2}{\partial x_3} + \frac{\partial u_3}{\partial x_2} \right) & \frac{\partial u_3}{\partial x_3} \end{bmatrix}.$$

(b) Cylindrical coordinates:

$$[U] = \begin{bmatrix} \frac{\partial u_r}{\partial r} & \frac{1}{2} \left(\frac{1}{r} \frac{\partial u_r}{\partial \theta} - \frac{u_\theta}{r} + \frac{\partial u_\theta}{\partial r} \right) & \frac{1}{2} \left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \right) \\ U_{21} = U_{12} & \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r} & \frac{1}{2} \left(\frac{\partial u_\theta}{\partial z} + \frac{1}{r} \frac{\partial u_z}{\partial \theta} \right) \\ U_{31} = U_{13} & U_{32} = U_{23} & \frac{\partial u_z}{\partial z} \end{bmatrix}.$$

(c) Spherical coordinates:

$$[U] = \begin{bmatrix} \frac{\partial u_r}{\partial r} & \frac{1}{2} \left(\frac{1}{r} \frac{\partial u_r}{\partial \theta} - \frac{u_\theta}{r} + \frac{\partial u_\theta}{\partial r} \right) & \frac{1}{2} \left(\frac{1}{r \sin \theta} \frac{\partial u_r}{\partial \phi} - \frac{u_\phi}{r} + \frac{\partial u_\phi}{\partial r} \right) \\ U_{21} = U_{12} & \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r} & \frac{1}{2} \left(\frac{1}{r \sin \theta} \frac{\partial u_\theta}{\partial \phi} - \frac{u_\phi \cot \theta}{r} + \frac{1}{r} \frac{\partial u_\phi}{\partial \theta} \right) \\ U_{31} = U_{13} & U_{32} = U_{23} & \frac{1}{r \sin \theta} \frac{\partial u_\phi}{\partial \phi} + \frac{u_r}{r} + \frac{u_\theta \cot \theta}{r} \end{bmatrix}.$$

Example. A unit cube with edges parallel to the coordinate axes is given the displacement field

$$u_1 = kx_1, \quad u_2 = u_3 = 0, \quad k = 10^{-4}. \quad (191)$$

Find the increase in length of the diagonal AB (see Figure 15) (a) using the infinitesimal strain tensor U and (b) by geometry.

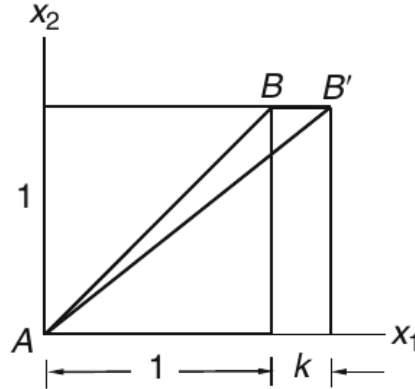


Figure 15: A unit cube with edges parallel to the coordinate axes

Solution. (a) We have

$$[U] = \begin{bmatrix} k & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (ii)$$

The diagonal was originally in the direction $\mathbf{n} = (\mathbf{e}_1 + \mathbf{e}_2)/\sqrt{2}$. Its unit elongation is

$$U_{nn} = \mathbf{n} \cdot U \cdot \mathbf{n} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} k & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \frac{k}{2}. \quad (\text{iii})$$

Since $AB = \sqrt{2}$,

$$\Delta AB = U_{nn} AB = \frac{k}{2} \sqrt{2}. \quad (\text{iv})$$

(b) Geometrically,

$$\Delta AB = AB' - AB = \sqrt{1 + (1 + k)^2} - \sqrt{2} = \sqrt{2} \left(1 + k + \frac{k^2}{2} \right)^{1/2} - \sqrt{2}. \quad (\text{v})$$

Now, to first order in k ,

$$\left(1 + k + \frac{k^2}{2} \right)^{1/2} = 1 + \frac{1}{2} \left(k + \frac{k^2}{2} \right) + \dots \simeq 1 + \frac{k}{2}. \quad (\text{vi})$$

Therefore, in agreement with part (a),

$$\Delta AB \simeq \frac{k}{2} \sqrt{2}. \quad (\text{vii})$$

2.2.7 Principal Axes of Strain

Since the strain tensor U is symmetric, there exist three mutually perpendicular directions $\mathbf{n}_1$, $\mathbf{n}_2$, and $\mathbf{n}_3$ with respect to which its matrix is diagonal:

$$[U]_{\mathbf{n}_i} = \begin{bmatrix} U_1 & 0 & 0 \\ 0 & U_2 & 0 \\ 0 & 0 & U_3 \end{bmatrix}. \quad (192)$$

Geometrically, infinitesimal line elements along these directions remain mutually perpendicular after deformation, to first order in the strain. They are called the *principal directions*. The unit elongations U_1 , U_2 , and U_3 are the eigenvalues of U , or the *principal strains*. They include the maximum and minimum normal strains among all directions through the point.

For a given strain tensor, the principal strains are found from its characteristic equation,

$$\lambda^3 - I_1 \lambda^2 + I_2 \lambda - I_3 = 0, \quad (193)$$

where

$$I_1 = U_{11} + U_{22} + U_{33}, \quad (194)$$

$$I_2 = \begin{vmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{vmatrix} + \begin{vmatrix} U_{22} & U_{23} \\ U_{32} & U_{33} \end{vmatrix} + \begin{vmatrix} U_{11} & U_{13} \\ U_{31} & U_{33} \end{vmatrix}, \quad (195)$$

and

$$I_3 = \det[U_{ij}]. \quad (196)$$

The coefficients I_1 , I_2 , and I_3 are the *principal scalar invariants* of the strain tensor.

Example. Let's calculate the strain tensor for $\vec{u} = \alpha(y, x, 0)$ with $0 < \alpha \ll 1$ and determine the principal directions of strain and the change in length scales along these directions.

With $u_x = \alpha y$, $u_y = \alpha x$, $u_z = 0$, the displacement gradient is

$$\nabla \vec{u} = \begin{pmatrix} \partial_x u_x & \partial_y u_x & \partial_z u_x \\ \partial_x u_y & \partial_y u_y & \partial_z u_y \\ \partial_x u_z & \partial_y u_z & \partial_z u_z \end{pmatrix} = \begin{pmatrix} 0 & \alpha & 0 \\ \alpha & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Hence

$$U = \frac{1}{2}(\nabla \vec{u} + (\nabla \vec{u})^T) = \begin{pmatrix} 0 & \alpha & 0 \\ \alpha & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

In the xy -subspace, $U_{2 \times 2} = \begin{pmatrix} 0 & \alpha \\ \alpha & 0 \end{pmatrix}$ has eigenvalues and unit eigenvectors

$$\lambda_1 = +\alpha, \quad \vec{n}_1 = \frac{1}{\sqrt{2}}(1, 1, 0), \quad \lambda_2 = -\alpha, \quad \vec{n}_2 = \frac{1}{\sqrt{2}}(1, -1, 0),$$

and a third eigenvalue $\lambda_3 = 0$ with $\vec{n}_3 = (0, 0, 1)$.

The normal strain in the direction of a unit vector $\vec{n}$ is $\varepsilon_n = \vec{n} \cdot \vec{U} \vec{n}$. Along the principal directions,

$$\left. \frac{\Delta \ell}{\ell} \right|_{\vec{n}_1} = +\alpha \quad (\text{dilation along } x = y), \quad \left. \frac{\Delta \ell}{\ell} \right|_{\vec{n}_2} = -\alpha \quad (\text{contraction along } x = -y),$$

and $\Delta \ell / \ell = 0$ along $\vec{n}_3 = \hat{z}$.

Thus, the material is stretched along the diagonal $x = y$ by a fractional amount α and compressed by the same amount along the orthogonal diagonal $x = -y$; there is no strain in the z -direction (see Fig. 16).

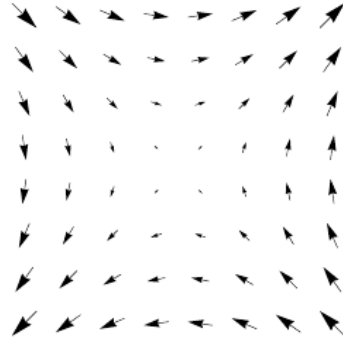


Figure 16: Arrow plot of the two-dimensional Lagrangian linear displacement field $\vec{u} = (y, x, 0)$ in the square $-1 < x < 1$ and $-1 < y < 1$. The material is dilated along one diagonal and contracted along the other. These are the principal directions of strain everywhere. Figure taken from Ref. [3].

2.3 Geometrical Meaning of the Strain Tensor

The strain tensor contains all the relevant information about local changes in geometric relationships, such as lengths of material elements and angles between them. Other geometric quantities, such as curves, surfaces, and volumes, are also affected under deformation.

2.3.1 Lengths and Angles

To define the projection of a tensor u_{ij} along two directions $\vec{a}$ and $\vec{b}$, we introduce

$$U_{ab} = \hat{a} \cdot \vec{U} \cdot \hat{b} = \frac{\vec{a} \cdot \vec{U} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}. \quad (197)$$

The change in the scalar product is

$$\delta(\vec{a} \cdot \vec{b}) = 2 |\vec{a}| |\vec{b}| U_{ab}. \quad (198)$$

Setting $\vec{b} = \vec{a}$ gives the change in length:

$$\frac{\delta|\vec{a}|}{|\vec{a}|} = U_{aa}. \quad (199)$$

Thus, the diagonal projection U_{aa} represents the *fractional change of length* in the direction of $\vec{a}$.

For the angle ϕ between $\vec{a}$ and $\vec{b}$, since $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \phi$, one finds

$$\delta\phi = \phi - \phi_0 = \frac{(U_{aa} + U_{bb}) \cos \phi - 2u_{ab}}{\sin \phi}. \quad (200)$$

For orthogonal vectors ($\phi = 90^\circ$),

$$\delta\phi = -2U_{ab}. \quad (201)$$

The off-diagonal projections of the strain tensor thus determine the change in angle between actually orthogonal needles.

2.3.2 Infinitesimal Elements

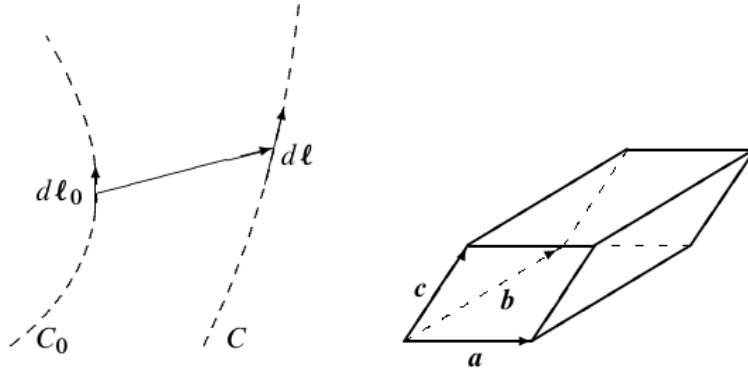


Figure 17: Left: A line element is stretched and rotated by the displacement that changes the curve from C_0 to C . Right : Three infinitesimal needles span a parallelepiped with volume $dV = \vec{a} \times \vec{b} \cdot \vec{c}$. Figure taken from Ref. [3].

Curve element. A curve element behaves like a small needle (see Fig. 17). Under a displacement, the change from $d\vec{\ell}_0$ to $d\vec{\ell}$ is

$$\delta(d\vec{\ell}) \equiv d\vec{\ell} - d\vec{\ell}_0 = d\vec{\ell} \cdot \nabla \vec{u} = \nabla \vec{u}^T \cdot d\vec{\ell}. \quad (202)$$

Volume Element. For a tiny parallelepiped spanned by three infinitesimal vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$, the volume element is $dV = \vec{a} \times \vec{b} \cdot \vec{c}$ (see Fig. 17). Its variation under displacement is

$$\delta(dV) \equiv dV - dV_0 = \nabla \cdot \vec{u} dV \quad (203)$$

$$= dV \sum_i \nabla_i u_i = dV U_{ii}. \quad (204)$$

Hence, the divergence of the displacement field which corresponds to the trace of $\vec{U}$, $\nabla \cdot \vec{u} = \text{Tr}(\vec{U})$, gives the fractional change of volume.

The change in density ρ of a material particle, with constant mass dM , satisfies

$$\delta\rho = -\rho \nabla \cdot \vec{u}. \quad (205)$$

Thus, when $\nabla \cdot \vec{u} = \text{Tr}(\vec{U}) = 0$, both the volume and density remain unchanged.

Thus, in general,

$$e = U_{ii} = \frac{\partial u_i}{\partial x_i} = \text{div } \mathbf{u}. \quad (206)$$

This unit volume change is known as *dilatation*. In terms of displacements, we have:

In rectangular Cartesian coordinates,

$$e = \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} + \frac{\partial u_3}{\partial x_3}. \quad (207)$$

In cylindrical coordinates,

$$e = \frac{\partial u_r}{\partial r} + \frac{1}{r} \left(\frac{\partial u_\theta}{\partial \theta} + u_r \right) + \frac{\partial u_z}{\partial z}. \quad (208)$$

In spherical coordinates,

$$e = \frac{\partial u_r}{\partial r} + \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{2u_r}{r} + \frac{1}{r \sin \theta} \frac{\partial u_\phi}{\partial \phi} + \frac{u_\theta \cot \theta}{r}. \quad (209)$$

Surface Element. For a surface element $d\vec{S} = \vec{a} \times \vec{b}$, we find from the volume relation that

$$\delta(d\vec{S}) = (\nabla \cdot \vec{u} \mathbf{1} - \nabla \vec{u}) \cdot d\vec{S}. \quad (210)$$

Both the magnitude and direction of $d\vec{S}$ are modified by the deformation, following a rule different from that of the curve element.

2.4 Displacement Work

Deforming a body requires work, part of which is stored as elastic potential energy, while another part is dissipated as heat due to internal friction. No real material is perfectly elastic. For instance, when squeezing a rubber ball, the stored energy is released when the force is removed, but a steel ball bouncing on a floor eventually stops because energy is lost to internal friction and air resistance. In continuum mechanics, energy relations are best derived by *following the work*—that is, tracking the work done by forces within the body.

Consider a volume V of material not in mechanical equilibrium. Let the effective force acting on an element of volume dV be $d\vec{F} = \vec{f}^* dV$. To keep all material points in fixed, non-equilibrium positions, we must apply an external distribution of *virtual forces*

$$\vec{f}' = -\vec{f}^*. \quad (211)$$

If the body is then displaced infinitesimally by $\delta\vec{u}(\vec{x})$, the work done by the virtual forces is

$$\delta W = \int_V \vec{f}' \cdot \delta\vec{u} dV = - \int_V \vec{f}^* \cdot \delta\vec{u} dV \quad (212)$$

$$= - \int_V \vec{f} \cdot \delta\vec{u} dV - \int_V (\nabla \cdot \vec{\sigma}^T) \cdot \delta\vec{u} dV, \quad (213)$$

where $\vec{f}^* = \vec{f} + \nabla \cdot \vec{\sigma}^T$ was used. Then performing an integration by parts in the second term gives

$$\int_V \sum_{ij} (\nabla_j \sigma_{ij}) \delta u_i dV = \int_V \sum_{ij} \nabla_j (\sigma_{ij} \delta u_i) dV - \int_V \sum_{ij} \sigma_{ij} \nabla_j \delta u_i dV \quad (214)$$

$$= \int_V \nabla \cdot (\sigma^T \cdot \delta u) dV - \int_V \sum_{ij} \sigma_{ij} \nabla_j \delta u_i dV \quad (215)$$

$$= \oint_S (\sigma^T \cdot \delta u) \cdot d\vec{S} - \int_V \sum_{ij} \sigma_{ij} \nabla_j \delta u_i dV. \quad (216)$$

The surface term vanishes because $\delta\vec{u} = 0$ at the boundary S . The second integral can, by virtue of the symmetry of the tensor σ_{ij} , be written as

$$\int_V \sum_{ij} \sigma_{ij} \nabla_j \delta u_i dV = \frac{1}{2} \int_V \sum_{ij} \sigma_{ij} (\nabla_j \delta u_i + \nabla_i \delta u_j) dV \quad (217)$$

$$= \frac{1}{2} \int_V \sum_{ij} \sigma_{ij} \delta (\nabla_j u_i + \nabla_i u_j) dV \quad (218)$$

$$= \int_V \sum_{ij} \sigma_{ij} \delta U_{ij} dV \quad (219)$$

$$= \int_V \text{Tr}[\vec{\sigma} \cdot \delta\vec{U}] dV \equiv \int_V \vec{\sigma} : \delta\vec{U} dV, \quad (220)$$

where

$$\delta U_{ij} = \frac{1}{2} (\nabla_i \delta u_j + \nabla_j \delta u_i) \quad (221)$$

is the infinitesimal change in the strain tensor.

The virtual work becomes

$$\delta W = - \int_V \vec{f} \cdot \delta\vec{u} dV + \int_V \vec{\sigma} : \nabla \delta\vec{u} dV \quad (222)$$

$$= - \int_V \vec{f} \cdot \delta\vec{u} dV + \int_V \vec{\sigma} : \delta\vec{U} dV. \quad (223)$$

where $\vec{\sigma} : \nabla \delta \vec{u} = \sum_{ij} \sigma_{ij} \nabla_j \delta u_i$. **The first term represents work against body forces, and the second corresponds to work against internal stresses.** The portion of virtual work associated with internal stresses defines the infinitesimal work of deformation:

$$\delta W_{\text{deform}} = \int_V \vec{\sigma} : \nabla \delta \vec{u} dV. \quad (224)$$

Thus, the deformation work represents the contribution to the body's *deformation energy*.

2.5 Thermodynamics of deformation

2.5.1 Concepts from Thermodynamics

The first law expresses energy conservation. In specific (per unit mass) form:

$$d\varepsilon = dq + dw. \quad (225)$$

Restricting to mechanical work due to volume changes,

$$dw = -p d\mathcal{V}, \quad (226)$$

so that

$$d\varepsilon = dq - p d\mathcal{V}. \quad (227)$$

Introducing entropy s and temperature T , **the second law** implies

$$dq \leq T ds, \quad (228)$$

with equality for reversible processes. Hence, for reversible transformations,

$$d\varepsilon = T ds - p d\mathcal{V}. \quad (229)$$

This identifies temperature and pressure as

$$T = \left(\frac{\partial \varepsilon}{\partial s} \right)_{\mathcal{V}}, \quad (230)$$

$$p = - \left(\frac{\partial \varepsilon}{\partial \mathcal{V}} \right)_s. \quad (231)$$

Since $d\varepsilon$ is an exact differential, the Maxwell relation follows:

$$\left(\frac{\partial T}{\partial \mathcal{V}} \right)_s = - \left(\frac{\partial p}{\partial s} \right)_{\mathcal{V}}. \quad (232)$$

Moreover, there exists an equation of state

$$\varepsilon = \varepsilon(s, \mathcal{V}),$$

so that the thermodynamic state can be specified by two independent variables among ε , s , and $\mathcal{V}$.

The variables ε , s , and $\mathcal{V}$ are *specific extensive quantities*, derived from the extensive variables E_{int} , S , and V . Extensive quantities scale with system size, while specific quantities remove this dependence and characterize intrinsic material properties. In contrast, temperature T and pressure p are *intensive variables*. From the first law,

$$d\varepsilon = T ds - p d\mathcal{V}, \quad (233)$$

we identify the conjugate pairs:

$$(T, s), \quad (p, \mathcal{V}).$$

This also shows that T and p are not independent variables, but are determined from the equation of state.

Common thermodynamic processes are defined by constraints on these variables:

- **Adiabatic:** $dq = 0$
- **Incompressible:** $d\mathcal{V} = 0$
- **Isentropic:** $ds = 0$
- **Isothermal:** $dT = 0$
- **Isobaric:** $dp = 0$

Legendre Transforms

Instead of describing a thermodynamic system in terms of the variables $(s, \mathcal{V})$, one can equivalently use the intensive variables (T, p) by means of Legendre transforms. Starting from

$$d\varepsilon = T ds - p d\mathcal{V}, \quad (234)$$

new thermodynamic potentials are defined by subtracting the conjugate pairs.

Gibbs free energy

$$g = \varepsilon - Ts + p\mathcal{V}, \quad (235)$$

$$dg = -s dT + \mathcal{V} dp. \quad (236)$$

Enthalpy

$$h = \varepsilon + p\mathcal{V}, \quad (237)$$

$$dh = T ds + \mathcal{V} dp. \quad (238)$$

Helmholtz free energy

$$f = \varepsilon - Ts, \quad (239)$$

$$df = -s dT - p d\mathcal{V}. \quad (240)$$

These potentials correspond to different choices of independent variables:

$$\varepsilon(s, \mathcal{V}), \quad f(T, \mathcal{V}), \quad h(s, p), \quad g(T, p).$$

Since these are exact differentials, they imply Maxwell relations:

$$\left(\frac{\partial s}{\partial p}\right)_T = -\left(\frac{\partial \mathcal{V}}{\partial T}\right)_p, \quad (241)$$

$$\left(\frac{\partial T}{\partial p}\right)_s = -\left(\frac{\partial \mathcal{V}}{\partial s}\right)_p, \quad (242)$$

$$\left(\frac{\partial s}{\partial \mathcal{V}}\right)_T = -\left(\frac{\partial p}{\partial T}\right)_\mathcal{V}. \quad (243)$$

2.5.2 Deformation regimes

In solid continuum mechanics, deformation regimes are classified according to the magnitude of the deformation and the material response (see Fig. 18). Broadly, three regimes are distinguished:

1. Infinitesimal (small) deformations.

In this regime, displacements and strains are very small compared to unity. The linearized strain tensor is valid, geometric nonlinearities such as large rotations or nonlinear displacement gradients are neglected, and the principle of superposition applies. This approximation underlies most problems in classical elasticity, such as stress analysis in beams and plates.

2. Finite (large) deformations.

Here, displacements or strains are not negligible, and nonlinear kinematics must be employed. Quantities such as the deformation gradient tensor and the Green–Lagrange strain are used. Large rotations, shear, and finite strains are fully accounted for, and the governing equations are nonlinear. Applications include rubber elasticity, biomechanics of soft tissues, and geomechanics.

3. Extreme (highly nonlinear) deformations.

This regime involves very large strains, often accompanied by material nonlinearities or instabilities. The material response may be anisotropic or rate-dependent, and phenomena such as plasticity, viscoelasticity, viscoplasticity, damage, or fracture must be considered. Examples include crash simulations, metal forming at large strains, and failure analysis.

In summary: the *small deformation regime* corresponds to linear elasticity, the *finite deformation regime* requires nonlinear kinematics but can still be elastic or elastic–plastic, while the *extreme regime* involves very large strains typically coupled with inelastic effects such as plasticity, fracture, or damage.

2.5.3 First law

Besides elastic deformations, we shall also suppose that the process of deformation occurs so slowly that thermodynamic equilibrium is established in the body at every instant, in accordance with the external conditions. This assumption is almost always justified in practice. The process will then be thermodynamically reversible.

In what follows we shall take all such thermodynamic quantities as the entropy $\mathcal{S}$, the internal energy $\mathcal{E}$, etc., relative to unit volume of the body, and not relative to unit mass as in fluid mechanics.

The unit volumes before and after deformation must be distinguished since they generally contain different amounts of matter. The thermodynamic quantities are therefore referred to the

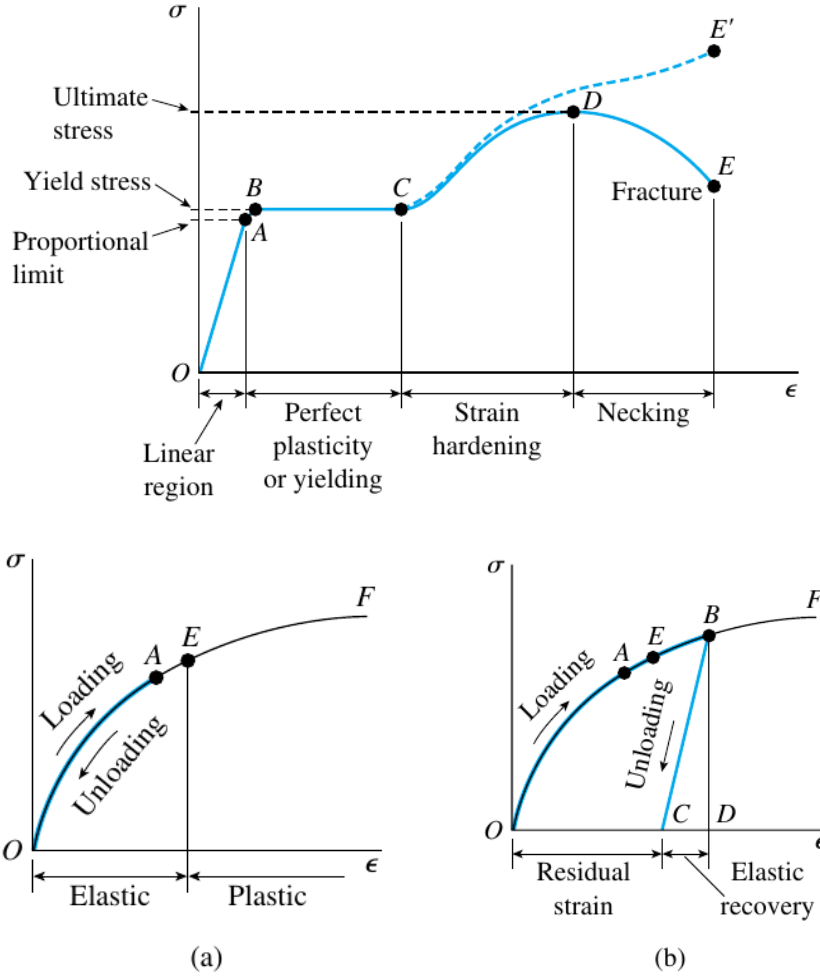


Figure 18: Stress-strain diagrams illustrating (a) elastic behavior, and (b) partially elastic behavior.

unit volume of the **undeformed body**. The total internal energy of the body is obtained by integrating $\mathcal{E}$ over the undeformed volume.

An infinitesimal change in the internal energy is given by the difference between the heat acquired by the unit volume and the work done by internal stresses $\delta R = dW/dV$. For a reversible process, $Td\mathcal{S}$ represents the heat, and δR the work. Thus,

$$d\mathcal{E} = Td\mathcal{S} + \delta R. \quad (244)$$

Using $dW = \sum_{ij} \sigma_{ij} \delta U_{ij} dV$, we obtain

$$d\mathcal{E} = Td\mathcal{S} + \sum_{ij} \sigma_{ij} \nabla_j \delta u_i \quad (245)$$

$$= Td\mathcal{S} + \sum_{ij} \sigma_{ij} \delta U_{ij}. \quad (246)$$

This is the **fundamental thermodynamic relation** for deformed bodies.

In the case of hydrostatic compression, $\sigma_{ij} = -p\delta_{ij}$, leading to

$$\sum_{ij} \sigma_{ij} \nabla_j \delta u_i = -p \sum_i \nabla_i \delta u_i = -p \nabla \cdot \delta \vec{u} = -p \frac{\delta(dV)}{dV}. \quad (247)$$

Therefore, this reduces to the familiar form

$$dE = T dS - p \delta(dV). \quad (248)$$

2.6 Helmholtz and Gibbs Free Energies

In the thermodynamic description of elastic bodies, it is convenient to introduce thermodynamic potentials whose natural variables correspond to the physical quantities held fixed in a given situation. Two such potentials are the **Helmholtz free energy** and the **Gibbs free energy**.

2.6.1 Helmholtz Free Energy

The Helmholtz free energy is defined as

$$\mathcal{F} = \mathcal{E} - T\mathcal{S}, \quad (249)$$

where $\mathcal{E}$ is the internal energy per unit volume, T the temperature, and $\mathcal{S}$ the entropy per unit volume. Its differential form at constant temperature is

$$d\mathcal{F} = -\mathcal{S} dT + \sum_{ik} \sigma_{ik} dU_{ik}. \quad (250)$$

Thus $\mathcal{F} = \mathcal{F}(T, U_{ik})$ is naturally a function of the temperature and the strain tensor. Differentiation with respect to U_{ik} gives the stress tensor:

$$\sigma_{ik} = \left(\frac{\partial \mathcal{E}}{\partial U_{ik}} \right)_{\mathcal{S}} = \left(\frac{\partial \mathcal{F}}{\partial U_{ik}} \right)_T. \quad (251)$$

This makes $\mathcal{F}$ particularly useful in elasticity, where one often specifies the strain and seeks the corresponding stress.

2.6.2 Gibbs Free Energy

The Gibbs free energy is defined as

$$\Phi = \mathcal{E} - T\mathcal{S} - \sum_{ik} \sigma_{ik} U_{ik} = \mathcal{F} - \sum_{ik} \sigma_{ik} U_{ik}. \quad (252)$$

Its differential is

$$d\Phi = -\mathcal{S} dT - \sum_{ik} U_{ik} d\sigma_{ik}, \quad (253)$$

which shows that Φ is naturally a function of temperature and stress:

$$\Phi = \Phi(T, \sigma_{ik}).$$

Differentiating with respect to σ_{ik} yields the strain tensor:

$$U_{ik} = - \left(\frac{\partial \Phi}{\partial \sigma_{ik}} \right)_T. \quad (254)$$

Helmholtz and Gibbs free energies are fundamental thermodynamic potentials that quantify the amount of energy in a system available to do useful work, but they apply under different physical constraints. The Helmholtz free energy is the appropriate potential for systems held at constant temperature and volume, making it useful in contexts such as elasticity where the deformation (and therefore volume change) is controlled. The Gibbs free energy, on the other hand, applies to systems at constant temperature and pressure, and is particularly relevant for processes where the external pressure is fixed, such as phase transitions or chemical reactions. Both potentials encode the balance between internal energy and entropy, but their distinct natural variables determine which one is most convenient for describing a given physical situation.

The two potentials are related by a Legendre transform exchanging the conjugate variables (U_{ik}, σ_{ik}) . Both potentials encode the elastic response of the material and provide a systematic way to derive constitutive relations such as Hooke's law from thermodynamic principles.

2.7 Exercises

1. Consider the displacement field

$$u_1 = k(2X_1^2 + X_1X_2), \quad u_2 = kX_2^2, \quad u_3 = 0, \quad k = 10^{-4}. \quad (255)$$

- (a) Find the unit elongations and the change in angle for two material elements $d\mathbf{X}^{(1)} = dX_1\mathbf{e}_1$ and $d\mathbf{X}^{(2)} = dX_2\mathbf{e}_2$ that emanate from the particle at $\mathbf{X} = \mathbf{e}_1 + \mathbf{e}_2$.
- (b) Sketch the deformed positions of these two elements.

2. In rectangular Cartesian coordinates, the strain tensor at a point is

$$[U] = 10^{-4} \begin{bmatrix} 5 & 3 & 0 \\ 3 & 4 & -1 \\ 0 & -1 & 2 \end{bmatrix}. \quad (256)$$

- (a) What is the unit elongation in the direction $2\mathbf{e}_1 + 2\mathbf{e}_2 + \mathbf{e}_3$?
- (b) What is the change in angle between two lines that are perpendicular in the undeformed state and point in the directions $2\mathbf{e}_1 + 2\mathbf{e}_2 + \mathbf{e}_3$ and $3\mathbf{e}_1 - 6\mathbf{e}_3$?

3. For the strain tensor given in the problem above:

- (a) Find the unit elongation in the direction $3\mathbf{e}_1 - 4\mathbf{e}_2$.
- (b) Find the change in angle between two elements in the directions $3\mathbf{e}_1 - 4\mathbf{e}_3$ and $4\mathbf{e}_1 + 3\mathbf{e}_3$.

4. (a) Determine the principal scalar invariants of the strain tensor on the left. (b) Show that the matrix on the right cannot represent the same state of strain:

$$[U] = 10^{-4} \begin{bmatrix} 5 & 3 & 0 \\ 3 & 4 & -1 \\ 0 & -1 & 2 \end{bmatrix}, \quad 10^{-4} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 2 \end{bmatrix}. \quad (257)$$

5. The matrix of an infinitesimal strain tensor is

$$[U] = \begin{bmatrix} k_1 x_2 & 0 & 0 \\ 0 & -k_2 x_2 & 0 \\ 0 & 0 & -k_2 x_2 \end{bmatrix}. \quad (258)$$

- (a) Find the location of the particles that undergo no volume change.
- (b) What relation must hold between k_1 and k_2 so that no material element changes its volume?

6. Check whether the following strain field satisfies the compatibility conditions:

$$[U] = k \begin{bmatrix} x_1 + x_2 & x_1 & x_2 \\ x_1 & x_2 + x_3 & x_3 \\ x_2 & x_3 & x_1 + x_3 \end{bmatrix}, \quad k = 10^{-4}. \quad (259)$$

7. Check whether the following strain field satisfies the compatibility conditions:

$$[U] = k \begin{bmatrix} x_1^2 & x_2^2 + x_3^2 & x_1 x_3 \\ x_2^2 + x_3^2 & 0 & x_1 \\ x_1 x_3 & x_1 & x_2^2 \end{bmatrix}, \quad k = 10^{-4}. \quad (260)$$

8. Does the displacement field

$$u_1 = \sin x_1, \quad u_2 = x_1^3 x_2, \quad u_3 = \cos x_3 \quad (261)$$

correspond to a compatible strain field?

9. Given the strain field

$$U_{12} = U_{21} = k x_1 x_2, \quad k = 10^{-4}, \quad (262)$$

with all other $U_{ij} = 0$:

- (a) Check the compatibility equations for this field.
- (b) By attempting to integrate the strain field, show that no continuous displacement field produces it.

10. Given the strain components

$$U_{11} = \frac{1}{\alpha} f(x_2, x_3), \quad U_{22} = U_{33} = -\frac{\nu}{\alpha} f(x_2, x_3), \quad U_{12} = U_{13} = U_{23} = 0, \quad (263)$$

show that compatibility requires $f(x_2, x_3)$ to be linear in x_2 and x_3 .

3 Hooke's law

To apply the general thermodynamic formalism to a specific case, we must express the free energy $\mathcal{F}$ of the body as a function of the strain tensor U_{ik} . Assuming small deformations, the free energy can be expanded in powers of U_{ik} . We mainly restrict our attention to isotropic bodies.

3.1 Isotropic bodies: free energy and elastic coefficients

For a deformed body at constant temperature, we take the undeformed state as the reference configuration, in the absence of external forces and at the same temperature. When $U_{ik} = 0$, the internal stresses vanish, i.e. $\sigma_{ik} = 0$. Hence, since $\sigma_{ik} = \partial F / \partial U_{ik}$, there is no linear term in the expansion of F in powers of U_{ik} .

As $\mathcal{F}$ is a scalar quantity, its expansion must involve scalar combinations of U_{ik} . Two independent second-order scalars can be formed:

$$\left(\sum_i U_{ii} \right)^2 \quad \text{and} \quad \sum_{ik} U_{ik}^2. \quad (264)$$

Thus, retaining terms up to second order, we obtain

$$\mathcal{F} = \mathcal{F}_0 + \frac{1}{2} \lambda \left(\sum_i U_{ii} \right)^2 + \mu \sum_{ik} U_{ik}^2 \quad (265)$$

$$= \mathcal{F}_0 + \frac{1}{2} \lambda (\text{Tr} U)^2 + \mu \text{Tr}(U^2) \quad (266)$$

where λ and μ are the **Lamé coefficients**. This is the general expression for the free energy of an isotropic elastic body.

A deformation in which $\sum_i U_{ii} = 0$ corresponds to no change in volume but a change in shape, called a **pure shear**. Conversely, a deformation for which $U_{ik} = \text{constant} \times \delta_{ik}$ represents a change in volume without a change in shape, called a **hydrostatic compression**.

3.2 Decomposition of the strain tensor

Any deformation can be written as the sum of a pure shear and a hydrostatic compression:

$$U_{ik} = \left(U_{ik} - \frac{1}{3} \delta_{ik} \sum_l U_{ll} \right) + \frac{1}{3} \delta_{ik} \sum_l U_{ll}. \quad (267)$$

The first term represents a pure shear, and the second term a hydrostatic compression. Substituting this decomposition into $\mathcal{F}$, we obtain

$$\mathcal{F} = \mu \sum_{ik} \left(U_{ik} - \frac{1}{3} \delta_{ik} \sum_l U_{ll} \right)^2 + \frac{1}{2} K \left(\sum_l U_{ll} \right)^2, \quad (268)$$

where K is the **bulk modulus** (modulus of hydrostatic compression), and μ is the **shear modulus** (modulus of rigidity). The bulk modulus is related to the Lamé coefficients by

$$K = \lambda + \frac{2}{3} \mu. \quad (269)$$

Thermodynamic stability requires that the free energy be a minimum at equilibrium, i.e.

$$K > 0, \quad \mu > 0. \quad (270)$$

3.3 Stress–strain Relation

From the thermodynamic relation $\sigma_{ik} = (\partial\mathcal{F}/\partial U_{ik})_T$, we compute:

$$d\mathcal{F} = K \sum_l U_{ll} \sum_n dU_{nn} + 2\mu \sum_{ik} (U_{ik} - \frac{1}{3}\delta_{ik} \sum_l U_{ll}) d(U_{ik} - \frac{1}{3}\delta_{ik} \sum_n U_{nn}) \quad (271)$$

$$\begin{aligned} &= K \sum_l U_{ll} \sum_n dU_{nn} + 2\mu \sum_{ik} (U_{ik} - \frac{1}{3}\delta_{ik} \sum_l U_{ll}) \sum_{ik} dU_{ik} \\ &\quad - 2\mu \sum_{ik} (U_{ik} - \frac{1}{3}\delta_{ik} \sum_l U_{ll}) \frac{1}{3}\delta_{ik} \sum_n dU_{nn} \end{aligned} \quad (272)$$

$$= \left[K \sum_l U_{ll} \sum_{ik} \delta_{ik} + 2\mu \sum_{ik} (U_{ik} - \frac{1}{3}\delta_{ik} \sum_l U_{ll}) \right] dU_{ik} \quad (273)$$

$$= \left[(K - \frac{2}{3}\mu) \sum_l U_{ll} \sum_{ik} \delta_{ik} + 2\mu \sum_{ik} U_{ik} \right] dU_{ik} \quad (274)$$

$$= \left[\lambda \sum_l U_{ll} \sum_{ik} \delta_{ik} + 2\mu \sum_{ik} U_{ik} \right] dU_{ik}. \quad (275)$$

Hence, the stress tensor is

$$\sigma_{ik} = K \sum_l U_{ll} \delta_{ik} + 2\mu \left(U_{ik} - \frac{1}{3}\delta_{ik} \sum_l U_{ll} \right) \quad (276)$$

$$= \lambda \sum_l U_{ll} \delta_{ik} + 2\mu U_{ik}. \quad (277)$$

This expression determines the stress tensor in terms of the strain tensor for an isotropic body. It shows that, if the deformation is a pure shear or a pure hydrostatic compression, the relation between σ_{ik} and U_{ik} is determined only by the modulus of rigidity or of hydrostatic compression, respectively.

3.4 Hooke's Law

Taking the trace to the first line of the previous expression, we find

$$\sum_i \sigma_{ii} = 3K \sum_l U_{ll}, \quad \Rightarrow \quad \sum_l U_{ll} = \frac{\sum_i \sigma_{ii}}{3K}. \quad (278)$$

Substituting back we obtain the inverse relation:

$$U_{ik} = \frac{\delta_{ik}}{9K} \sum_l \sigma_{ll} + \frac{1}{2\mu} \left(\sigma_{ik} - \frac{1}{3}\delta_{ik} \sum_l \sigma_{ll} \right), \quad (279)$$

which expresses the strain tensor in terms of the stress tensor. From Eq. (279), we see that U_{ik} is a linear function of σ_{ik} ; this proportionality defines **Hooke's law** for small deformations.

Example. For an isotropic Hookean material: (a) show that the principal directions of stress and strain coincide; (b) find a relation between their principal values.

Solution.

- (a) Let $\mathbf{n}_1$ be an eigenvector of the strain tensor U , so that $U\mathbf{n}_1 = U_1\mathbf{n}_1$. Hooke's law gives

$$\boldsymbol{\sigma}\mathbf{n}_1 = 2\mu U\mathbf{n}_1 + \lambda e\mathbf{n}_1 = (2\mu U_1 + \lambda e)\mathbf{n}_1. \quad (280)$$

Thus, $\mathbf{n}_1$ is also a principal direction of stress.

- (b) Let U_1, U_2, U_3 be the principal strains. Since $e = U_1 + U_2 + U_3$, the corresponding principal stresses are

$$\sigma_1 = 2\mu U_1 + \lambda(U_1 + U_2 + U_3), \quad (281)$$

$$\sigma_2 = 2\mu U_2 + \lambda(U_1 + U_2 + U_3), \quad (282)$$

$$\sigma_3 = 2\mu U_3 + \lambda(U_1 + U_2 + U_3). \quad (283)$$

Example. For an isotropic Hookean material: (a) find a relation between the first invariants of stress and strain; (b) use it to express strain in terms of stress.

Solution.

- (a) Contracting the indices in Hooke's law, $\sigma_{ij} = 2\mu U_{ij} + \lambda e\delta_{ij}$, gives

$$\sigma_{kk} = (2\mu + 3\lambda)U_{kk} = (2\mu + 3\lambda)e. \quad (284)$$

- (b) Hence,

$$e = \frac{\sigma_{kk}}{2\mu + 3\lambda}. \quad (285)$$

Substituting this into Hooke's law and solving for U , we obtain

$$U = \frac{1}{2\mu}\boldsymbol{\sigma} - \frac{\lambda\sigma_{kk}}{2\mu(2\mu + 3\lambda)}I. \quad (286)$$

Example. A material is called *incompressible* if its volume does not change under any state of stress. Show that, for an incompressible isotropic linearly elastic solid with finite Young's modulus E_Y , (a) Poisson's ratio is $\nu = 1/2$, (b) the shear modulus is $\mu = E_Y/3$, and (c) $k \rightarrow \infty$, $\lambda \rightarrow \infty$, while $k - \lambda = 2\mu/3$.

Solution.

- (a) Contracting the inverse form of Hooke's law gives

$$U_{kk} = \frac{1}{E_Y} [(1 + \nu)\sigma_{kk} - 3\nu\sigma_{kk}] = \frac{1 - 2\nu}{E_Y}\sigma_{kk}. \quad (287)$$

Since $U_{kk} = e$ is the change in volume per unit volume, it must vanish under any stress in an incompressible material. As E_Y is finite, $\nu = 1/2$.

- (b) Using $E_Y = 2\mu(1 + \nu)$, we obtain

$$E_Y = 3\mu, \quad \text{hence} \quad \mu = \frac{E_Y}{3}. \quad (288)$$

(c) For a hydrostatic stress $\sigma_{ij} = \sigma\delta_{ij}$, the bulk modulus is

$$k = \frac{\sigma}{e} = \frac{E_Y}{3(1-2\nu)}. \quad (289)$$

Thus $k \rightarrow \infty$ as $\nu \rightarrow 1/2$. Since

$$k = \lambda + \frac{2\mu}{3}, \quad (290)$$

we also have $\lambda \rightarrow \infty$, whereas

$$k - \lambda = \frac{2\mu}{3} \quad (291)$$

remains finite.

Example. Obtain the displacement functions by integrating the strain–displacement relations for the strain components given by

$$u_1 = \left(\frac{\sigma}{E_Y}\right) x_1, \quad u_2 = -\nu \left(\frac{\sigma}{E_Y}\right) x_2, \quad u_3 = -\nu \left(\frac{\sigma}{E_Y}\right) x_3. \quad (292)$$

Solution. The relations

$$\frac{\partial u_1}{\partial x_1} = \frac{\sigma}{E_Y}, \quad \frac{\partial u_2}{\partial x_2} = -\frac{\nu\sigma}{E_Y}, \quad \frac{\partial u_3}{\partial x_3} = -\frac{\nu\sigma}{E_Y}$$

give

$$u_1 = \frac{\sigma}{E_Y} x_1 + f_1(x_2, x_3), \quad u_2 = -\frac{\nu\sigma}{E_Y} x_2 + f_2(x_1, x_3), \quad u_3 = -\frac{\nu\sigma}{E_Y} x_3 + f_3(x_1, x_2). \quad (293)$$

Here $f_1(x_2, x_3)$, $f_2(x_1, x_3)$, and $f_3(x_1, x_2)$ are integration functions. Substituting these expressions into

$$\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} = 0, \quad \frac{\partial u_1}{\partial x_3} + \frac{\partial u_3}{\partial x_1} = 0, \quad \frac{\partial u_2}{\partial x_3} + \frac{\partial u_3}{\partial x_2} = 0,$$

We obtain

$$\begin{aligned} \frac{\partial f_1(x_2, x_3)}{\partial x_2} &= -\frac{\partial f_2(x_1, x_3)}{\partial x_1} = g_1(x_3), \\ \frac{\partial f_1(x_2, x_3)}{\partial x_3} &= -\frac{\partial f_3(x_1, x_2)}{\partial x_1} = g_2(x_2), \\ \frac{\partial f_2(x_1, x_3)}{\partial x_3} &= -\frac{\partial f_3(x_1, x_2)}{\partial x_2} = g_3(x_1). \end{aligned} \quad (ii)$$

Here g_1 , g_2 , and g_3 are integration functions. Integrating gives

$$f_1 = g_1(x_3)x_2 + g_4(x_3), \quad f_1 = g_2(x_2)x_3 + g_6(x_2), \quad (iii)$$

$$f_2 = g_1(x_3)x_1 + g_5(x_3), \quad f_2 = g_3(x_1)x_3 + g_8(x_1), \quad (iv)$$

$$-f_3 = g_2(x_2)x_1 + g_7(x_2), \quad -f_3 = g_3(x_1)x_2 + g_9(x_1). \quad (v)$$

From (iii),

$$g_1(x_3) = a_1x_3 + b_1, \quad g_2(x_2) = a_1x_2 + b_2, \quad g_4(x_3) = b_2x_3 + c_2, \quad g_6(x_2) = b_1x_2 + c_2. \quad (vi)$$

From (iv) and (vi),

$$g_3(x_1) = -a_1x_1 + b_3, \quad g_8(x_1) = -b_1x_1 + c_3, \quad -g_5(x_3) = b_3x_3 + c_3. \quad (\text{vii})$$

From (v), (vi), and (vii),

$$a_1 = 0, \quad g_9(x_1) = b_2x_1 + c_4, \quad g_7(x_2) = b_3x_2 + c_4. \quad (\text{viii})$$

Thus,

$$f_1 = b_1x_2 + b_2x_3 + c_2, \quad f_2 = b_3x_3 - b_1x_1 + c_3, \quad f_3 = -b_2x_1 - b_3x_2 - c_4. \quad (\text{ix})$$

Consequently,

$$\begin{aligned} u_1 &= \frac{\sigma}{E_Y}x_1 + b_1x_2 + b_2x_3 + c_2, \\ u_2 &= -\frac{\nu\sigma}{E_Y}x_2 + b_3x_3 - b_1x_1 + c_3, \\ u_3 &= -\frac{\nu\sigma}{E_Y}x_3 - b_2x_1 - b_3x_2 - c_4. \end{aligned} \quad (\text{x})$$

The terms

$$u_1 = b_1x_2 + b_2x_3 + c_2, \quad u_2 = b_3x_3 - b_1x_1 + c_3, \quad u_3 = -b_2x_1 - b_3x_2 - c_4$$

describe a rigid body motion: their displacement gradient is antisymmetric.

3.5 Homogeneous deformations

To see how Hooke's law works for continuous systems, we now turn to the extremely simple case of a *static uniform deformation* in which the strain tensor U_{ij} takes the same value everywhere in a body at all times. Hooke's law then ensures that the stress tensor is likewise constant throughout the body, so that all its derivatives vanish, $\nabla_k \sigma_{ij} = 0$. From the condition for mechanical equilibrium,

$$f_i + \sum_{j=1}^3 \nabla_j \sigma_{ij} = 0,$$

it follows that $f_i = 0$, so that uniform deformation necessarily excludes body forces. Conversely, in the presence of body forces such as gravity or electromagnetism, there must always be non-uniform deformation of an isotropic material.

Furthermore, at the boundary of a uniformly deformed body, the stress vector

$$\vec{\sigma} \cdot \hat{n} = \left\{ \sum_j \sigma_{ij} n_j \right\} \quad (11)$$

is required to be continuous, and this puts strong restrictions on the form of the external forces that may act on the surface of the body. Uniform deformation is for this reason only possible under very special circumstances, but when it applies, the displacement field is nearly trivial.

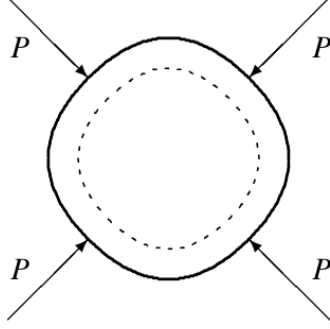


Figure 19: A body made from isotropic, homogenous material subject to a uniform external pressure will be uniformly compressed. Figure taken from Ref. [3].

3.5.1 Hydrostatic Compression

In a fluid at rest with a constant pressure p , the stress tensor is

$$\sigma_{ik} = -p\delta_{ik} \quad (12)$$

everywhere. If a solid body made from isotropic material is immersed into this fluid, the natural guess is that the pressure will also be p inside the body. Since $\sum_l \sigma_{ll} = -3p$, the strain becomes

$$U_{ik} = -\frac{p}{3K}\delta_{ik}. \quad (13)$$

On the other hand, the displacement field can be obtained from

$$U_{xx} = \nabla_x u_x = -\frac{p}{3K}, \quad (14)$$

$$U_{yy} = \nabla_y u_y = -\frac{p}{3K}, \quad (15)$$

$$U_{zz} = \nabla_z u_z = -\frac{p}{3K}. \quad (16)$$

Integrating directly we are able to find a particular solution for the displacement field:

$$u_x = -\frac{p}{3K}x, \quad (17)$$

$$u_y = -\frac{p}{3K}y, \quad (18)$$

$$u_z = -\frac{p}{3K}z. \quad (19)$$

The most general solution is obtained by adding an arbitrary infinitesimal rigid body displacement to this expression.

We emphasize that this result was obtained by making a reasonable guess for the form of the stress tensor inside the body. Although such a guess could, in principle, be incorrect, it is in fact justified by a uniqueness theorem. The theorem ensures—much like the corresponding uniqueness theorem in electrostatics—that if the mechanical equilibrium equations and the boundary conditions are satisfied by a proposed solution, then the solution is unique (up to an arbitrary small rigid body displacement). Thus, in any elastostatic problem, one may freely add a small rigid body motion to the solution without affecting the stresses.

3.5.2 Uniform stretching

Consider a rod along the z -axis, with forces applied uniformly over its ends (the rod is only pulled (or pushed) from its ends). On the lateral surface (the cylindrical surface), there is no force applied.

Let the force per unit area be p . Since the deformation is homogeneous, σ_{ik} is constant. Since there is no lateral force on the sides of the rod ($n_z = 0$), we have that

$$\sigma \cdot \hat{n} = \sum_k \sigma_{ik} n_k = 0, \quad (20)$$

where $\hat{n}$ is the outward normal vector to the side surface. Because the sides are parallel to the z -axis, the outward normal $\hat{n}$ has:

$$\hat{n} = (n_x, n_y, 0) = (n_1, n_2, 0). \quad (21)$$

It follows that

$$\sigma_{11} n_1 + \sigma_{12} n_2 = 0, \quad (22)$$

$$\sigma_{12} n_1 + \sigma_{22} n_2 = 0, \quad (23)$$

$$\sigma_{13} n_1 + \sigma_{23} n_2 = 0. \quad (24)$$

Therefore $\sigma_{ik} = 0$ except σ_{zz} . On the end surface we have

$$\sum_k \sigma_{3k} n_k = p, \quad (25)$$

from which $\sigma_{zz} = p$.

From the general strain-stress relation,

$$U_{xx} = U_{yy} = -\frac{1}{3} \left(\frac{1}{2\mu} - \frac{1}{3K} \right) p, \quad (26)$$

$$U_{zz} = \frac{1}{3} \left(\frac{1}{3K} + \frac{1}{\mu} \right) p. \quad (27)$$

Young's modulus

The component U_{zz} gives the relative lengthening of the rod. The coefficient of p is called the coefficient of extension, and its reciprocal is the modulus of extension or Young's modulus, E . The longitudinal extension is

$$U_{zz} = \frac{\sigma_{zz}}{E} = \frac{p}{E}, \quad (28)$$

where

$$E = \frac{9K\mu}{3K + \mu} \quad (29)$$

is Young's modulus.

Young's modulus is measured in the same units as pressure, and typical values for metals are, like the bulk modulus, of the order of 10^{11} Pa = 100 GPa = 1 Mbar. In the same way as the bulk modulus is a measure of the incompressibility of a material, Young's modulus is a measure of the *instretchability*. The larger it is, the harder it is to stretch the material. In order to obtain a large

strain $U_{xx} \approx 100\%$, one would have to apply stresses of magnitude $\sigma_{xx} \approx E$. Such strains are, of course, not permitted in the theory of small deformations, but Young's modulus nevertheless sets the scale. The fact that the yield stress for metals is roughly a thousand times smaller than Young's modulus shows that for metals the elastic strain can never become larger than 10^{-3} . This in turn justifies the assumption of small displacement gradients underlying the use of Cauchy's strain tensor.

Poisson's ratio

On the other hand, the components U_{xx} and U_{yy} give the relative compression of the rod in the transverse direction. The ratio of the transverse compression to the longitudinal extension is called the Poisson's ratio ν :

$$U_{xx} = -\nu U_{zz}. \quad (30)$$

Using the Lamé constants,

$$\nu = \frac{1}{2} \frac{3K - 2\mu}{3K + \mu}. \quad (31)$$

Since $K > 0$ and $\mu > 0$,

$$-1 \leq \nu \leq \frac{1}{2}. \quad (32)$$

Poisson's ratio is also a material parameter characterizing isotropic materials. It is dimensionless, and it is always positive for natural materials. Typical values for metals lie around 0.30.

Finally, the relative volume change is

$$U_{ii} = \frac{p}{3K}. \quad (33)$$

The free energy of a stretched rod is

$$F = \frac{p^2}{2E}. \quad (34)$$

The displacement field is then obtained by integrating the relations

$$\nabla_x u_x = U_{xx}, \quad \nabla_y u_y = U_{yy}, \quad \nabla_z u_z = U_{zz},$$

which yields the particular solution

$$u_z = \frac{p}{E} z, \quad (35)$$

$$u_x = -\nu \frac{p}{E} x, \quad (36)$$

$$u_y = -\nu \frac{p}{E} y. \quad (37)$$

This displacement describes a simple dilatation along the z -axis and a corresponding compression toward the z -axis in the xy -plane.

It is often convenient to express the elastic constants in terms of Young's modulus E and Poisson's ratio ν . In these variables, the Lamé coefficients become

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}, \quad (38)$$

$$\mu = \frac{E}{2(1+\nu)}, \quad (39)$$

$$K = \frac{E}{3(1-2\nu)}. \quad (40)$$

The free energy of a deformed isotropic body can then be written as

$$F = \frac{E}{2(1+\nu)} \left(\sum_{ik} U_{ik}^2 + \frac{\nu}{1-2\nu} \left(\sum_l U_{ll} \right)^2 \right). \quad (41)$$

The stress tensor in terms of the strain tensor takes the form

$$\sigma_{ik} = \frac{E}{1+\nu} \left(U_{ik} + \frac{\nu}{1-2\nu} \sum_l u_{ll} \delta_{ik} \right). \quad (42)$$

Conversely, the strain tensor in terms of the stress tensor is

$$U_{ik} = \frac{1}{E} \left[(1+\nu)\sigma_{ik} - \nu \sum_l \sigma_{ll} \delta_{ik} \right]. \quad (43)$$

If the rod-like spring is clamped on the sides by a rigid material, the boundary conditions become

$$u_y = u_x = 0 \quad \text{on the sides.}$$

In this case, the only non-vanishing constant strain is U_{zz} , and the solution is obtained in the same manner as above .

3.5.3 Unilateral Compression

Let us now consider the compression of a rod whose sides are fixed so that they cannot move laterally. The external forces responsible for the compression act on the ends of the rod and along its length, taken to be the z -axis. Such a deformation is called a *unilateral compression*. Since the rod deforms only in the z -direction, the only nonzero component of the strain tensor is U_{zz} .

From Eq. (42) for the stress tensor in terms of the strain tensor, we obtain

$$\sigma_{xx} = \sigma_{yy} = \frac{E}{(1+\nu)(1-2\nu)} U_{zz}, \quad (44)$$

$$\sigma_{zz} = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)} U_{zz}. \quad (45)$$

Denoting the compressive force per unit area (stress) by p so that $\sigma_{zz} = p$ (negative for compression), we solve for the longitudinal strain:

$$U_{zz} = p \frac{(1+\nu)(1-2\nu)}{E(1-\nu)}. \quad (46)$$

The components σ_{xx} and σ_{yy} describe the transverse stresses generated by the lateral constraint. Substituting Eq. (46) into Eq. (45) gives

$$\sigma_{xx} = \sigma_{yy} = \frac{p\nu}{1-\nu}. \quad (47)$$

Finally, using the general elastic energy formula

$$F = \frac{1}{2}\sigma_{ik}u_{ik},$$

and observing that only σ_{zz} and u_{zz} are nonzero, we obtain the free energy of the compressed rod:

$$F = \frac{p^2(1+\nu)(1-2\nu)}{2E(1-\nu)}. \quad (48)$$

3.5.4 Uniform Shear

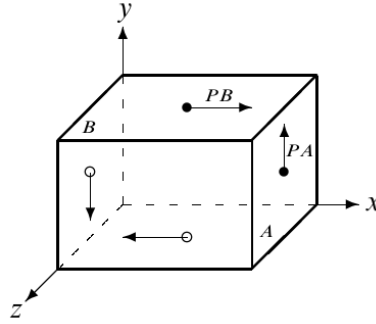


Figure 20: Clamped rectangular slab under constant shear stress $\sigma_{xy} = \sigma_{yx} = P$. The upper clamp is acted upon by an external force $\mathcal{F}_x = PB$, where B is the area of the clamp, while the force on the lower is $\mathcal{F}_x = -PB$. The symmetry of the stress tensor demands a clamp force $\mathcal{F}_y = PA$ on the right-hand side and a force $\mathcal{F}_y = -PA$ on the left-hand side, where A is the area of that clamp. Figure taken from Ref. [3].

Let's consider a clamped rectangular slab of homogeneous, isotropic material subjected to shear stress along one side (here, the x -direction). As argued there, the shear stress

$$\sigma_{xy} = P$$

must be constant throughout the material. What was not appreciated earlier is that the symmetry of the stress tensor requires that

$$\sigma_{yx} = P$$

everywhere as well. Consequently, shearing forces must act on the ends of the slab, while the remaining sides are free (see the figure 20).

Assuming there are no other stresses, the only nonzero strain component is

$$U_{xy} = \frac{P}{2\mu},$$

and using the relation

$$2U_{xy} = \nabla_y u_x + \nabla_x u_y,$$

we obtain a particular solution for the displacement field:

$$u_x = \frac{P}{\mu} y, \quad (49)$$

$$u_y = 0, \quad (50)$$

$$u_z = 0. \quad (51)$$

In these coordinates the displacement in the x -direction vanishes for $y = 0$ and increases linearly with y . Each infinitesimal “needle” of the material is not only sheared but also rotated by a small angle

$$\phi = \frac{1}{2} (\nabla_x u_y - \nabla_y u_x) = -\frac{P}{2\mu} \quad (52)$$

about the z -axis.

3.6 Navier–Cauchy Equation

Basic elastostatics (the theory of static elastic deformation) concerns deformations in bodies with highly symmetric non-exceptional shapes, subject to simple combinations of external forces. The bodies are assumed to be made from an isotropic and homogeneous linear elastic material, but even with all these assumptions it still takes some work to find the analytic solutions, when it can be done at all. The fundamental equations of elastostatics are obtained from the results of the preceding chapters. They consist of:

$$f_i + \sum_j \nabla_j \sigma_{ij} = 0, \quad \text{mechanical equilibrium,} \quad (53)$$

$$\sigma_{ij} = 2\mu U_{ij} + \lambda \delta_{ij} \sum_k U_{kk}, \quad \text{Hooke's law,} \quad (54)$$

$$U_{ij} = \frac{1}{2} (\nabla_i u_j + \nabla_j u_i), \quad \text{Cauchy's strain tensor.} \quad (55)$$

We shall use these equations with a time-independent external body force. In the presence of gravity, for example, the body force is

$$\vec{f} = \rho \vec{g}.$$

Other forces of electromagnetic origin can also appear, such as those generated by an inhomogeneous electric field acting on a dielectric material.

Substituting the second and third lines into the equilibrium condition, we obtain in index notation

$$\sum_j \nabla_j \sigma_{ij} = 2\mu \sum_j \nabla_j U_{ij} + \lambda \nabla_i \sum_j U_{jj} \quad (56)$$

$$= \mu \sum_j \nabla_j^2 u_i + (\lambda + \mu) \nabla_i \sum_j \nabla_j u_j. \quad (57)$$

Rewriting the second line in vector notation yields the equilibrium equation in its final form:

$$\vec{f} + \mu \nabla^2 \vec{u} + (\lambda + \mu) \nabla (\nabla \cdot \vec{u}) = \vec{0}. \quad (58)$$

This is known as *Navier's equation of equilibrium* or the *Navier–Cauchy equilibrium equation*. Although this equation is compact, it is often more convenient in analytic work to use the original system.

Because the displacement gradients are assumed to be small, all nonlinear terms are ignored. The displacement field is expressed as a function of the original coordinates of the undeformed material, using the Lagrangian representation.

The linearity of the equilibrium equations allows *superposition* of solutions. For instance, if a body is both compressed and stretched uniformly, the total displacement is simply the sum of the two individual displacements.

The boundary conditions are often implicit in the mere posing of an elastostatics problem. Typically, a part of the body surface is “glued” to a hard surface where the displacement has to vanish, and where the environment automatically provides the external reaction forces necessary to balance the surface stresses. On the remaining part of the body surface, explicit external forces implement the “user control” of the deformation. In regions where the external forces vanish, the body surface is said to be free. For the body to remain at rest, the total external force and the total external moment of force must always vanish.

3.6.1 The Equations of Equilibrium in a gravitational field

In terms of the Young’s modulus and Poisson’s ratio the equation becomes

$$\nabla^2 \vec{u} + \frac{1}{1-2\nu} \nabla(\nabla \cdot \vec{u}) = -\rho \vec{g} \frac{2(1+\nu)}{E}. \quad (59)$$

It is sometimes helpful to use the vector identity

$$\nabla(\nabla \cdot \vec{u}) = \nabla^2 \vec{u} + \nabla \times \nabla \times \vec{u}. \quad (60)$$

Then

$$\nabla(\nabla \cdot \vec{u}) - \frac{1-2\nu}{2(1-\nu)} \nabla \times \nabla \times \vec{u} = -\rho \vec{g} \frac{(1+\nu)(1-2\nu)}{E(1-\nu)}. \quad (61)$$

3.6.2 Surface-Loaded Bodies

A particularly important case occurs when the deformation is caused not by body forces but by forces applied only at the boundary. The equilibrium equation then becomes

$$(1-2\nu)\nabla^2 \vec{u} + \nabla(\nabla \cdot \vec{u}) = 0, \quad (62)$$

or, using the identity above,

$$2(1-\nu)\nabla(\nabla \cdot \vec{u}) - (1-2\nu)\nabla \times \nabla \times \vec{u} = 0. \quad (63)$$

The external forces appear in the solution only through the boundary conditions. Taking the divergence and using the identity

$$\nabla \cdot \nabla \equiv \nabla^2, \quad (64)$$

we obtain

$$\nabla^2(\nabla \cdot \vec{u}) = 0, \quad (65)$$

i.e. $\nabla \cdot \vec{u}$ (the volumetric strain) is a harmonic function. Taking the Laplacian to Eq. (62) we further obtain

$$\nabla^2 \nabla^2 \vec{u} = 0, \quad (66)$$

demonstrating that, in equilibrium, the displacement vector satisfies the *biharmonic equation*. This remains valid in a uniform gravitational field, but not for general external forces varying in space.

3.6.3 Estimates

When dealing with partial differential equations, it is often useful to obtain rough estimates for the magnitude of a solution. Such estimates aim only at capturing the correct order of magnitude of the fields, though particular circumstances in a specific problem may invalidate them. If that occurs—or if high precision is needed—analytic or numerical solutions are required.

Consider an elastic body subjected to surface stresses of typical magnitude P , and assume that gravitational effects on the deformation can be ignored. A rough estimate for the stresses inside the body is then

$$|\sigma_{ij}| \sim P. \quad (67)$$

Since the elastic moduli λ , μ , E , and K are of similar size, the strain is of order

$$|U_{ij}| \sim \frac{P}{E}. \quad (68)$$

As deformation arises from gradients of the displacement field, the typical variation in displacement across a body of characteristic size L is

$$|\Delta u_i| \sim L |U_{ij}| \sim \frac{LP}{E}. \quad (69)$$

Example: deformation of the legs of a chair

A person of mass $M \approx 70$ kg sits on a chair supported by four wooden legs whose total cross-sectional area is $A \approx 30$ cm² and length $L \approx 50$ cm. The stress on the legs is

$$P \approx \frac{Mg_0}{A} \approx 230 \text{ kPa}. \quad (70)$$

With $E \approx 10^9$ Pa for wood, the strain is

$$|U_{ij}| \sim \frac{P}{E} \approx 2.3 \times 10^{-4}, \quad (71)$$

and the maximal displacement is

$$|\Delta u_i| \sim \frac{PL}{E} \approx 0.1 \text{ mm}. \quad (72)$$

If gravity dominates, mechanical equilibrium implies a stress variation over height L of

$$|\Delta \sigma_{ij}| \approx \rho g L. \quad (73)$$

The corresponding strain variation is

$$|\Delta U_{ij}| \sim \frac{\rho g L}{E}. \quad (74)$$

Since U_{ij} is dimensionless, define the gravitational deformation scale

$$D \sim \frac{E}{\rho g}. \quad (75)$$

Thus,

$$|\Delta U_{ij}| \sim \frac{L}{D}. \quad (76)$$

Gravitationally induced displacement variations over height L satisfy

$$|\Delta u_i| \sim L |\Delta U_{ij}| \sim \frac{L^2}{D}. \quad (77)$$

Example: gravitational settling of a tall building.

Consider a building of height $L = 413 \text{ m}$ and ground area $A = 63 \times 63 \text{ m}^2$, with average density $\rho = 100 \text{ kg/m}^3$. Suppose its weight is carried by steel columns of total cross-section fA with $f = 1\%$. The total mass is

$$M = \rho AL = 1.6 \times 10^8 \text{ kg}. \quad (78)$$

The stress on the supports is

$$P = \frac{Mg_0}{fA} \approx 400 \text{ bar}. \quad (79)$$

With $E = 2 \times 10^{11} \text{ Pa}$ for steel, the deformation range is

$$|\Delta U_{ij}| \approx \frac{P}{E} \approx \frac{\rho g_0 L}{fE}, \quad (80)$$

so the gravitational deformation scale becomes

$$D \approx \frac{fE}{\rho g_0} \approx 2000 \text{ km}. \quad (81)$$

Thus the total settling of the building is

$$\frac{L^2}{D} \approx 8 \text{ cm}. \quad (82)$$

3.6.4 Saint-Venant's principle.

When an elastic body in mechanical equilibrium is subjected to an additional static load that acts only within a small, localized region, the resulting deformation does not propagate far into the body. Saint-Venant's principle states that *the deformation produced by a localized force distribution with zero net force and zero net moment decays rapidly and does not extend far beyond the linear size of the region where the forces are applied*. Numerical examples show that such effects typically fade within a few characteristic lengths of the loaded region.

The principle is extremely useful in engineering: as long as the total force and total moment applied to a structure are unchanged, the detailed manner in which these loads are distributed locally has negligible influence on the deformation at distant regions. Everyday situations illustrate this, such as the fact that standing with feet together or apart affects the floor only near the feet, since the total weight and moment remain unchanged.

Although intuitively reasonable, Saint-Venant's principle has been difficult to prove rigorously for arbitrary geometries. Proofs exist for certain regular shapes, while counterexamples appear in others. Despite the long-standing debate, engineers routinely rely on the principle because of its strong practical validity.

3.7 Solids at rest

3.7.1 Uniform settling

An infinitely extended slab of homogeneous and isotropic elastic material placed on a horizontal surface is a kind of “elastic sea”, which like the fluid sea may be assumed to have the same properties everywhere in a horizontal plane. In a flat-Earth coordinate system, where gravity is given by $\vec{g} = (0, 0, -g_0)$, we expect a uniformly vertical displacement, which only depends on the z -coordinate,

$$\vec{u} = (0, 0, u_z(z)) = u_z(z) \hat{e}_z.$$

In order to realize this “elastic sea” in a finite system, it must be surrounded by fixed, vertical, and slippery walls (see Fig. 21). The vertical walls forbid horizontal but allow vertical displacement, and at the bottom, $z = 0$, we place a horizontal supporting surface that forbids vertical displacement. At the top, $z = h$, the elastic material is left free to move without any external forces acting on it.

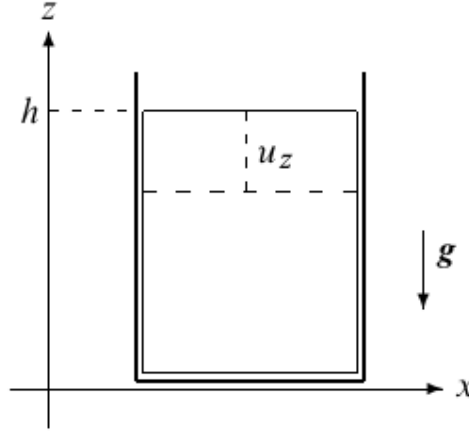


Figure 21: Elastic “sea” of material undergoing a downward displacement because of gravity. The container has fixed, slippery walls. Figure taken from Ref. [3].

The only non-vanishing strain is $u_{zz} = \nabla_z u_z$. From the explicit form of Hooke’s law, we obtain the non-vanishing stresses

$$\sigma_{xx} = \sigma_{yy} = \lambda U_{zz}, \quad \sigma_{zz} = (\lambda + 2\mu) U_{zz}. \quad (83)$$

The first Cauchy’s equilibrium equation simplifies in this case to

$$\nabla_z \sigma_{zz} = \rho_0 g_0. \quad (84)$$

where ρ_0 is the constant mass density of the undeformed material. Using the boundary condition $\sigma_{zz} = 0$ at $z = h$, this equation may be integrated immediately to

$$\sigma_{zz} = -\rho_0 g_0 (h - z). \quad (85)$$

The vertical pressure $p_z = -\sigma_{zz} = \rho_0 g_0 (h - z)$ is positive and rises linearly with depth $h - z$, just as in the fluid sea. It balances everywhere the full weight of the material above, but this was expected since there are no shear stresses to distribute the vertical load. The horizontal pressures

$p_x = p_y = p_z \lambda / (\lambda + 2\mu)$ are also positive but smaller than the vertical, because both λ and μ are positive in normal materials. The horizontal pressures are eventually balanced by the stiffness of the fixed vertical walls surrounding the elastic sea.

The strain

$$u_{zz} = \nabla_z u_z = \frac{\sigma_{zz}}{\lambda + 2\mu} = -\frac{\rho_0 g_0}{\lambda + 2\mu} (h - z) \quad (9.8)$$

is negative, corresponding to a compression.

Characteristic length scale

The variation in stress over the vertical size L of a body is $|\Delta\sigma_{ij}| \approx \rho g L$. The corresponding variation in strain becomes $|\Delta u_{ij}| \sim L \rho g / E$ for non-exceptional materials. Since u_{ij} is dimensionless, it is convenient to define the *gravitational deformation scale*,

$$D \sim \frac{E}{\rho g}, \quad (86)$$

so that $|\Delta u_{ij}| \sim L/D$. The length D characterizes the scale for major gravitational deformation (of order unity), and small deformations require $L \ll D$. Finally, the gravitationally induced variation in the displacement over a vertical distance L is estimated to be of magnitude $|\Delta u_i| \sim L |\Delta u_{ij}| \sim L^2/D \ll L$.

The characteristic length scale for major deformation is in this case chosen to be

$$D = \frac{\lambda + 2\mu}{\rho_0 g_0} = \frac{1 - \nu}{(1 + \nu)(1 - 2\nu)} \frac{E}{\rho_0 g_0}. \quad (9.9)$$

Integrating the strain with the boundary condition $u_z = 0$ for $z = 0$, we finally obtain

$$u_z = \frac{h^2 - (h - z)^2}{2D}. \quad (9.10)$$

The displacement is always negative, largest in magnitude at the top, $z = h$, and vanishes at the bottom. It varies quadratically with height h (see Fig. 22).

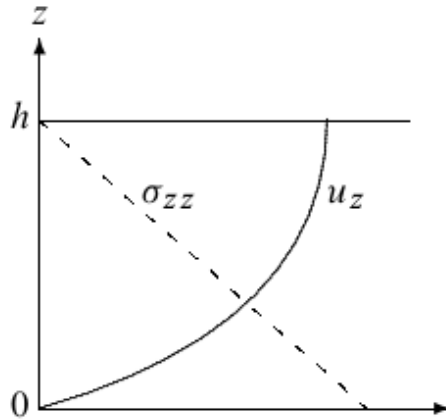


Figure 22: Sketch of the displacement (solid curve) and stress (dashed) for the elastic “elastic sea in a box”. Figure taken from Ref. [3].

Displacement field in the shear-free approximation

In the absence of shear stresses ($\sigma_{xy} = \sigma_{xz} = \sigma_{yz} = 0$), the equilibrium equations give

$$\nabla_x \sigma_{xx} = 0, \quad \nabla_y \sigma_{yy} = 0, \quad \nabla_z \sigma_{zz} = -\rho_0 g_0. \quad (87)$$

The solutions are

$$\sigma_{xx} = 0, \quad \sigma_{yy} = 0, \quad \sigma_{zz} = -\rho_0 g_0 (h - z). \quad (88)$$

Using the inverse Hooke's law for an isotropic elastic medium, the non-vanishing strain components are

$$u_{xx} = u_{yy} = \nu \frac{\rho_0 g_0}{E} (h - z), \quad (89)$$

$$u_{zz} = -\frac{\rho_0 g_0}{E} (h - z). \quad (90)$$

Introducing the characteristic deformation length

$$D = \frac{E}{\rho_0 g_0}, \quad (91)$$

we obtain

$$u_{xx} = u_{yy} = \frac{\nu}{D} (h - z), \quad u_{zz} = -\frac{h - z}{D}. \quad (92)$$

The strain-displacement relations give

$$u_{xx} = \frac{\partial u_x}{\partial x}, \quad u_{yy} = \frac{\partial u_y}{\partial y}, \quad u_{zz} = \frac{\partial u_z}{\partial z}. \quad (93)$$

Integrating each equation with respect to its corresponding coordinate,

$$u_x = \frac{\nu}{D} (h - z)x + f(y, z), \quad (94)$$

$$u_y = \frac{\nu}{D} (h - z)y + g(x, z), \quad (95)$$

$$u_z = -\frac{hz - z^2/2}{D} + F(x, y), \quad (96)$$

where f , g , and F are arbitrary functions to be determined.

Since $\sigma_{xz} = 0$, Hooke's law implies $u_{xz} = 0$. Therefore,

$$2u_{xz} = \frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} = 0. \quad (97)$$

Substituting the displacement components,

$$-\frac{\nu x}{D} + \frac{\partial f}{\partial z} + \frac{\partial F}{\partial x} = 0. \quad (98)$$

Thus,

$$\frac{\partial F}{\partial x} = \frac{\nu x}{D} - \frac{\partial f}{\partial z}. \quad (99)$$

The left-hand side depends only on x and y , whereas $\partial f/\partial z$ depends only on y and z . Consequently, $\partial f/\partial z$ must be independent of z . We can therefore write

$$f(y, z) = z a(y) + b(y). \quad (100)$$

The previous equation becomes

$$\frac{\partial F}{\partial x} = \frac{\nu x}{D} - a(y). \quad (101)$$

Integrating with respect to x ,

$$F(x, y) = \frac{\nu x^2}{2D} - x a(y) + G(y). \quad (102)$$

Similarly, $\sigma_{yz} = 0$ implies

$$2u_{yz} = \frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y} = 0. \quad (103)$$

Substitution gives

$$-\frac{\nu y}{D} + \frac{\partial g}{\partial z} + \frac{\partial F}{\partial y} = 0. \quad (104)$$

Using the expression obtained for F ,

$$\frac{\partial g}{\partial z} = \frac{\nu y}{D} + x a'(y) - G'(y). \quad (105)$$

Since g does not depend on y , the right-hand side must be independent of y . Differentiating with respect to y ,

$$\frac{\nu}{D} + x a''(y) - G''(y) = 0. \quad (106)$$

Because this must hold for arbitrary x ,

$$a''(y) = 0, \quad G''(y) = \frac{\nu}{D}. \quad (107)$$

Hence,

$$a(y) = \alpha y + \beta, \quad (108)$$

$$G(y) = \frac{\nu y^2}{2D} + \gamma y + \delta, \quad (109)$$

where $\alpha, \beta, \gamma, \delta$ are constants. It follows that

$$\frac{\partial g}{\partial z} = \alpha x - \gamma, \quad (110)$$

and therefore

$$g(x, z) = z(\alpha x - \gamma) + c(x). \quad (111)$$

The remaining condition is

$$2u_{xy} = \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} = 0. \quad (112)$$

Using the expressions for f and g ,

$$\frac{\partial u_x}{\partial y} = \alpha z + b'(y), \quad (113)$$

$$\frac{\partial u_y}{\partial x} = \alpha z + c'(x). \quad (114)$$

Thus,

$$2\alpha z + b'(y) + c'(x) = 0. \quad (115)$$

Since this equation must hold for all x, y, z , we obtain

$$\alpha = 0, \quad (116)$$

$$b'(y) = \Omega, \quad (117)$$

$$c'(x) = -\Omega, \quad (118)$$

where Ω is a constant. Therefore,

$$b(y) = \Omega y + C_x, \quad (119)$$

$$c(x) = -\Omega x + C_y. \quad (120)$$

Collecting the results, the displacement components are

$$u_x = \frac{\nu}{D}(h - z)x + \beta z + \Omega y + C_x, \quad (121)$$

$$u_y = \frac{\nu}{D}(h - z)y - \gamma z - \Omega x + C_y, \quad (122)$$

$$u_z = -\frac{hz - z^2/2}{D} + \frac{\nu(x^2 + y^2)}{2D} - \beta x + \gamma y + \delta. \quad (123)$$

The terms containing β , γ , and Ω describe infinitesimal rigid rotations, while C_x , C_y , and δ correspond to rigid translations. These motions do not contribute to the strain tensor.

We may therefore choose the reference frame such that

$$\beta = \gamma = \Omega = C_x = C_y = 0. \quad (124)$$

The remaining displacement field is

$$u_x = \nu \frac{(h - z)x}{D}, \quad (125)$$

$$u_y = \nu \frac{(h - z)y}{D}, \quad (126)$$

$$u_z = -\frac{hz - z^2/2}{D} + \nu \frac{x^2 + y^2}{2D} + \delta. \quad (127)$$

Since

$$hz - \frac{z^2}{2} = \frac{h^2 - (h - z)^2}{2}, \quad (128)$$

and renaming the arbitrary constant $\delta = K$, we finally obtain

$$\begin{aligned} u_x &= \nu \frac{(h - z)x}{D}, \\ u_y &= \nu \frac{(h - z)y}{D}, \\ u_z &= -\frac{h^2 - (h - z)^2}{2D} + \nu \frac{x^2 + y^2}{2D} + K. \end{aligned} \quad (129)$$

Differentiating the resulting displacement field,

$$\frac{\partial u_x}{\partial x} = \frac{\nu(h - z)}{D}, \quad (130)$$

$$\frac{\partial u_y}{\partial y} = \frac{\nu(h - z)}{D}, \quad (131)$$

$$\frac{\partial u_z}{\partial z} = -\frac{h - z}{D}, \quad (132)$$

which reproduces the prescribed normal strains. The shear strains also vanish:

$$2u_{xz} = -\frac{\nu x}{D} + \frac{\nu x}{D} = 0, \quad (133)$$

$$2u_{yz} = -\frac{\nu y}{D} + \frac{\nu y}{D} = 0, \quad (134)$$

$$2u_{xy} = 0. \quad (135)$$

The horizontal displacement represents an expansion that is largest at the bottom $z = 0$ and vanishes at the upper surface $z = h$. However, at the bottom surface,

$$u_z(x, y, 0) = \nu \frac{x^2 + y^2}{2D} + K. \quad (136)$$

No choice of the constant K can make this displacement vanish for all x and y . Thus, although the solution satisfies the equilibrium equations and the shear-free conditions, it cannot satisfy the displacement boundary condition imposed by a rigid horizontal support.

3.7.2 Uniform pure bending

Uniform pure bending describes the deformation of a beam into a circular arc under the action of terminal couples (moments) without shear forces or body forces (see Fig. 23). In pure bending, the stresses applied at the ends do not stretch or compress the beam on average, but only produce bending moments. The bending is assumed to be *uniform*, meaning that stresses and strains are identical along every cross-section of the beam.

For this to occur, an initially straight beam of length L must deform into a circular arc of radius R , with each transverse ray also becoming part of this circle. Because every ray follows the same curvature, it is sufficient to study a small slice of the beam to understand the deformation of the entire beam.

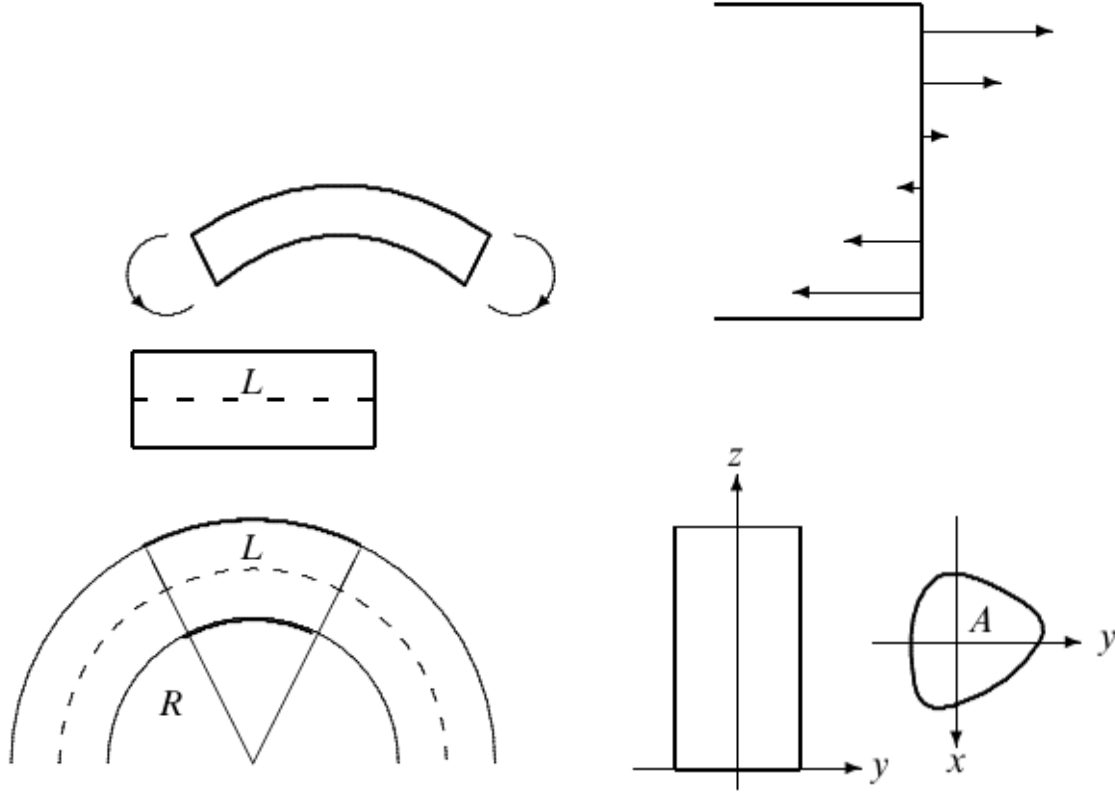


Figure 23: Top left: Bending a beam by wrenching it at the ends. Top right: A bending couple may be created by applying normal stresses only to a terminal. Bottom left: A uniformly bent beam is a part of a circular ring. Bottom right: The unbent beam is aligned with the z -axis. Its cross-section, A , in the xy -plane is the same for all z . Figures taken from Ref. [3].

Non-uniform bending can later be described by assembling slices with different radii of curvature. By Saint-Venant's principle and the linearity of elasticity, the behavior of beams under various end loads can be constructed by superposition of displacement fields.

In a Cartesian coordinate system, we align the undeformed beam with the z -axis, and put the terminal cross-sections at $z = 0$ and $z = L$. The length L of the beam may be chosen as small as we please. The cross-section A in the xy -plane may be of arbitrary shape, but we may position the coordinate system in the xy -plane with its origin coinciding with the *area centroid*, such that the area integral over the coordinates vanishes,

$$\int_A x dA = \int_A y dA = 0.$$

Finally, we require that the central ray after bending becomes part of a circle in the yz -plane with radius R and its center on the y -axis at $y = R$. The radius R is obviously the length scale for major deformation, and must be assumed large compared to the transverse dimensions of the beam.

3.7.3 Shear-free bending

What precisely happens in the beam when it is bent depends on the way the actual stresses are distributed on its terminals, although by Saint-Venant's principle the details should only matter

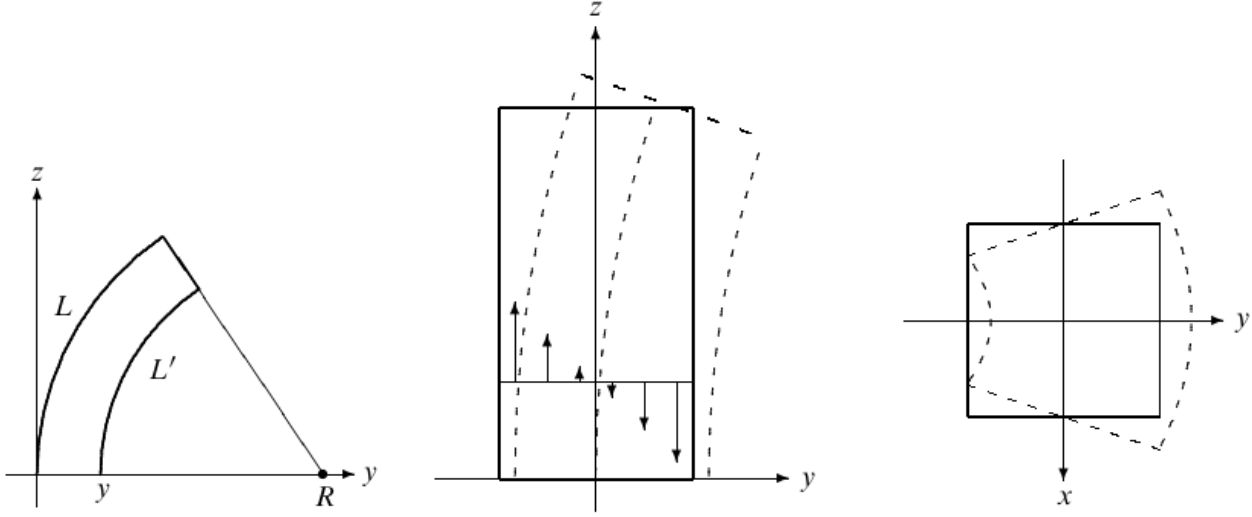


Figure 24: Left: The length of the arc at y must satisfy $L'/(R - y) = L/R$. Center: Sketch of the bending of a beam. The arrows show the strain U_{zz} . Right: Sketch of the deformation in the xy -plane of a beam with quadratic cross-section. This deformation may easily be observed by bending a rubber eraser. Figures taken from Ref. [3].

near the terminals. In the simplest case we may view the beam as a loose bundle of thin elastic strings that do not interact with each other, but are stretched or compressed individually according to their position in the beam without generating shear stresses.

Let us fix the central string so that it does not change its length L when bent into a circle of radius R . A simple geometric construction shows that a nearby ray in position y will change its length to L' , satisfying $(R - y)/L' = R/L$ (see Fig. 24). Thus the beam experiences a longitudinal strain,

$$U_{zz} = \frac{\delta L}{L} = \frac{L' - L}{L} = -\frac{y}{R}. \quad (137)$$

For negative y the material of the beam is being stretched, while for positive y it is being compressed. For the strain to be small, we must have $|y| \ll R$ everywhere in the beam.

Under the assumption that the bending is done without shear and that there are no forces acting on the sides of the beam, it follows as in the preceding section that $\sigma_{xx} = \sigma_{yy} = 0$. The only non-vanishing stress is $\sigma_{zz} = EU_{zz}$, and the non-vanishing strains are as before found from the inverted Hooke's law,

$$U_{xx} = U_{yy} = -\nu U_{zz} = \nu \frac{y}{R}, \quad (138)$$

where ν is Poisson's ratio. This shows that the material is being stretched horizontally and compressed vertically for $y > 0$ and conversely for $y < 0$.

Using $U_{ij} = \frac{1}{2}(\nabla_i u_j + \nabla_j u_i)$ and requiring that the central ray is only bent, not stretched, a particular solution is found to be

$$u_x = \nu \frac{xy}{R}, \quad u_y = \frac{z^2}{2R} + \nu \frac{y^2 - x^2}{2R}, \quad u_z = -\frac{yz}{R}.$$

The second term in u_y is, like in the preceding section, forced upon us by the requirement of no shear stresses (and strains). For displacement gradients to be small, all dimensions of the beam have to be small compared to R . Note that the beam's actual dimensions do not appear in the displacement field, which is therefore a generic solution for pure bending of any beam.

3.7.4 Total force

The only non-vanishing stress component is

$$\sigma_{zz} = E u_{zz} = -\frac{E}{R} y. \quad (139)$$

It is a tension for negative y , and we consequently expect the material of the beam to first break down at the most distant point of the cross-section opposite the direction of bending, as common experience also tells us.

The total force acting on a cross-section vanishes,

$$\mathcal{F}_z = \int_A \sigma_{zz} dS_z = -\frac{E}{R} \int_A y dA = 0. \quad (140)$$

because the origin of the coordinate system is chosen to coincide with the centroid of the beam cross-section.

3.7.5 Total moment

The moments of the longitudinal stress in any cross-section are

$$\mathcal{M}_x = \int_A y \sigma_{zz} dS_z = -\frac{E}{R} \int_A y^2 dA. \quad (141)$$

$$\mathcal{M}_y = -\int_A x \sigma_{zz} dS_z = \frac{E}{R} \int_A xy dA. \quad (142)$$

$$\mathcal{M}_z = 0. \quad (143)$$

The component $\mathcal{M}_x$ orthogonal to the bending plane is called the *bending moment*. The integral

$$I = \int_A y^2 dA \quad (144)$$

is the ‘moment of inertia’ (or *area moment*) of the beam cross section around the direction of $\mathcal{M}_x$. The component $\mathcal{M}_y$ vanishes only if

$$\int_A xy dA = 0. \quad (145)$$

This will, for example, be the case if the beam has circular cross-section, is mirror symmetric under reflection in either axis, or more generally if the axes coincide with the *principal directions* of the cross-section. There are always two orthogonal principal directions for any shape of cross-section.

The magnitude of the bending moment $\mathcal{M}_b = \mathcal{M}_x$ is known as the *Bernoulli–Euler law*,

$$\frac{1}{R} = \frac{\mathcal{M}}{EI}. \quad (146)$$

The product EI is called the *flexural rigidity* or *bending stiffness* of the beam. The larger it is, the larger is the moment required to bend it with a given radius of curvature. The unit of flexural rigidity is $\text{Pa m}^4 = \text{N m}^2$.

Note that constant shear or normal stresses, if present, do not contribute to the cross-section moment, and in many engineering applications the Bernoulli–Euler law combined with linearity and Saint-Venant’s principle is enough to give a reasonable idea of how much a beam is deformed by external loads.

3.7.6 Moments of inertia of common beam cross-sections

Rectangular beam A rectangular beam with sides $2a$ (along x) and $2b$ (along y) has moment of inertia

$$I = \int_{-a}^a dx \int_{-b}^b y^2 dy = \frac{4}{3}ab^3. \quad (147)$$

This shows that bending resistance increases much faster with the thickness $2b$ in the bending direction.

Elliptic beam An elliptical beam with major axes $2a$ and $2b$ has moment of inertia

$$I = \int_{-a}^a dx \int_{-b\sqrt{1-x^2/a^2}}^{b\sqrt{1-x^2/a^2}} y^2 dy = \frac{4}{3}ab^3 \int_0^1 (1-t^2)^{3/2} dt = \frac{\pi}{4}ab^3. \quad (148)$$

For a circular beam of radius a , this becomes $I = \frac{\pi}{4}a^4$, the same for all bending directions.

Circular pipe For a circular pipe with inner radius a and outer radius b ,

$$I = \int_{a \leq r \leq b} y^2 dA = \int_{r \leq b} y^2 dA - \int_{r \leq a} y^2 dA = \frac{\pi}{4}(b^4 - a^4). \quad (149)$$

Thus the moment of inertia of a pipe is the difference between two circular beams. Because of the fourth-power dependence on radius, flexural rigidity is highly sensitive to pipe diameter.

4 Waves

Elastic vibrations arise when a solid body undergoes small displacements about an equilibrium configuration. A familiar example is the ringing of a bell, where small-amplitude oscillations of the material propagate into the surrounding air as sound waves.

All solids are elastic to some extent and can support vibrations. In ideal elastic materials, these oscillations would persist indefinitely. In reality, energy is dissipated through internal friction (heat) and radiation (e.g. sound emission), so vibrations eventually decay.

Any small-amplitude vibration can be decomposed into a superposition of independent *normal modes*, each with a well-defined frequency. In finite bodies, the spectrum of these frequencies is discrete, while at high frequencies it may become approximately continuous.

Alternatively, elastic vibrations can be interpreted as superpositions of propagating *elastic waves*, whose properties are determined by the material and boundary conditions.

4.1 Elastic Vibrations and Elastodynamics

We describe the state of a deformable solid using a displacement field

$$\vec{u}(\vec{x}, t),$$

which gives the displacement of a material point from its equilibrium position $\vec{x}$.

The velocity and acceleration fields are:

$$\vec{v}(\vec{x}, t) = \frac{\partial \vec{u}}{\partial t}, \quad (150)$$

$$\vec{a}(\vec{x}, t) = \frac{\partial^2 \vec{u}}{\partial t^2}. \quad (151)$$

We assume *small deformations*, so that linear elasticity applies and differences between Eulerian and Lagrangian descriptions can be neglected.

4.1.1 Equation of Motion (Navier Equation)

Applying Newton's second law to a continuous medium leads to Navier's equation of motion:

$$\rho \frac{\partial^2 \vec{u}}{\partial t^2} = \vec{f} + \mu \nabla^2 \vec{u} + (\lambda + \mu) \nabla (\nabla \cdot \vec{u}), \quad (152)$$

where:

- ρ is the mass density,
- $\vec{f}$ is the body force density,
- λ and μ are the Lamé coefficients.

In the absence of body forces ($\vec{f} = 0$), this equation governs the propagation of elastic disturbances in solids.

At material interfaces:

$$[\vec{u}] = 0, \quad (153)$$

$$[\boldsymbol{\sigma} \cdot \hat{n}] = 0, \quad (154)$$

i.e. displacement and traction must be continuous.

At a free surface:

$$\boldsymbol{\sigma} \cdot \hat{n} = 0.$$

4.1.2 Driving Forces and Dissipation

Elastic vibrations can be driven by:

- surface forces (e.g. wind, impacts),
- body forces (e.g. gravity),
- electromagnetic interactions (e.g. piezoelectric effects).

Dissipation is always present in real materials and converts mechanical energy into heat. Sustained oscillations require continuous energy input. Dissipation is nevertheless so small in most elastic materials that it, to a very good approximation, can be omitted, as it is in Navier's equation of motion.

4.1.3 Decomposition of Displacement Fields

An arbitrary vector field $\vec{u}$ can be decomposed into longitudinal and transverse components,

$$\vec{u} = \vec{u}_L + \vec{u}_T, \quad (155)$$

where

$$\nabla \times \vec{u}_L = 0, \quad \nabla \cdot \vec{u}_T = 0. \quad (156)$$

The decomposition is not necessarily unique unless appropriate boundary conditions are specified.

Since the longitudinal component is irrotational, it can be written as the gradient of a scalar potential,

$$\vec{u}_L = \nabla \psi. \quad (157)$$

Taking the divergence of the decomposition gives

$$\begin{aligned} \nabla \cdot \vec{u} &= \nabla \cdot \vec{u}_L + \nabla \cdot \vec{u}_T \\ &= \nabla^2 \psi. \end{aligned} \quad (158)$$

Therefore, the scalar potential can be obtained by solving Poisson's equation

$$\nabla^2 \psi = \nabla \cdot \vec{u}. \quad (159)$$

Once ψ is known, the longitudinal and transverse components are constructed as

$$\vec{u}_L = \nabla \psi, \quad (160)$$

$$\vec{u}_T = \vec{u} - \nabla \psi. \quad (161)$$

Indeed,

$$\nabla \cdot \vec{u}_T = \nabla \cdot \vec{u} - \nabla^2 \psi = 0, \quad (162)$$

while

$$\nabla \times \vec{u}_L = \nabla \times \nabla \psi = 0. \quad (163)$$

4.1.4 Wave Equations in Elastic Media

The longitudinal component therefore describes waves for which the displacement is associated with local compression and expansion of the medium, whereas the transverse component describes shear waves. The longitudinal and transverse degrees of freedom propagate independently in a homogeneous isotropic elastic medium.

Substituting this decomposition into Navier's equation (with $\vec{f} = 0$), we obtain two independent wave equations. Inserting $\vec{u} = \vec{u}_L + \vec{u}_T$ in the same equation we obtain

$$0 = -\rho \frac{\partial^2 \vec{u}_L}{\partial t^2} + \mu \nabla^2 \vec{u}_L + (\lambda + \mu) \nabla (\nabla \cdot \vec{u}_L) - \rho \frac{\partial^2 \vec{u}_T}{\partial t^2} + \mu \nabla^2 \vec{u}_T + (\lambda + \mu) \nabla (\nabla \cdot \vec{u}_T) \quad (164)$$

$$0 = -\rho \frac{\partial^2 \vec{u}_L}{\partial t^2} + \mu \nabla^2 \vec{u}_L + (\lambda + \mu) \nabla (\nabla \cdot \vec{u}_L) - \rho \frac{\partial^2 \vec{u}_T}{\partial t^2} + \mu \nabla^2 \vec{u}_T. \quad (165)$$

Transverse (shear) waves. Taking the rotational and knowing that $\nabla \times u_L = 0$ we arrive to

$$0 = \nabla \times \left[-\rho \frac{\partial^2 \vec{u}_T}{\partial t^2} + \mu \nabla^2 \vec{u}_T \right]. \quad (166)$$

In consequence Taking the rotational and knowing that $\nabla \times u_L = 0$ we arrive to

$$0 = -\rho \frac{\partial^2 \vec{u}_T}{\partial t^2} + \mu \nabla^2 \vec{u}_T. \quad (167)$$

Longitudinal (compressional) waves Using $\nabla(\nabla \cdot \vec{u}) = \nabla^2 \vec{u} + \nabla \times (\nabla \times \vec{u})$, and taking the divergence and knowing that $\nabla \cdot u_T = 0$ we obtain

$$0 = \nabla \cdot \left[-\rho \frac{\partial^2 \vec{u}_L}{\partial t^2} + (\lambda + 2\mu) \nabla^2 \vec{u}_L \right], \quad (168)$$

from which it follows that

$$0 = -\rho \frac{\partial^2 \vec{u}_L}{\partial t^2} + (\lambda + 2\mu) \nabla^2 \vec{u}_L. \quad (169)$$

These correspond to two distinct types of elastic waves. The associated phase velocities are:

$$c_L = \sqrt{\frac{\lambda + 2\mu}{\rho}}, \quad (170)$$

$$c_T = \sqrt{\frac{\mu}{\rho}}. \quad (171)$$

Typically:

$$c_L > c_T,$$

since longitudinal waves involve compression as well as shear.

- **Longitudinal waves** involve volume changes ($\nabla \cdot \vec{u} \neq 0$) and are associated with pressure variations.
- **Transverse waves** involve shear deformation ($\nabla \cdot \vec{u} = 0$) and do not change the volume.

This distinction is crucial in geophysics:

- *P*-waves (primary) are longitudinal and travel faster,
- *S*-waves (secondary) are transverse and travel slower,
- *S*-waves cannot propagate in fluids.

A useful parameter is the ratio:

$$\frac{c_T}{c_L} = \sqrt{\frac{\mu}{\lambda + 2\mu}}, \quad (172)$$

which depends on the material properties and is related to Poisson's ratio. These results provide the foundation for the study of elastic waves in solids.

Material	c_L	q
Aluminum	6.4	0.48
Titanium	6.1	0.51
Iron	5.9	0.54
Nickel	5.8	0.52
Magnesium	5.8	0.54
Quartz	5.5	0.63
Tungsten	5.2	0.55
Copper	4.7	0.49
Silver	3.7	0.45
Gold	3.6	0.33
Lead	2.1	0.33

Table 1: Longitudinal sound speed c_L (in km s^{-1}) and the ratio of transverse to longitudinal speed $q = c_T/c_L$ for various isotropic materials. The lightest and hardest materials generally have the largest longitudinal sound speed.

Earthquake wave types . In earthquakes (see Fig. 25), longitudinal pressure waves are denoted P (for *primary*), because they arrive first due to the higher longitudinal phase velocity in any material. Typically they move at speeds of 4 to 7 km s^{-1} in the Earth’s crust. Transverse shear waves move at roughly half the speed and thus arrive later at the seismometer. They are for this reason denoted S (for *secondary*). In fluid material, such as the Earth’s liquid core, shear waves cannot propagate far. Besides these body waves, earthquakes are also accompanied by *surface waves*.

4.2 Harmonic vibrations

Because Navier’s equation is linear (for constant material parameters), any time-dependent displacement field can be expanded as a superposition of harmonic modes. We therefore consider solutions with a single angular frequency ω . We assume no body forces: $\vec{f} = 0$.

4.2.1 Harmonic Ansatz

A harmonic displacement field satisfies:

$$\frac{\partial^2 \vec{u}}{\partial t^2} = -\omega^2 \vec{u}. \quad (173)$$

The general real solution can be written as:

$$\vec{u}(\vec{x}, t) = \vec{u}_1(\vec{x}) \cos \omega t + \vec{u}_2(\vec{x}) \sin \omega t. \quad (174)$$

It is convenient to introduce a complex amplitude:

$$\vec{u}(\vec{x}) = \vec{u}_1(\vec{x}) + i\vec{u}_2(\vec{x}), \quad (175)$$

so that

$$\vec{u}(\vec{x}, t) = \text{Re} [\vec{u}(\vec{x}) e^{-i\omega t}] , \quad \frac{\partial \vec{u}}{\partial t} = \omega \text{Im} [\vec{u}(\vec{x}) e^{-i\omega t}] . \quad (176)$$

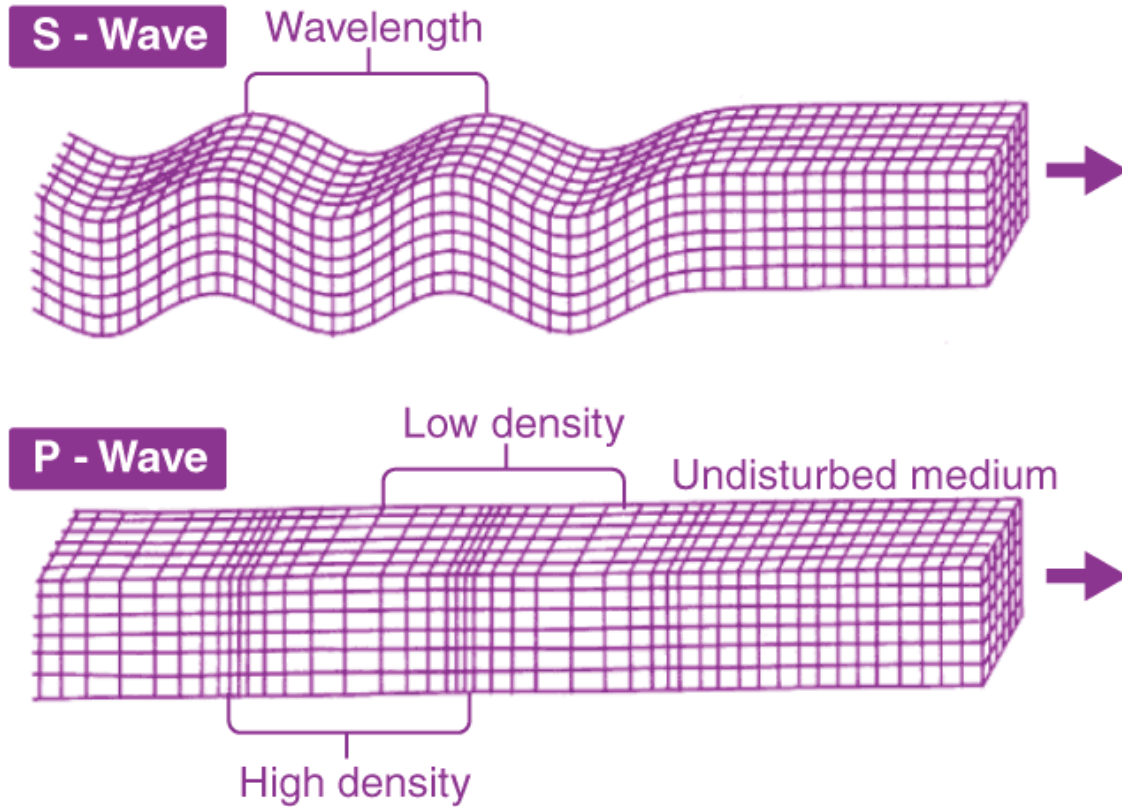


Figure 25: S waves also called secondary waves and shear waves, are the second waves to hit the seismographs. They are transverse waves, which means that the motion is perpendicular to the direction of wave propagation. S waves can only travel through solids, and scientists have successfully mapped the earth's interior by studying the routes of these waves.

Substituting into Navier's equation:

$$\rho \frac{\partial^2 \vec{u}}{\partial t^2} = \mu \nabla^2 \vec{u} + (\lambda + \mu) \nabla (\nabla \cdot \vec{u}), \quad (177)$$

we obtain the time-independent equation:

$$\mu \nabla^2 \vec{u} + (\lambda + \mu) \nabla (\nabla \cdot \vec{u}) = -\rho \omega^2 \vec{u}. \quad (178)$$

The above equation is an eigenvalue equation for the tensor differential operator

$$[\mu \delta_{ij} \nabla^2 + (\lambda + \mu) \partial_i \partial_j] u_j = -\rho \omega^2 u_i, \quad (179)$$

with eigenvalue $-\rho \omega^2$.

4.2.2 Plane Wave Solutions

Far from boundaries, solutions can be approximated by plane waves:

$$\vec{u}(\vec{x}, t) = \vec{a} e^{i(\vec{k} \cdot \vec{x} - \omega t)}. \quad (180)$$

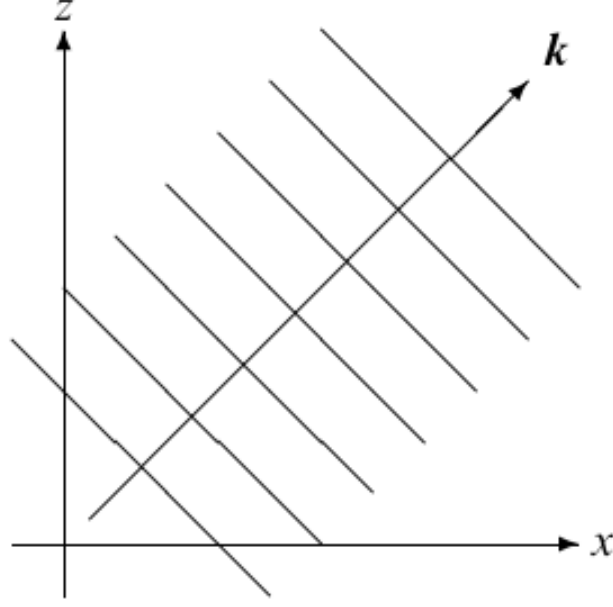


Figure 26: A plane wave has constant phase on planes orthogonal to the wave vector $\vec{k}$ (here with $k_y = 0$). The phase is spatially periodic with wavelength $\lambda = \frac{2\pi}{|\vec{k}|}$.

Substituting into the equation yields:

$$\mu k^2 \vec{a} + (\lambda + \mu) \vec{k} (\vec{k} \cdot \vec{a}) = \rho \omega^2 \vec{a}. \quad (181)$$

For a plane elastic wave, the eigenvalue problem admits one longitudinal mode and two degenerate transverse modes.

Longitudinal waves

For the longitudinal mode, the displacement is parallel to the direction of propagation. Its eigenvalue is

$$\rho \omega^2 = (\lambda + 2\mu) k^2, \quad (182)$$

and the displacement can be written as

$$\vec{u}_L(\vec{x}, t) = a \hat{k} e^{i(\vec{k} \cdot \vec{x} - \omega t)}, \quad \vec{k} = \frac{\omega}{c_L} \hat{k}, \quad (183)$$

where a is the amplitude and

$$c_L = \sqrt{\frac{\lambda + 2\mu}{\rho}}. \quad (184)$$

Transverse plane waves

The other two eigenvectors are transverse to the direction of propagation and have the same eigenvalue,

$$\rho \omega^2 = \mu k^2. \quad (185)$$

A general transverse plane wave can therefore be written as

$$\vec{u}_T(\vec{x}, t) = a \hat{t} e^{i(\vec{k} \cdot \vec{x} - \omega t)}, \quad \vec{k} = \frac{\omega}{c_T} \hat{k}, \quad (186)$$

where

$$\hat{t} \cdot \hat{k} = 0, \quad c_T = \sqrt{\frac{\mu}{\rho}}. \quad (187)$$

Thus, for a given frequency ω and propagation direction $\hat{k}$, longitudinal and transverse waves have different wave numbers,

$$k_L = \frac{\omega}{c_L}, \quad k_T = \frac{\omega}{c_T}, \quad (188)$$

and consequently different wavelengths.

4.2.3 Wave intensity

Elastic waves transport energy through the material. The rate at which the elastic stresses do work across a surface defines an energy flux. The corresponding intensity vector is

$$\vec{I} = -\vec{v} \cdot \boldsymbol{\sigma}, \quad (189)$$

where

$$\vec{v} = \frac{\partial \vec{u}}{\partial t} \quad (190)$$

is the material velocity. The magnitude of $\vec{I}$ is the wave intensity.

For an isotropic linear elastic medium, Hooke's law is

$$\sigma_{ij} = \lambda U_{kk} \delta_{ij} + 2\mu U_{ij}, \quad (191)$$

where the strain tensor is

$$U_{ij} = \frac{1}{2} (\partial_i u_j + \partial_j u_i). \quad (192)$$

Therefore, in terms of the displacement field,

$$\sigma_{ij} = \lambda (\nabla \cdot \vec{u}) \delta_{ij} + \mu (\partial_i u_j + \partial_j u_i). \quad (193)$$

In tensor notation, this can be written as

$$\boldsymbol{\sigma} = \lambda (\nabla \cdot \vec{u}) \mathbf{1} + \mu [\nabla \vec{u} + (\nabla \vec{u})^T]. \quad (194)$$

For a harmonic plane wave,

$$\vec{u} = \vec{a} e^{i(\vec{k} \cdot \vec{x} - \omega t)}, \quad (195)$$

we have

$$\nabla \cdot \vec{u} = i(\vec{k} \cdot \vec{a})e^{i(\vec{k} \cdot \vec{x} - \omega t)}, \quad (196)$$

$$\partial_i u_j = i k_i a_j e^{i(\vec{k} \cdot \vec{x} - \omega t)}. \quad (197)$$

Hence, the stress tensor becomes

$$\sigma_{ij} = i \left[\lambda (\vec{k} \cdot \vec{a}) \delta_{ij} + \mu (k_i a_j + k_j a_i) \right] e^{i(\vec{k} \cdot \vec{x} - \omega t)}. \quad (198)$$

For a harmonic plane wave with real polarization vector $\vec{a}$,

$$\vec{u} = \vec{a} \cos \phi, \quad \phi = \vec{k} \cdot \vec{x} - \omega t, \quad (199)$$

the material velocity is

$$\vec{v} = \omega \vec{a} \sin \phi. \quad (200)$$

Using Hooke's law for an isotropic elastic medium, the contraction between the velocity and the stress tensor can be written as

$$\vec{v} \cdot \boldsymbol{\sigma} = -\omega \left[(\lambda + \mu)(\vec{a} \cdot \vec{k})\vec{a} + \mu a^2 \vec{k} \right] \sin^2 \phi. \quad (201)$$

Since the average over one oscillation period is

$$\langle \sin^2 \phi \rangle = \frac{1}{2}, \quad (202)$$

the time-averaged intensity follows directly.

For a longitudinal wave,

$$\vec{a} = a \hat{k}, \quad \vec{k} = \frac{\omega}{c_L} \hat{k}, \quad (203)$$

which gives

$$\langle \vec{I}_L \rangle = \frac{1}{2} \rho \omega^2 a^2 c_L \hat{k}. \quad (204)$$

For a transverse wave,

$$\vec{a} \cdot \vec{k} = 0, \quad \vec{k} = \frac{\omega}{c_T} \hat{k}, \quad (205)$$

and therefore

$$\langle \vec{I}_T \rangle = \frac{1}{2} \rho \omega^2 a^2 c_T \hat{k}. \quad (206)$$

Thus, in both cases the average energy flux points along the direction of wave propagation.

Elastic waves carry both kinetic and elastic deformation energy. For either a longitudinal or a transverse harmonic plane wave, the average total energy density is

$$\langle \varepsilon \rangle = \frac{1}{2} \rho \omega^2 a^2. \quad (207)$$

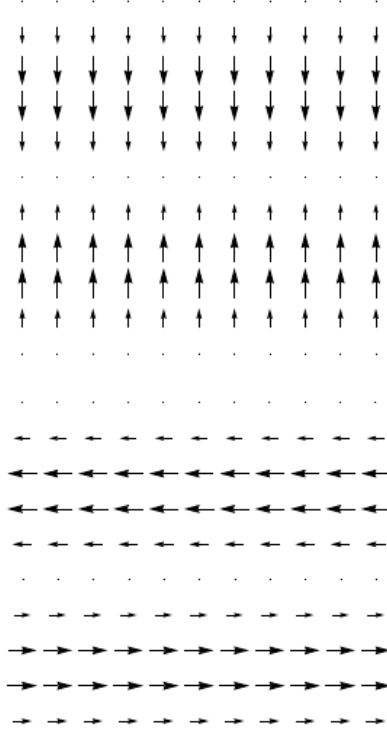


Figure 27: Single wavelength of longitudinal and transverse plane waves moving toward the top of the page. Top: Longitudinal wave; the displacement oscillates in the direction of motion. Bottom: Transverse wave; the displacement oscillates orthogonally to the direction of motion.

The average intensities can consequently be expressed in the particularly simple form

$$\langle \vec{I}_L \rangle = \langle \varepsilon \rangle c_L \hat{k}, \quad (208)$$

$$\langle \vec{I}_T \rangle = \langle \varepsilon \rangle c_T \hat{k}. \quad (209)$$

Hence, the energy carried by a longitudinal or transverse plane wave is transported, on average, along the propagation direction with the corresponding phase velocity c_L or c_T .

- Longitudinal waves involve compression and propagate faster.
- Transverse waves involve shear and propagate more slowly.
- Both carry equal energy density (for equal amplitude), but transport it at different speeds.

These harmonic solutions form the basis for:

- normal modes in finite bodies,
- reflection and transmission at interfaces,
- surface wave phenomena,
- applications in seismology and astrophysics.

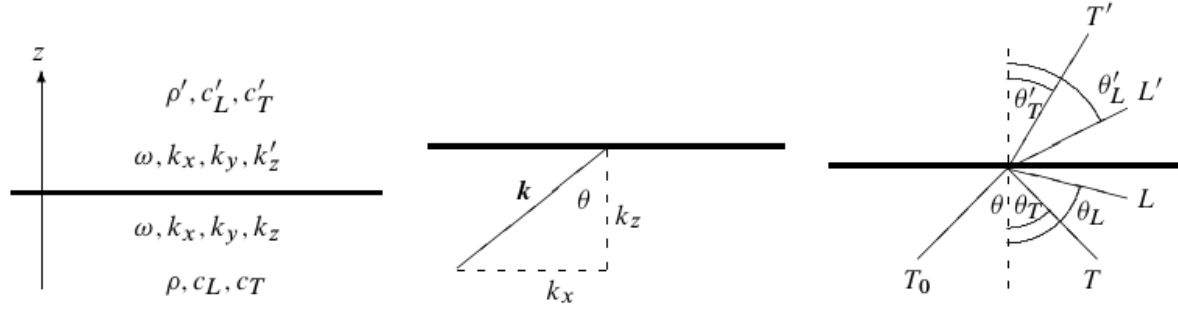


Figure 28: Left: A plane interface between two media. The material properties are different on the two sides of the interface, but the the boundary conditions demand that the frequency and the wavenumbers components along the interface are the same. Center: Geometry for determining the angle of incidence between the direction of propagation and the normal to the interface. Right: An incident transverse wave T_0 produces a reflected transverse wave having the same angle with the normal, $\Theta = \theta$, but may also produce a longitudinal wave with larger angle.

4.2.4 Reflection and refraction

Consider two homogeneous and isotropic semi-infinite elastic media separated by the plane $z = 0$. The longitudinal and transverse wave velocities are c_L, c_T in one medium and c'_L, c'_T in the other.

A plane wave incident on the interface generally produces both a reflected wave in the original medium and a refracted wave in the second medium. Moreover, even if the incident wave is purely longitudinal or purely transverse, the reflected and refracted waves can contain both longitudinal and transverse components. This phenomenon is known as *mode conversion*.

Boundary conditions

At the interface, the displacement and the traction must be continuous. Taking the interface to be the xy -plane, these conditions are

$$u'_x = u_x, \quad u'_y = u_y, \quad u'_z = u_z, \quad (210)$$

$$\sigma'_{xz} = \sigma_{xz}, \quad \sigma'_{yz} = \sigma_{yz}, \quad \sigma'_{zz} = \sigma_{zz}. \quad (211)$$

The remaining components of the stress tensor are not required to be continuous.

Because the system is invariant under translations in time and parallel to the interface, all incident, reflected, and refracted plane-wave components must have the same frequency and the same components of the wave vector parallel to the interface. Thus,

$$k'_x = k_x, \quad k'_y = k_y, \quad \omega' = \omega. \quad (212)$$

Consequently, the incident, reflected, and refracted wave vectors lie in the plane defined by the incident wave vector and the normal to the interface. The problem can therefore be taken, without loss of generality, to lie in the xz -plane, with

$$k_y = 0. \quad (213)$$

Snell's law

Let θ be the angle between the direction of propagation and the normal to the interface. For a plane wave,

$$\sin \theta = \frac{k_x}{|\vec{k}|}. \quad (214)$$

Since the phase velocity is $c = \frac{\omega}{|\vec{k}|}$, we can write

$$\sin \theta = \frac{k_x c}{\omega}. \quad (215)$$

Because k_x and ω are common to all wave components at the interface, two components with phase velocities c_1 and c_2 satisfy

$$\frac{\sin \theta_2}{\sin \theta_1} = \frac{c_2}{c_1}. \quad (216)$$

This is Snell's law for elastic waves.

The relation applies to longitudinal and transverse waves, both for reflection and refraction. In particular, since generally

$$c_L > c_T, \quad (217)$$

the longitudinal and transverse components produced by mode conversion propagate at different angles.

A reflected wave of the same type as the incident wave has the same phase velocity and therefore the familiar result

$$\theta_{\text{reflected}} = \theta_{\text{incident}}. \quad (218)$$

In contrast, the propagation angles of the other reflected and refracted components depend on their longitudinal or transverse character and on the elastic properties of the two media.

4.3 Surface waves

At a material interface it is possible to have waves that propagate along the interface while decaying exponentially away from it. Such waves do not propagate into the bulk of the material and are therefore called *surface waves*.

In an elastic medium, two important examples are Rayleigh waves and Love waves. Both are particularly relevant in seismology, where they can be generated when seismic bulk waves reach the surface of the Earth.

4.3.1 Rayleigh waves

Consider an isotropic elastic medium occupying the half-space $z < 0$, with a free surface at $z = 0$. We look for a surface wave propagating along the x -direction, with a common phase factor

$$e^{i(kx - \omega t)}. \quad (219)$$

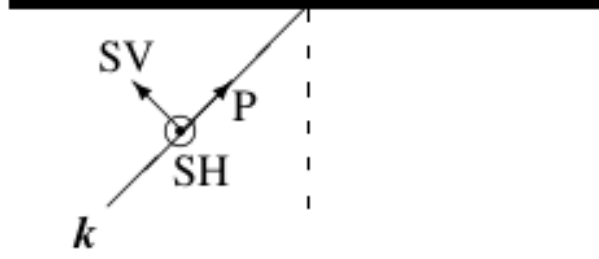


Figure 29: Incident elastic wave. The polarization of a P-wave is parallel to the wave vector. The polarization of an SV-wave lies in the propagation plane. The polarization of an SH-wave points out of the paper.

Recall that in an infinite isotropic elastic medium there are longitudinal and transverse waves. For a wave propagating in the xz -plane, a transverse wave can have two independent polarizations.

The first one has its displacement along the y -direction, perpendicular to the plane defined by the direction of propagation and the normal to the surface. This polarization is conventionally called a horizontally polarized transverse wave, or an *SH wave*. The second transverse polarization has its displacement in the xz -plane. It is conventionally called a vertically polarized transverse wave, or an *SV wave*. Thus, the three possible bulk-wave polarizations can be classified as

$$P : \quad \vec{u}_L \parallel \vec{k}, \quad (220)$$

$$SV : \quad \vec{u}_T \perp \vec{k}, \quad \vec{u}_T \text{ in the } xz\text{-plane}, \quad (221)$$

$$SH : \quad \vec{u}_T \parallel \hat{e}_y. \quad (222)$$

The letters P and S refer to the longitudinal and transverse seismic waves, respectively. The transverse S waves are further classified as *SV* and *SH* according to their polarization relative to the vertical plane containing the direction of propagation.

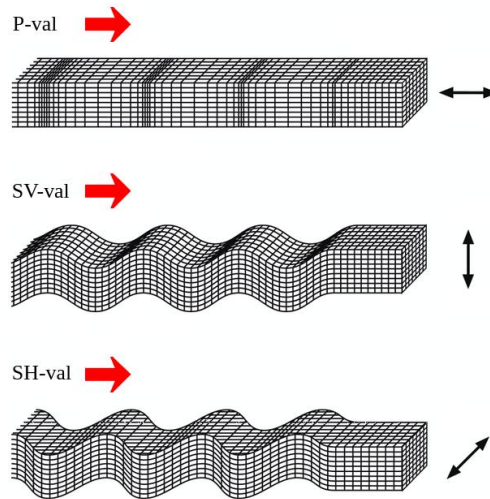


Figure 30: P, SV and SH waves.

For a Rayleigh wave, the displacement takes place entirely in the xz -plane. Consequently, the *SH* component is absent. The surface wave must instead be constructed as a superposition of a

longitudinal P wave and a transverse SV wave. We therefore seek a superposition of the form

$$\vec{u} = \vec{u}_T + \vec{u}_L, \quad (223)$$

where both components propagate along the surface and decay into the material. We write:

$$\vec{u} = A_T (k_T, 0, k) e^{-ik_T z} e^{i(kx - \omega t)} + A_L (k, 0, -k_L) e^{-ik_L z} e^{i(kx - \omega t)}. \quad (224)$$

Since there are no continuity requirements on the displacement field, the boundary conditions reduce in this case to the vanishing of the stress vector,

$$\vec{\sigma} \cdot \hat{n}_z|_{z=0} = \sum_j \sigma_{ij} n_{zj}|_{z=0} = 0. \quad (225)$$

From this,

$$\sigma_{xz} = \sigma_{yz} = \sigma_{zz} = 0, \quad \text{for } z = 0. \quad (226)$$

The non-vanishing strains at $z = 0$ are

$$u_{xx} = \nabla_x u_x = ik k_T A_T + ik^2 A_L, \quad (227)$$

$$u_{zz} = \nabla_z u_z = -ik k_T A_T + ik_L^2 A_L, \quad (228)$$

$$2u_{xz} = \nabla_x u_z + \nabla_z u_x = i(k^2 - k_T^2) A_T - 2ik k_L A_L. \quad (229)$$

From these we obtain the non-vanishing stress components:

$$\sigma_{xz} = 2\mu u_{xz} = i\mu [(k^2 - k_T^2) A_T - 2ik k_L A_L], \quad (230)$$

$$\begin{aligned} \sigma_{zz} &= (2\mu + \lambda) u_{zz} + \lambda u_{xx} \\ &= -i\mu [2k k_T A_T + (k^2 - k_T^2) A_L]. \end{aligned} \quad (231)$$

In the last step we have used the relation

$$\rho\omega^2 = (\lambda + 2\mu)(k_L^2 + k^2) \quad (232)$$

to eliminate λ .

For a surface wave, the displacement must vanish as $z \rightarrow -\infty$. Consequently, the wave-vector components normal to the surface must be imaginary. We write

$$k_T = i\kappa_T, \quad k_L = i\kappa_L, \quad (233)$$

with

$$\kappa_T > 0, \quad \kappa_L > 0. \quad (234)$$

The z dependence then becomes

$$e^{\kappa_T z}, \quad e^{\kappa_L z}, \quad (235)$$

which indeed vanishes for $z \rightarrow -\infty$.

For the transverse and longitudinal bulk waves, the dispersion relations give

$$k^2 + k_T^2 = \frac{\omega^2}{c_T^2}, \quad (236)$$

$$k^2 + k_L^2 = \frac{\omega^2}{c_L^2}. \quad (237)$$

Using $k_T = i\kappa_T$ and $k_L = i\kappa_L$, we obtain

$$\kappa_T = \sqrt{k^2 - \frac{\omega^2}{c_T^2}}, \quad (238)$$

$$\kappa_L = \sqrt{k^2 - \frac{\omega^2}{c_L^2}}. \quad (239)$$

Defining the phase velocity along the surface as

$$c = \frac{\omega}{k}, \quad (240)$$

these expressions can be written as

$$\kappa_T = k \sqrt{1 - \frac{c^2}{c_T^2}}, \quad (241)$$

$$\kappa_L = k \sqrt{1 - \frac{c^2}{c_L^2}}. \quad (242)$$

For the wave to decay exponentially into the material, both κ_T and κ_L must be real. Since

$$c_T < c_L, \quad (243)$$

the stronger condition is

$$c < c_T < c_L. \quad (244)$$

Thus, a Rayleigh wave must propagate more slowly than both transverse and longitudinal bulk waves.

Free-surface boundary conditions

At a free surface the traction must vanish,

$$\vec{t} = \boldsymbol{\sigma} \cdot \hat{e}_z = 0. \quad (245)$$

Since the Rayleigh wave has no y component, the relevant conditions at $z = 0$ are

$$\sigma_{xz} = 0, \quad \sigma_{zz} = 0. \quad (246)$$

Applying these conditions to the superposition of the transverse and longitudinal waves gives two coupled equations for their amplitudes,

$$(k^2 - \kappa_T^2) A_T - 2k\kappa_L A_L = 0, \quad (247)$$

$$2k\kappa_T A_T + (k^2 - \kappa_L^2) A_L = 0. \quad (248)$$

This is a homogeneous system for A_T and A_L . A nonzero displacement therefore exists only if its determinant vanishes. This gives

$$(k^2 + \kappa_T^2)^2 = 4k^2 \kappa_T \kappa_L. \quad (249)$$

This condition determines the phase velocity of the Rayleigh wave.

Rayleigh equation

Using

$$c = \frac{\omega}{k}, \quad (250)$$

together with the expressions for κ_T and κ_L , the previous condition becomes

$$\left(2 - \frac{c^2}{c_T^2}\right)^2 = 4\sqrt{\left(1 - \frac{c^2}{c_T^2}\right)\left(1 - \frac{c^2}{c_L^2}\right)}. \quad (251)$$

It is convenient to introduce the dimensionless ratios

$$\xi = \frac{c}{c_T}, \quad q = \frac{c_T}{c_L}. \quad (252)$$

The Rayleigh equation then takes the form

$$(2 - \xi^2)^2 = 4\sqrt{(1 - \xi^2)(1 - q^2\xi^2)}. \quad (253)$$

Squaring this equation and solving for q gives

$$q = \sqrt{\frac{16 - 24\xi^2 + 8\xi^4 - \xi^6}{16(1 - \xi^2)}}. \quad (254)$$

The physically relevant solution satisfies

$$0 < \xi < 1, \quad (255)$$

in agreement with the condition $c < c_T$. For typical elastic materials,

$$q = \frac{c_T}{c_L} \sim 0.5, \quad (256)$$

and consequently ξ is close to unity. Thus, the Rayleigh-wave velocity is only slightly smaller than the transverse-wave velocity.

For $q = 0$, the maximum value is approximately

$$\xi_0 \simeq 0.9553. \quad (257)$$

An approximate expression valid near $\xi = 1$ is

$$\xi \simeq 1 - \frac{2}{43 - 64q^2}. \quad (258)$$

For $q = 0.5$, this approximation is accurate to better than one percent.

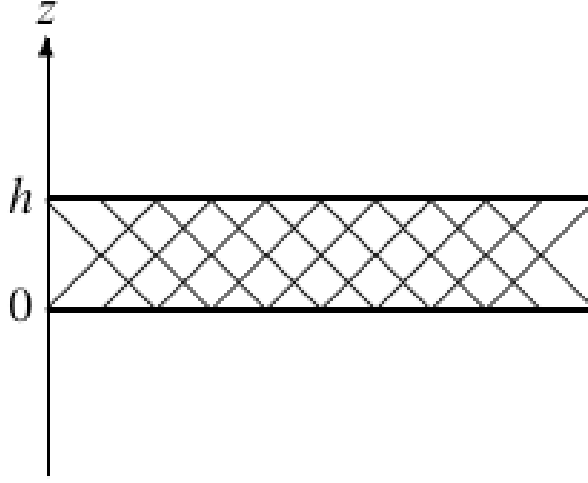


Figure 31: Love waves may arise in a layer of thickness h on top of a half-space $z < 0$ of other material.

Physical interpretation

A Rayleigh wave is therefore neither a purely longitudinal nor a purely transverse wave. It is a particular superposition of a longitudinal P wave and a vertically polarized transverse SV wave. Both components are evanescent in the direction normal to the surface,

$$\vec{u} \sim e^{i(kx - \omega t)} (e^{\kappa_T z} + e^{\kappa_L z}), \quad (259)$$

and consequently the motion becomes progressively weaker with depth.

The displacement has both horizontal and vertical components. During the passage of a Rayleigh wave, material elements therefore execute an approximately elliptical motion in the vertical plane containing the direction of propagation. There is no horizontal displacement perpendicular to the direction of propagation.

Rayleigh waves are especially important in seismology. Earthquakes generate longitudinal P waves and transverse S waves in the Earth's interior. When these bulk waves reach the free surface, they can excite Rayleigh waves. Since

$$c_R < c_T < c_L, \quad (260)$$

Rayleigh waves arrive after the corresponding P and S bulk waves at a seismometer located sufficiently far from the source.

The existence of Rayleigh waves illustrates an important consequence of boundary conditions: although longitudinal and transverse waves propagate independently in an infinite isotropic elastic medium, a free surface couples them and permits a new type of elastic normal mode localized near the boundary.

4.3.2 Love waves

One might expect the existence of surface waves with SH polarization, either at a free surface or at the interface between two elastic media. However, such waves cannot exist in a homogeneous elastic half-space. Love waves become possible when the elastic properties of the material vary with depth.

In the simplest case, a surface layer with different elastic properties from the underlying material acts as a *waveguide* for horizontally polarized transverse waves. The displacement is perpendicular to both the direction of propagation and the vertical direction. Thus, for propagation along the x -direction,

$$\vec{u} = u_y(x, z, t) \hat{e}_y. \quad (261)$$

An important difference with Rayleigh waves is that Love waves are *dispersive*: their phase velocity depends on the wavelength, or equivalently on the frequency. If c'_T denotes the transverse-wave velocity in the surface layer and c_T that in the underlying bulk material, with

$$c'_T < c_T, \quad (262)$$

the phase velocity c of a Love wave lies between these two velocities,

$$\boxed{c'_T < c < c_T}. \quad (263)$$

The slower surface layer therefore traps the transverse wave and guides its propagation along the surface.

Since Love waves propagate faster than Rayleigh waves in the surface layer, they arrive earlier at a seismometer when both originate from the same seismic event. Their motion is entirely horizontal and transverse to the direction of propagation, producing a strong shearing motion of the surface.

This horizontal shear can be particularly damaging to buildings. For this reason, Love waves are among the most destructive seismic waves.

4.4 Applications

4.4.1 Characteristic frequencies of radial vibrations of an elastic sphere

Let's determine the characteristic frequencies of radial vibrations of an elastic sphere with radius R .

We take spherical polar coordinates, with the origin at the centre of the sphere. For radial vibrations, $\vec{u}$ is along the radius, and is a function of r and t only. Hence $\nabla \times \vec{u} = 0$. We define the displacement “potential” ϕ by

$$\vec{u} = \nabla \phi = \frac{\partial \phi}{\partial r} \hat{r}.$$

The equation of motion, expressed in terms of ϕ , is just the wave equation

$$c_L^2 \nabla^2 \phi = \ddot{\phi},$$

or, for oscillations periodic in time ($\propto e^{-i\omega t}$),

$$\nabla^2 \phi = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = -k^2 \phi, \quad k = \frac{\omega}{c_L}. \quad (264)$$

The solution which is finite at the origin is

$$\phi = \frac{A}{r} \sin(kr) \quad (\text{time factor omitted}).$$

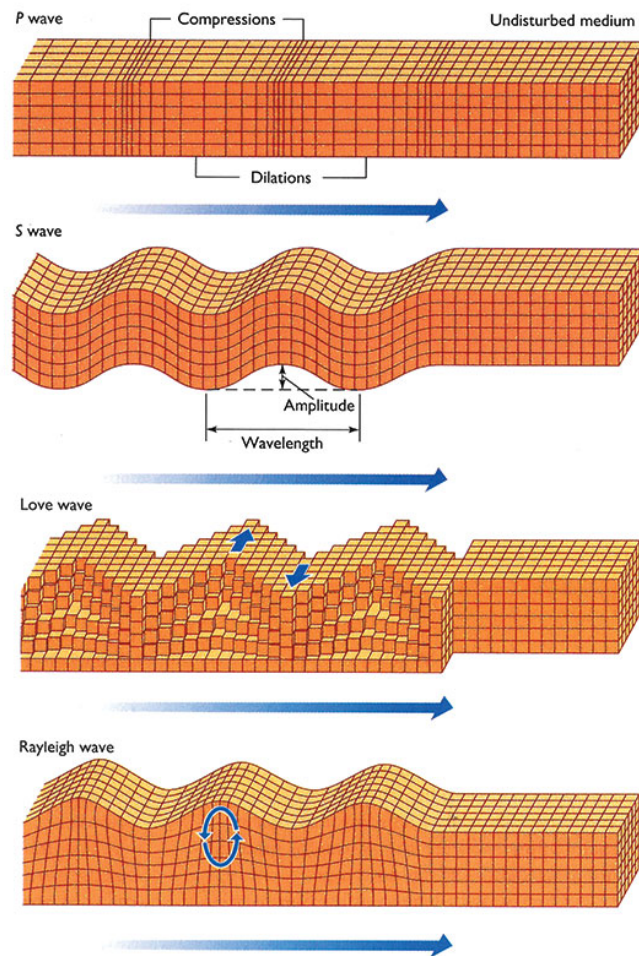


Figure 32: Types of Seismic Waves. Because P waves travel fastest of the four wave types, they arrive at a location prior to surface waves, which tend to be more destructive. P waves thus serve as a kind of early-warning system in advance of subsequent, more severe shaking.

The radial stress is

$$\sigma_{rr} = \rho \left\{ (c_L^2 - 2c_T^2) \sum_i u_{ii} + 2c_T^2 u_{rr} \right\} \quad (265)$$

$$= \rho \left\{ (c_L^2 - 2c_T^2) \nabla^2 \phi + 2c_T^2 \frac{d^2 \phi}{dr^2} \right\}. \quad (266)$$

Using the wave equation, this becomes

$$\frac{\sigma_{rr}}{\rho} = -\omega^2 \phi - 4c_T^2 \frac{1}{r} \frac{d\phi}{dr}. \quad (267)$$

The boundary condition $\sigma_{rr}(R) = 0$ leads to the equation

$$\frac{\tan(kR)}{kR} = \frac{1}{1 - (kR c_T / 2c_L)^2}. \quad (268)$$

The roots of this equation determine the characteristic frequencies

$$\omega = kc_L$$

of the radial vibrations.

Connection with Stellar Radial Oscillations

The radial vibrations of an elastic sphere provide a useful analogy for radial oscillations of stars. In both cases, one studies small perturbations around a spherical equilibrium configuration and obtains a discrete set of normal-mode frequencies.

For the elastic sphere, the restoring forces are elastic stresses, and the radial displacement is described by

$$\vec{u}(r, t) = u_r(r, t) \hat{r}.$$

The characteristic frequencies are determined by the boundary condition that the radial stress vanish at the free surface:

$$\sigma_{rr}(R) = 0.$$

In stellar oscillations, the analogous variable is the radial fluid displacement,

$$\vec{\xi}(r, t) = \xi_r(r, t) \hat{r},$$

and the restoring force is mainly pressure, together with gravity. The free-surface condition is usually expressed as the vanishing of the Lagrangian pressure perturbation at the stellar surface:

$$\Delta p(R) = 0.$$

Thus, the correspondence is schematically:

$$\text{elastic sphere:} \quad \sigma_{rr}(R) = 0,$$

$$\text{fluid star:} \quad \Delta p(R) = 0.$$

The main difference is that an elastic solid supports both longitudinal and transverse waves, whereas an ideal fluid does not support shear stresses. Therefore, in a fluid star,

$$\mu = 0,$$

and only compressional modes remain.

In the elastic sphere, radial modes are purely longitudinal because

$$\nabla \times \vec{u} = 0.$$

Similarly, radial stellar pulsations are compressional modes involving alternating expansion and contraction of the star.

The mode spectrum is discrete in both cases because the system has a finite radius and boundary conditions select only certain allowed wavelengths. In the elastic sphere, the frequencies satisfy a transcendental equation such as

$$\frac{\tan(kR)}{kR} = \frac{1}{1 - (kR c_T / 2c_L)^2}, \quad \omega = kc_L.$$

In stellar oscillations, the corresponding eigenvalue problem is more complicated because the equilibrium density, pressure, and sound speed depend on radius:

$$\rho_0 = \rho_0(r), \quad p_0 = p_0(r), \quad c_s = c_s(r).$$

Nevertheless, the basic idea is the same: radial normal modes are standing compressional waves constrained by regularity at the center and a surface boundary condition.

Thus, the elastic sphere is a useful prototype for understanding:

- normal modes in spherical systems,
- quantization of oscillation frequencies by boundary conditions,
- radial pulsations as standing compressional waves,
- the role of surface conditions in determining eigenfrequencies.

4.4.2 Radial vibrations of a spherical cavity in an infinite elastic medium

Now we are going to determine the frequency of radial vibrations of a spherical cavity in an infinite elastic medium for which $c_L \gg c_T$.

In an infinite medium, radial oscillations of the cavity are accompanied by the emission of longitudinal sound waves, leading to loss of energy and hence to damping of the oscillations. When $c_L \gg c_T$ (i.e. $K \gg \mu$), this emission is weak, and we can speak of characteristic frequencies of oscillations with a small coefficient of damping.

We seek a solution of Eq. (264) of previous problem in the form of an outgoing spherical wave,

$$\phi = \frac{Ae^{ikr}}{r}, \quad k = \frac{\omega}{c_L}. \quad (269)$$

Using Eq. (267), the boundary condition $\sigma_{rr}(R) = 0$ yields

$$\left(\frac{kRc_T}{c_L} \right)^2 = 4(1 - ikR). \quad (270)$$

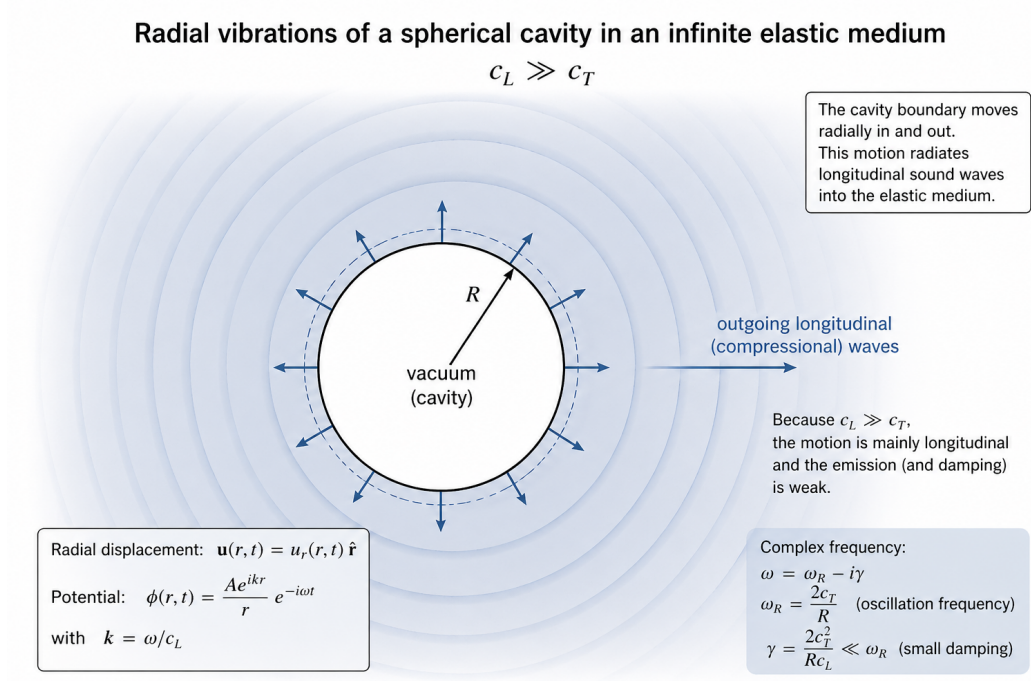


Figure 33: Radial vibrations of a spherical cavity in an infinite elastic medium

Hence, for $c_L \gg c_T$, we obtain

$$\omega = \frac{2c_T}{R} \left(1 - i \frac{c_T}{c_L} \right). \quad (271)$$

The real part of ω gives the characteristic frequency of oscillation; the imaginary part gives the damping coefficient.

In an incompressible medium ($c_L \rightarrow \infty$), there would be no damping. These vibrations are specifically due to the shear resistance of the medium ($\mu \neq 0$). It should be noticed that they have

$$kR = \frac{2c_T}{c_L} \ll 1,$$

i.e. the corresponding wavelength is large compared with R .

It is interesting to compare this with the result for vibrations of an elastic sphere, where with $c_L \gg c_T$ the first characteristic frequency is given by

$$kR = \pi.$$

4.5 Problems

1. Derive the form of Navier's equation of motion when the Lamé coefficients depend on position.
2. Show that the eigenvalues ω^2 of the amplitude equation for free waves Eq.(178) are real and positive when the boundary conditions specify the vanishing of the displacement field or of the stress vector.

3. Show that an arbitrary vector field may be resolved into (not necessarily unique) longitudinal and transverse components, and that the longitudinal component may be chosen to be a gradient. Show that the individual components of a mixed field may always be chosen to satisfy Navier's equation of motion individually.
4. Show that the time average of a product of the real parts of two complex oscillating quantities is

$$\langle \text{Re}[Ae^{-i\omega t}] \text{Re}[Be^{-i\omega t}] \rangle = \frac{1}{2} \text{Re}[A^*B]. \quad (272)$$

5. Calculate the

- (a) kinetic and
- (b) elastic energy densities

for the generic wave Eq. (180) with complex amplitude and complex wave vector.

6. (a) Show that for complex amplitude $\vec{a}$ and wave vector $\vec{k}$, the time average of the intensity is

$$\langle \vec{I} \rangle = \frac{1}{2} \omega \text{Re} \left[\vec{a}^* \cdot (\mu(\vec{k}\vec{a} + \vec{a}\vec{k}) + \lambda \vec{a} \cdot \vec{k} \mathbf{1}) \right] e^{-2\text{Im}[\vec{k}] \cdot \vec{x}}. \quad (273)$$

5 Applications of the Navier–Cauchy Equation

The Navier–Cauchy equation is the basic field equation for small deformations of an isotropic elastic solid:

$$\rho \ddot{\mathbf{u}} = (\lambda + \mu) \nabla(\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u} + \mathbf{f}, \quad (274)$$

where $\mathbf{u}(\mathbf{x}, t)$ is the displacement field, ρ the mass density, λ and μ the Lamé parameters, and $\mathbf{f}$ a body-force density (e.g. gravity).

In astronomy and planetary science, Eq. (274) is used at a wide range of scales:

- seismic waves and normal modes of the Earth and other planets,
- tidal deformation and lithospheric flexure in rocky planets and icy moons,
- elastic oscillations of dense stars (white dwarfs, neutron stars),
- estimates of crustal breaking and “starquakes” in magnetars.

5.1 Elastic Waves and Seismology

If body forces vary slowly in time, or can be neglected on wave time-scales, we set $\mathbf{f} = 0$ in Eq. (274). For a homogeneous material (ρ, λ, μ constant), we look for plane-wave solutions of the form

$$\mathbf{u}(\mathbf{x}, t) = \mathbf{A} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)},$$

where $\mathbf{k}$ is the wavevector, ω the angular frequency and $\mathbf{A}$ a constant polarization vector.

Inserting this ansatz into Eq. (274) (with $\mathbf{f} = 0$) and simplifying, one finds two independent types of waves:

1. Longitudinal (P) waves:

$$\mathbf{A} \parallel \mathbf{k}, \quad v_P = \frac{\omega}{k} = \sqrt{\frac{\lambda + 2\mu}{\rho}}.$$

2. Transverse (S) waves:

$$\mathbf{A} \perp \mathbf{k}, \quad v_S = \sqrt{\frac{\mu}{\rho}}.$$

The speeds v_P and v_S depend directly on the elastic constants and the density of the material.

In terrestrial seismology these relations are used in two directions:

- Given λ, μ, ρ for a rock, we can predict v_P and v_S .
- Given measured v_P and v_S from seismic waves, we can *invert* for λ, μ and infer the composition and state (solid vs liquid) of the interior.

The absence of S -waves in the Earth’s outer core, for example, indicates that this region is fluid.

5.2 Elastic Deformation of Planetary Bodies

In a self-gravitating planet or moon, the body force is due to gravity. For small deformations around a hydrostatic reference state, the static version of Eq. (274) can be written as

$$(\lambda + \mu) \nabla(\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u} + \rho \nabla \Phi = 0, \quad (275)$$

where Φ is a perturbing potential (e.g. the tidal potential produced by a companion). Solutions of this equation in spherical geometry lead to the *Love numbers* h_l, k_l that quantify how much a planet deforms under tidal forcing. These parameters are routinely used to constrain the thickness and rigidity of icy shells in moons such as Europa and Enceladus.

Another important application is lithospheric flexure: the bending of a planetary lithosphere in response to volcanic loads, impact basins or ice caps. In the thin-shell limit, the Navier–Cauchy equation reduces to a fourth order equation for the deflection $w(x, y)$ of the lithosphere, with an effective flexural rigidity proportional to Eh^3 (where E is Young’s modulus and h the plate thickness).

5.3 Astrophysical Elasticity: Neutron Star Crusts

At nuclear densities, matter in the outer layers of neutron stars forms a crystalline solid. The same basic Eq. (274) describes small elastic disturbances of this crust, though with extremely large μ ($\sim 10^{30}$ Pa) and extreme densities ($\rho \sim 10^{17}$ kg m $^{-3}$).

Shear waves in the crust have speed

$$v_S = \sqrt{\frac{\mu}{\rho}} \sim 10^8 \text{ m s}^{-1}, \quad (276)$$

and global torsional oscillations of the crust are one possible explanation for quasi-periodic oscillations (QPOs) observed in the X-ray tails of magnetar flares. Additionally, slow build-up of elastic strain in the crust is thought to lead to “starquakes”, which may be associated with glitches in pulsar timing and bursts of high-energy radiation.

5.4 Worked Examples

Example 1: Inferring elastic moduli from seismic speeds. Suppose seismic observations in a homogeneous layer of a terrestrial planet give

$$v_P = 8.0 \text{ km s}^{-1}, \quad v_S = 4.5 \text{ km s}^{-1},$$

and the density is $\rho = 3.3 \times 10^3$ kg m $^{-3}$. Estimate the Lamé parameters λ and μ , and the Poisson ratio ν .

Solution. First, from $v_S^2 = \mu/\rho$,

$$\mu = \rho v_S^2 = 3.3 \times 10^3 (4.5 \times 10^3)^2 \approx 6.7 \times 10^{10} \text{ Pa}.$$

Next, from $v_P^2 = (\lambda + 2\mu)/\rho$,

$$\lambda + 2\mu = \rho v_P^2 = 3.3 \times 10^3 (8.0 \times 10^3)^2 \approx 2.1 \times 10^{11} \text{ Pa},$$

so

$$\lambda \approx 2.1 \times 10^{11} - 2(6.7 \times 10^{10}) \approx 7.7 \times 10^{10} \text{ Pa}.$$

The Poisson ratio is

$$\nu = \frac{\lambda}{2(\lambda + \mu)} \approx \frac{7.7}{2(7.7 + 6.7)} \approx 0.27,$$

a typical value for silicate rocks.

Example 2: Simple tidal bulge estimate for a homogeneous planet. Consider a homogeneous, incompressible, self-gravitating sphere of radius R , density ρ and shear modulus μ , subjected to a small tidal potential of amplitude U_0 at its surface. Using scaling arguments from Eq. (275), estimate the vertical tidal displacement u_r at the surface.

Solution. Roughly, the elastic-restoring term scales as

$$\mu \frac{u_r}{R^2},$$

while the tidal body force density is $\rho \nabla U \sim \rho U_0/R$. Setting these of the same order at equilibrium,

$$\mu \frac{u_r}{R^2} \sim \rho \frac{U_0}{R} \quad \Rightarrow \quad u_r \sim \frac{\rho R}{\mu} U_0.$$

For Earth-like parameters ($\rho \sim 5 \times 10^3 \text{ kg m}^{-3}$, $\mu \sim 7 \times 10^{10} \text{ Pa}$, $R \sim 6.4 \times 10^6 \text{ m}$) and a tidal potential of order $U_0 \sim 1 \text{ m}^2 \text{ s}^{-2}$, the estimate gives

$$u_r \sim \frac{5 \times 10^3 \cdot 6.4 \times 10^6}{7 \times 10^{10}} \times 1 \sim 0.5 \text{ m},$$

consistent with the order of magnitude of solid Earth tides.

Example 3: Fundamental shear mode of a neutron star crust. Model the crust of a neutron star as a homogeneous elastic spherical shell of thickness H on top of a fluid core. For a rough estimate, take a slab of thickness H and assume that the fundamental standing shear wave has wavelength $\lambda \sim 2H$. Show that its frequency is

$$f_0 \sim \frac{v_S}{2H},$$

and evaluate f_0 for $H = 1 \text{ km}$ and $v_S = 10^8 \text{ m s}^{-1}$.

Solution. A standing shear wave with nodes at the two boundaries satisfies $\lambda = 2H$ and $f_0 = v_S/\lambda$. Thus

$$f_0 \sim \frac{v_S}{2H} = \frac{10^8}{2 \times 10^3} = 5 \times 10^4 \text{ Hz}.$$

More realistic models including spherical geometry and density gradients give frequencies of tens to hundreds of Hz for global torsional modes; the slab estimate above thus slightly overestimates the frequency but gives the correct scale.

5.5 Exercises

1. Plane waves in an elastic medium.

Starting from Eq. (274) with $\mathbf{f} = 0$ and constant λ, μ, ρ , derive the dispersion relations for longitudinal and transverse plane waves and show that the wave speeds are v_P and v_S given above.

Elastic waves in a planet

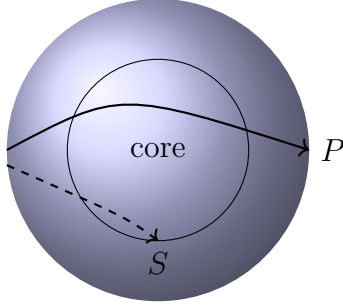


Figure 34: Schematic P and S waves in a layered planet. S -waves do not propagate through liquid regions.



Figure 35: Tidal deformation of a homogeneous planet.

2. Simple seismic travel time.

Consider a planet with a homogeneous mantle of thickness R in which v_P and v_S are constant. An earthquake occurs at depth h below the surface, directly “below” a seismometer.

- Derive expressions for the arrival times t_P and t_S .
- For $R = 3000$ km, $h = 100$ km, $v_P = 8$ km s⁻¹ and $v_S = 4.5$ km s⁻¹, compute t_P and t_S and the time delay $\Delta t = t_S - t_P$.

3. Tidal displacement of a homogeneous sphere (scaling).

Using Eq. (275), repeat the scaling argument of Example 2 and show that the dimensionless tidal Love number h_2 for a homogeneous incompressible sphere scales roughly as

$$h_2 \sim \frac{\rho g R}{\mu},$$

where g is the surface gravity. Discuss qualitatively what happens to h_2 when μ is very small (fluid planet) or very large (rigid body).

4. Torsional shear mode in a crustal shell.

Model a thin spherical elastic shell of thickness H and radius R as supporting purely tangential displacements $\mathbf{u}(\theta, \phi, t)$ with $\nabla \cdot \mathbf{u} = 0$ and no radial component. Show that these obey approximately a shear-wave equation on the sphere with speed v_S , and argue that the fundamental torsional mode has frequency of order

$$f \sim \frac{v_S}{2\pi R}.$$

Evaluate f for a neutron star with $R = 10$ km and $v_S = 10^8$ m s⁻¹.

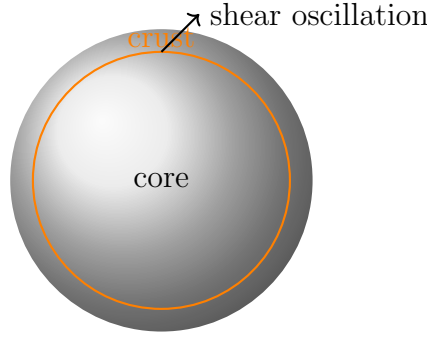


Figure 36: Neutron star with solid crust supporting shear waves.

5. Energy in a standing shear wave.

Consider a cube of side L filled with an isotropic elastic material. A standing shear wave in the x -direction has displacement

$$u_y(x, t) = A \sin(kx) \cos(\omega t), \quad u_x = u_z = 0.$$

- (a) Compute the shear strain u_{xy} and show that the time-averaged elastic energy stored in the cube is

$$\langle E_{\text{el}} \rangle = \frac{1}{4} \mu k^2 A^2 L^3.$$

- (b) Using the dispersion relation for shear waves, express this energy in terms of ρ , v_s , ω , A and L .

6 Solved problems

6.1 Long rod standing vertically in a gravitational field

Determine the deformation of a long rod (with length l) standing vertically in a gravitational field.

We take the z -axis along the axis of the rod, and the xy -plane in the plane of its lower end. The equations of equilibrium are

$$\frac{\partial \sigma_{ij}}{\partial x_i} = 0, \quad \frac{\partial \sigma_{iz}}{\partial x_i} = \rho g.$$

On the sides of the rod all components σ_{ik} except σ_{iz} must vanish, and on the upper end ($z = l$) $\sigma_{ix} = \sigma_{iy} = \sigma_{iz} = 0$.

The nonzero components consistent with equilibrium are

$$\sigma_{zz} = -\rho g(l - z), \quad (277)$$

with all other $\sigma_{ik} = 0$. From σ_{iz} we find U_{ik} to be

$$U_{xz} = U_{yz} = \frac{\sigma_{iz}}{E} = 0, \quad U_{zz} = \frac{\sigma_{zz}}{E}.$$

Integrating, we obtain the components of the displacement vector

$$u_x = \frac{\rho g(l - z)x}{E}, \quad u_y = \frac{\rho g(l - z)y}{E}, \quad u_z = \frac{\rho g}{2E} [l^2 - (l - z)^2] - \sigma(x^2 + y^2). \quad (278)$$

6.2 Hollow sphere

Determine the deformation of a hollow sphere (with external and internal radii R_2 and R_1) with a pressure p_1 inside and p_2 outside.

Using spherical polar coordinates, the displacement vector is radial:

$$\mathbf{u} = u_r(r) \hat{\mathbf{e}}_r,$$

and

$$\operatorname{div} \mathbf{u} = \frac{1}{r^2} \frac{d}{dr} (r^2 u_r) = \text{constant} \equiv 3a.$$

Thus

$$u_r = ar + \frac{b}{r^2}.$$

The components of the strain tensor are $U_{rr} = a - 2b/r^3$, $U_{\theta\theta} = U_{\phi\phi} = a + b/r^3$. The radial stress is

$$\sigma_{rr} = \frac{E}{(1+\sigma)(1-2\sigma)} [(1-\sigma)U_{rr} + 2\sigma U_{\theta\theta}] = \frac{E}{1-2\sigma} a - \frac{2E}{1+\sigma} \frac{b}{r^3}. \quad (279)$$

The constants a and b follow from the boundary conditions $\sigma_{rr} = -p_1$ at $r = R_1$ and $\sigma_{rr} = -p_2$ at $r = R_2$:

$$a = \frac{p_1 R_1^3 - p_2 R_2^3}{R_2^3 - R_1^3} \frac{1-2\sigma}{E}, \quad b = \frac{R_1^3 R_2^3 (p_1 - p_2)}{R_2^3 - R_1^3} \frac{1+\sigma}{2E}. \quad (280)$$

For a thin shell of thickness $h = R_2 - R_1$:

$$u = \frac{pR(1-\sigma)}{2Eh}, \quad \sigma_{\theta\theta} = \sigma_{\phi\phi} = \frac{pR}{2h}, \quad \bar{\sigma}_{rr} = \frac{p}{2}. \quad (281)$$

6.3 Solid sphere

Determine the deformation of a solid sphere (with radius R) in its own gravitational field.

With gravitational force $-gr/R$, the displacement satisfies:

$$\frac{E(1-\sigma)}{(1+\sigma)(1-2\sigma)} \frac{d}{dr} \left(\frac{1}{r^2} \frac{d}{dr} (r^2 u_r) \right) = \rho g \frac{r}{R}. \quad (282)$$

The finite solution satisfying $\sigma_{rr} = 0$ at $r = R$ is

$$u_r = -\frac{\rho g R(1-2\sigma)(1+\sigma)}{10E(1-\sigma)} r \left(\frac{3-\sigma}{1+\sigma} - \frac{r^2}{R^2} \right). \quad (283)$$

It should be noticed that the substance is compressed ($u_r < 0$) inside a spherical surface of radius $R\sqrt{(3-\sigma)/3(1+\sigma)}$ and stretched outside it ($u_r > 0$). The pressure at the centre of the sphere is

$$\frac{(3-\sigma)g\rho R}{10(1-\sigma)}.$$

6.4 Cylindrical pipe

Determine the deformation of a cylindrical pipe (with external and internal radii R_2 and R_1), with pressure p inside and no pressure outside.

Using cylindrical coordinates (r, θ, z) and radial displacement $u_r(r)$:

$$\frac{1}{r} \frac{d}{dr} (r u_r) = \text{constant} \equiv 2a.$$

Thus

$$u_r = ar + \frac{b}{r}.$$

The strain tensor gives

$$U_{rr} = a - \frac{b}{r^2}, \quad U_{\phi\phi} = \frac{u_r}{r} = a + \frac{b}{r^2}.$$

Boundary conditions $\sigma_{rr}(R_2) = 0$, $\sigma_{rr}(R_1) = -p$ give

$$a = \frac{pR_1^2}{R_2^2 - R_1^2} \frac{(1+\sigma)(1-2\sigma)}{E}, \quad b = \frac{pR_1^2 R_2^2}{R_2^2 - R_1^2} \frac{1+\sigma}{2E}. \quad (284)$$

The stresses become

$$\sigma_{rr} = \frac{pR_1^2}{R_2^2 - R_1^2} \left(1 - \frac{R_2^2}{r^2} \right), \quad \sigma_{\phi\phi} = \frac{pR_1^2}{R_2^2 - R_1^2} \left(1 + \frac{R_2^2}{r^2} \right), \quad \sigma_{zz} = \frac{2p\sigma R_1^2}{R_2^2 - R_1^2}. \quad (285)$$

6.5 Cylinder rotating uniformly

Determine the deformation of a cylinder rotating uniformly about its axis.

The centrifugal force $\rho\Omega^2 r$ acts outward. Displacement $u_r(r)$ satisfies

$$\frac{E(1-\sigma)}{(1+\sigma)(1-2\sigma)} \frac{d}{dr} \left(\frac{1}{r} \frac{d}{dr} (r u_r) \right) = -\rho\Omega^2 r. \quad (286)$$

Finite at $r = 0$ and $\sigma_{rr} = 0$ at $r = R$ yields

$$u_r = \frac{\rho\Omega^2(1+\sigma)(1-2\sigma)}{8E(1-\sigma)} \left[r(R^2 - \frac{2}{3}r^2) \right]. \quad (287)$$

6.6 Equations of equilibrium for an isotropic body

Derive the equations of equilibrium for an isotropic body (in the absence of body forces) in terms of the components of the stress tensor.

The required system of equations contains the three equations

$$\frac{\partial \sigma_{ik}}{\partial x_k} = 0. \quad (288)$$

and also the equations resulting from the fact that the six different components of U_{ik} are not independent quantities. To derive these equations, we first write down the system of differential relations satisfied by the components of the tensor U_{ik} . It is easy to see that the quantities

$$U_{ik} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_k} + \frac{\partial u_k}{\partial x_i} \right) \quad (289)$$

satisfy identically the relations

$$\frac{\partial^2 U_{ik}}{\partial x_l \partial x_m} + \frac{\partial^2 U_{lm}}{\partial x_i \partial x_k} = \frac{\partial^2 U_{il}}{\partial x_k \partial x_m} + \frac{\partial^2 U_{km}}{\partial x_i \partial x_l}. \quad (290)$$

Here there are only six essentially different relations, namely those corresponding to the following values of i, k, l, m : 1122, 1133, 2233, 1123, 2213, 3312. All these are retained if the above tensor equation is contracted with respect to l and m :

$$\nabla^2 U_{ik} + \frac{\partial^2 \sum_l U_{ll}}{\partial x_i \partial x_k} = \sum_l \frac{\partial^2 U_{il}}{\partial x_k \partial x_l} + \sum_l \frac{\partial^2 U_{kl}}{\partial x_i \partial x_l}. \quad (291)$$

Substituting here U_{ik} in terms of σ_{ik} we obtain the required equations:

$$(1 + \nu) \nabla^2 \sigma_{ik} + \frac{\partial^2 \sum_l \sigma_{ll}}{\partial x_i \partial x_k} = 0. \quad (292)$$

These equations remain valid in the presence of external forces constant throughout the body.

Contracting the above equation with respect to the suffixes i and k , we find that $\Delta \sigma_{ll} = 0$, i.e. σ_{ll} is a harmonic function. Taking the Laplacian of equation (3), we then find that $\Delta \Delta \sigma_{ik} = 0$, i.e. the components σ_{ik} are biharmonic functions.

6.7 General integral of the equations of equilibrium

Express the general integral of the equations of equilibrium (in the absence of body forces) in terms of an arbitrary biharmonic vector (B. G. Galerkin 1930).

We try

$$\vec{u} = \Delta \vec{f} + A \nabla (\nabla \cdot \vec{f}),$$

so that

$$\operatorname{div} \vec{u} = (1 + A) \Delta f.$$

Substituting in the equation of equilibrium gives

$$(1 - 2\sigma) \Delta \Delta f + [2(1 - \sigma)A + 1] \Delta \Delta f = 0. \quad (293)$$

Thus if f is biharmonic ($\Delta \Delta f = 0$):

$$\vec{u} = \Delta \vec{f} - \frac{1}{2(1 - \sigma)} \nabla (\nabla \cdot \vec{f}). \quad (294)$$

7 Problems

1. A material is called *incompressible* if there is no change of volume under any and all states of stresses. Show that for an incompressible isotropic linearly elastic solid with finite Young's modulus E , (a) Poisson's ratio $\nu = \frac{1}{2}$, (b) the shear modulus $\mu = E_Y/3$, and (c) $k \rightarrow \infty$, $\lambda \rightarrow \infty$ and $k - \lambda = 2\mu/3$.

2. If the components of strain at a point of structural steel are

$$U_{11} = 36 \times 10^{-6}, \quad U_{22} = 40 \times 10^{-6}, \quad U_{33} = 25 \times 10^{-6},$$

$$U_{12} = 12 \times 10^{-6}, \quad U_{23} = 0, \quad U_{13} = 30 \times 10^{-6},$$

find the stress components.

$$\lambda = 119.2 \text{ GPa}, \quad \mu = 79.2 \text{ GPa}.$$

3. An isotropic elastic body ($E_Y = 207 \text{ GPa}$, $\mu = 79.2 \text{ GPa}$) has a uniform state of stress given by

$$[\sigma] = \begin{bmatrix} 100 & 40 & 60 \\ 40 & -200 & 0 \\ 60 & 0 & 200 \end{bmatrix} \text{ MPa}.$$

(a) What are the strain components?

(b) What is the total change of volume for a five-centimeter cube of the material?

4. An isotropic elastic sphere ($E_Y = 207 \text{ GPa}$, $\mu = 79.2 \text{ GPa}$) of radius 5 cm is under the uniform stress field

$$[\sigma] = \begin{bmatrix} 6 & 2 & 0 \\ 2 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ MPa}.$$

Find the change of volume for the sphere.

5. Given the following displacement field in an isotropic linearly elastic solid:

$$u_1 = kX_3X_2, \quad u_2 = kX_3X_1, \quad u_3 = k(X_1^2 - X_2^2), \quad k = 10^{-4},$$

(a) Find the stress components.

(b) In the absence of body forces, is the state of stress a possible equilibrium stress field?

6. Given the following displacement field in an isotropic linearly elastic solid:

$$u_1 = kX_2X_3, \quad u_2 = kX_1X_3, \quad u_3 = kX_1X_2, \quad k = 10^{-4},$$

(a) Find the stress components.

(b) In the absence of body forces, is the state of stress a possible equilibrium stress field?

7. Given the following displacement field in an isotropic linearly elastic solid:

$$u_1 = kX_2X_3, \quad u_2 = kX_1X_3, \quad u_3 = k(X_1X_2 + X_3^2), \quad k = 10^{-4},$$

- (a) Find the stress components.
- (b) In the absence of body forces, is the state of stress a possible equilibrium stress field?

8. Given a motion

$$x_1 = X_1 + k(X_1 + X_2), \quad x_2 = X_2 + k(X_1 - X_2),$$

show that for a function $f(a, b) = ab$,

$$f(x_1, x_2) = f(X_1, X_2) + O(k), \quad (295)$$

$$\frac{\partial f(x_1, x_2)}{\partial x_1} = \frac{\partial f(X_1, X_2)}{\partial X_1} + O(k), \quad (296)$$

where $O(k) \rightarrow 0$ as $k \rightarrow 0$.

- 9. Do the previous problem for $f(a, b) = a^2 + b^2$.
- 10. Show that for any function $f(s)$, the displacement $u_1 = f(s)$ where $s = x_1 \pm c_L t$ satisfies the wave equation

$$\frac{\partial^2 u_1}{\partial t^2} = c_L^2 \frac{\partial^2 u_1}{\partial x_1^2}.$$

Calculate the ratio of the phase velocities c_L/c_T for the following Poisson's ratios: 1/3, 0.49, and 0.499.

- 11. Assume a displacement field that depends only on x_2 and t , i.e. $u_i = u_i(x_2, t)$. Obtain the differential equations that $u_i(x_2, t)$ must satisfy to be a possible motion in the absence of body forces.
- 12. Consider a linearly elastic medium. Assume the following form for the displacement field:

$$u_1 = \varepsilon [\sin \beta(x_3 - ct) + \alpha \sin \beta(x_3 + ct)], \quad u_2 = u_3 = 0.$$

- (a) What is the nature of this elastic wave (longitudinal, transverse, direction of propagation)?
- (b) Find the strains and stresses, and determine under what condition(s) the equations of motion are satisfied in the absence of body forces.
- (c) Suppose that there is a boundary at $x_3 = 0$ that is traction free. Under what condition(s) will the above motion satisfy this boundary condition for all time?
- (d) Suppose that there is a boundary at $x_3 = \ell$ that is also traction free. What further conditions will be imposed on the above motion to satisfy this boundary condition for all time?
- 13. Do the previous problem 12. if the boundary $x_3 = 0$ is fixed (no motion) and $x_3 = \ell$ is traction free.
- 14. Do the previous problem 12. if the boundaries $x_3 = 0$ and $x_3 = \ell$ are both rigidly fixed (no motion).

15. Do the previous problem 12. if the assumed displacement field is of the form:

$$u_3 = \varepsilon [\sin \beta(x_3 - ct) + \alpha \sin \beta(x_3 + ct)], \quad u_1 = u_2 = 0.$$

16. Do the previous problem 15. if the boundary $x_3 = 0$ is fixed (no motion) and $x_3 = \ell$ is traction free ($t = 0$).

17. Do the previous problem 15. if the boundaries $x_3 = 0$ and $x_3 = \ell$ are both rigidly fixed.

18. Consider the displacement field $u_i = u_i(x_1, x_2, x_3, t)$. In the absence of body forces:

- (a) obtain the governing equation for u_i for the case where the motion is equivoluminal.
- (b) obtain the governing equation for the dilatation e for the case where the motion is irrotational: $\partial u_i / \partial x_j = \partial u_j / \partial x_i$.

19. Write a displacement field for an infinite train of longitudinal waves propagating in the direction of $3\mathbf{e}_1 + 4\mathbf{e}_2$.

20. Write a displacement field for an infinite train of transverse waves propagating in the direction of $3\mathbf{e}_1 + 4\mathbf{e}_2$ and polarized in the x_1x_2 plane.

21. Solve for ε_2 and ε_3 in terms of ε_1 from the following two algebraic equations:

$$\varepsilon_2 \cos 2\alpha_1 + \varepsilon_3 \eta \sin 2\alpha_3 = \varepsilon_1 \cos 2\alpha_1, \quad (297)$$

$$\varepsilon_2 \sin 2\alpha_1 - (\varepsilon_3 / \eta) \cos 2\alpha_1 = -\varepsilon_1 \sin 2\alpha_1. \quad (298)$$

22. A transverse elastic wave of amplitude ε_1 is incident on a traction-free plane boundary. If the Poisson's ratio $\nu = 1/3$, determine the amplitudes and angles of reflection of the reflected waves for the following two incident angles: (a) $\alpha_1 = 0$ and (b) $\alpha_1 = 15^\circ$.

23. Referring to the reflection of plane elastic waves, consider a transverse elastic wave incident on a traction-free plane surface ($x_2 = 0$) with an angle α_1 with the x_2 -axis and polarized normal to the x_1x_2 plane of incidence. Show that the boundary condition at $x_2 = 0$ can be satisfied with only a reflected transverse wave that is similarly polarized. Determine the relation between amplitudes, wavelengths, and propagation directions.

24. Repeat the previous problem 23. for the case where the boundary $x_2 = 0$ is fixed.

25. A longitudinal elastic wave is incident on a fixed boundary $x_2 = 0$ with an incident angle α_1 . Show that in general there are two reflected waves (one longitudinal and one transverse), and determine the amplitude ratio of reflected to incident waves.

26. Repeat the previous problem 25. for the case where the boundary is traction free.

27. Verify that the thickness-stretch vibration

$$u_1 = (A \cos kx_1 + B \sin kx_1)(C \cos c_L kt + D \sin c_L kt)$$

satisfies the longitudinal wave equation

$$\frac{\partial^2 u_1}{\partial t^2} = c_L^2 \frac{\partial^2 u_1}{\partial x_1^2}.$$

- (a) Find the thickness-stretch vibration of a plate where $x_1 = 0$ is subjected to a forced displacement $u = (\alpha \cos \omega t) \mathbf{e}_1$ and $x_1 = \ell$ is free.
 - (b) Determine the resonance frequencies.
28. (a). Find the thickness-stretch vibration if $x_1 = 0$ is subjected to a traction $t = (\beta \cos \omega t) \mathbf{e}_1$ and $x_1 = \ell$ is fixed. (b). Find the resonance frequencies.
29. (a). Find the thickness-shear vibration if $x_1 = 0$ has a forced displacement $u = (\alpha \cos \omega t) \mathbf{e}_3$ and $x_1 = \ell$ is fixed. (b). Find the resonance frequencies.
30. (a). Find the thickness-shear vibration if $x_1 = 0$ has a forced displacement $u = \alpha(\cos \omega t \mathbf{e}_2 + \sin \omega t \mathbf{e}_3)$ and $x_1 = \ell$ is fixed. (b). Find the resonance frequencies.

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